FED. STD. NO. 162a
May 9, 1967

SUPERSEDING
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June 14, 1%5

FEDERAL STANDARD

HOSE, RUBBER,
VISUAL INSPECTION GUIDE FOR

This standard was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

U.S. Government Printing Office: 1967

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INFORMATION SHEET

ON

FEDERAL STANDARDS

This Federal Standard is issued in loose leaf form to permit the insertion or removal of new or revised pages.

All users of Federal Standards should keep them up to date by inserting revised or new pages as issued and removing superseded and canceled pages.

New and revised pages will be issued under Change Notices which will be numbered consecutively and will bear the date of issuance. Change Notices should be retained and filed in front of the Standard until such time as they are superseded by a reissue of the entire Standard.

This Information Sheet should be filed immediately behind the cover page of the Federal Standard.

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Fed. Std. No. 162a
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June 14, 1965

FEDERAL STANDARD

HOSE, RUBBER, VISUAL INSPECTION GUIDE FOR

Authority. This standard is issued pursuant to the Federal Property and Administrative Services Act of 1949, as amended, and its application to the purchase of the commodities referred to herein is mandatory on all Federal agencies.

S1. Scope. This standard covers visible defects in natural or synthetic rubber hose.

S2. Identification. Throughout this standard, the table and figures are identified by the possible defect under study and the classification of defect. When examined and tested in accordance with the procurement document, any defect listed in this standard which permits leakage shall be reclassified "critical" when the hose and couplings are intended for any of the following uses: divers air hose, high pressure hydraulic hose, outer space application, oxygen hose, steam hose, gasoline hose, or any application where leakage can result in injury or loss of life. References to classes in table I are identified as follows :

Class A—a very high quality user-sensitive hose, primarily divers air supply hose.

Class B—high quality hose, such as high pressure hydraulic hose where failure could result in loss of life.

Class C—all other types of hose.

Where a class is not indicated, the defect is intended for all classes.

S3. Purpose. The purpose of this standard is to:

(a) Provide photographs and descriptions of the defects commonly found in rubber hose together with a statement of their usual classification: "major defect or minor defect". "Critical defects" are determined by examination and tests.

(b) Facilitate inspection of rubber hose and increase efficiency of Government inspectors.

(c) Provide a training manual for new Government inspectors.

S4. Changes. When a Federal agency considers that this standard does not provide for its essential needs, written request for changing or adding to the standard, supported by adequate justification, shall be sent to the Administration. This justification shall explain wherein the standard does not provide for essential needs. The request shall be sent in duplicate to the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C., 20406. The Administration will determine the appropriate action to be taken and will notify the agency.

S5. Applicable document. The following standard, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this standard to the extent specified herein.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

S6. Quality assurance provisions. The sample lot, inspection level, and acceptable quality level (AQL) shall be as stated in the procurement specification and be processed in accordance with MIL-STD-105.

S7. Notes.

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S7.1 Incorporated data. This standard incorporates the requirement of MIL-STD-166, dated June 22, 1954.

S7.2 Grouping of defects. The defects in table I are grouped as follows:

(a) Fibre jacket.
(b) Wrap cured.
(c) Press cured.
(d) Miscellaneous.

S7.3 Definitions. The definitions listed in MIL-STD-105 for "critical defects", "major defects", and "minor defects" are applicable herein.

S8. Index.

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TABLE I. Classification of defects

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Cover: Braided, nonrubber covered				See reinforcement								
Knitted, nonrubber covered				See reinforcement								
Woven jacket, nonrubber covered:												
Abrasion	X	----	1	----	----	----	----	----	----	----	----	----
Discolored-mildew	X	----	Not shown.	----	----	----	----	----	----	----	----	----
Discolored-uneven color due to treatment	----	X	Not shown.	----	----	----	----	----	----	----	----	----
Friction strike through	X	----	2	----	----	----	----	----	----	----	----	----
Jacket insufficient	----	X	3	----	----	----	----	----	----	----	----	----
	X	----	4	----	----	----	----	----	----	----	----	----
	----	X	5	----	----	----	----	----	----	----	----	----
Knot, filler	X	----	6	----	----	----	----	----	----	----	----	----
Knot, filler	Not a defect.		7	----	----	----	----	----	----	----	----	----
Knotted yarn	X	----	8	----	----	----	----	----	----	----	----	----
Irregularity	----	X	8	----	----	----	----	----	----	----	----	----
Pick, missing or broken	X	----	9	----	----	----	----	----	----	----	----	----
Skips, warp	X	----	10	----	----	----	----	----	----	----	----	----
Snag	X	----	11	----	----	----	----	----	----	----	----	----
Twist, inner jacket	X	----	12	----	----	----	----	----	----	----	----	----
Rubber covered: Blister				Also see cover, loose								
Blister, under cover, large	----	----	----	X	----	13, 14, 15, and 16	X	----	13, 14, 15, and 16	X	----	13, 14, 15, and 16
Blister, small (see fig. 16 for size limit): Classes A and B	----	----	----	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Class C	----	----	----	----	X	Not shown.	----	X	Not shown.	----	X	Not shown.
Bloom, dusty, oily or waxy (as specified in the procurement document)	----	----	----	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Bloom, not specified in the procurement document (see discolored)	----	----	----	Not a defect.		Not shown.	Not a defect.		Not shown.	Not a defect.		Not shown.
Break (also see split)	----	----	----	X	----	Not shown. Similar to figure 17.	X	----	17	X	----	17
Buckle: (because interior contour will be affected)	----	----	----	----	----	----	----	----	----	X	----	18

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TABLE I. Classification of defects, cont'd.

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Cover: cont'd.												
Bulge, large:									Not shown.			Not shown.
Dynamic use (Air Force procurement only), and class A	----	----	----	X	----	19	X	----	Similar to	X	----	Similar to
Classes B and C	----	----	----	----	X	19	----	X	figure 19.	----	X	figure 19.
Bulge, small:						Not shown.						Not shown.
Dynamic use—service abrasion (Air Force procurement only)	----	----	----	X	----	Similar to	X	----	20	----	----	Similar to
Other use	----	----	----	Not a defect.		figure 20.	Not a defect.		20	Not a defect.		figure 20.
Bump:												
In contour	----	----	----	----	----	----	----	----	----	----	X	21
In nipple area	----	----	----	----	----	----	----	----	----	X	----	21
Crack	----	----	----	X	----	22	X	----	22	X	----	22
Dent	----	----	----	X	----	Not shown. Similar to figures 23 and 24.	X	----	23	X	----	24
Depressions:												
Large area	----	----	----	X	----	Not shown. Similar to figures 25 and 26.	X	----	25 and 26	X	----	Not shown. Similar to figures 25 and 26.
Deep	----	----	----	X	----	Not shown. Similar to figures 27 and 28.	X	----	27 and 28	X	----	27 and 28
Small and light, Class A	----	----	----	X	----	Not shown. Similar to figures 29, 30, 31, and 32.	X	----	29, 30, 31, and 32	X	----	29, 30, 31, and 32
Classes B and C	----	----	----	----	X	----	----	X	29, 30, 31, and 32	----	X	29, 30, 31, and 32
Small and light	----	----	----	----	X	Not shown. Similar to figures 33 and 34.	----	X	33 and 34	----	X	33 and 34
Discolored	----	----	----	Not a defect.		35	Not a defect.		35	Not a defect.		35
Dog leg	----	----	----	----	----	----	X	----	36	----	----	----
Groove, deep	----	----	----	X	----	Not shown. Similar to figure 37.	X	----	37	X	----	37
Rupture, large	----	----	----	X	----	Not shown. Similar to figure 38.	X	----	38	X	----	38

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Hole: reinforcement exposed	----	----	----	X	----	Not shown. Similar to figures 39, 40, and 41.	X	----	39, 40, and 41	X	----	39, 40, and 41
Hole, small, reinforcement not exposed	----	----	----	----	X	Not shown. Similar to figure 41.	----	X	41	----	X	41
Lap: Foldover, opening	----	----	----	X	----	42 and 43	X	----	43	X	----	Not shown. Similar to figures 42 and 43.
	----	----	----	X	----	44	X	----	Not shown. Similar to figure 44.	X	----	Not shown. Similar to figure 44.
Open lap in fabric	----	----	----	X	----	Not shown. Similar to figure 45.	X	----	Not shown. Similar to figure 45.	X	----	45
No opening and no lack of adhesion (similar to fig. 43 but no delamination)	----	----	----	Not a defect.		----	Not a defect.		----	Not a defect.		----
Lead: Burst	----	----	----	----	----	----	X	----	46	----	----	----
Charge mark (dent) (see also dog leg)	----	----	----	----	----	----	X	----	47	----	----	----
Charge mark	----	----	----	----	----	----	X	----	48	----	----	----
	----	----	----	----	----	----	----	X	48	----	----	----
	----	----	----	----	----	----	Not a defect.		48	----	----	----
Charge mark (also see machine stop): Class A	----	----	----	----	----	----	----	X	49	----	----	----
Classes B and C	----	----	----	----	----	----	Not a defect.		49	----	----	----
Flakes, lead: Class A	----	----	----	----	----	----	X	----	50 and 51	----	----	----
Class B	----	----	----	----	----	----	X	----	50	----	----	----
	----	----	----	----	----	----	----	X	51	----	----	----
Class C	----	----	----	----	----	----	Not a defect.		51	----	----	----
	----	----	----	----	----	----	----	X	50	----	----	----
Machine stop (also see charge mark): Class A	----	----	----	----	----	----	X	----	52	----	----	----
Class B	----	----	----	----	----	----	----	X	52	----	----	----
Class C	----	----	----	----	----	----	Not a defect.		52	----	----	----
Pop (also see lead, burst): Through body	----	----	----	----	----	----	X	----	53	----	----	----
Reinforcement exposed	----	----	----	----	----	----	X	----	54	----	----	----
Reinforcement, not exposed: Class A	----	----	----	----	----	----	X	----	55	----	----	----
Class B	----	----	----	----	----	----	----	X	55	----	----	----
Class C	----	----	----	----	----	----	Not a defect.		55	----	----	----

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TABLE I. Classification of defects, cont'd.

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Cover: cont'd.												
Slit or stripper cut	----	----	----	----	----	----	X	----	56	----	----	----
Stripper cut (into braided cover)	----	----	----	----	----	----	X	----	57	----	----	----
Light cover, under gage, or thickness below tolerance limit	----	----	----	X	----	Not shown. Similar to figures 58 and 59.	X	----	58 and 59	----	----	----
Light cover (roughness), thickness not below specified or not specified	----	----	----	----	X	Not shown. Similar to figures 60, 61, 62, and 63.	----	X	60, 61, 62, and 63	----	----	----
Loose (also see blister)	----	----	----	X	----	64	X	----	65	X	----	64 and 65
Lubricant, excess	----	----	----	----	----	----	Not a defect.		66	----	----	----
Lump, foreign material inclusion or lamination	----	----	----	----	----	----	X	----	67 and 68	----	----	----
Lump, small protrusion, no hole: Class A	----	----	----	----	----	----	X	----	69	----	----	----
Classes B and C	----	----	----	----	----	----	----	X	69	----	----	----
Missing (reinforcement exposed)	----	----	----	X	----	70 and 71	X	----	70 and 71	X	----	70 and 71
Patched cover, open edges	----	----	----	X	----	Not shown. Similar to figure 72.	X	----	72	X	----	72
Patched incorrect thickness (either high or low spot)	----	----	----	X	----	73	X	----	Not shown. Similar to figure 73.	X	----	Not shown. Similar to figure 73.
Patched-correct thickness and securely cured	----	----	----	Not a defect.		73 and 74	Not a defect.		Not shown. Similar to figures 73 and 74.	Not a defect.		Not shown. Similar to figures 73 and 74.
Porosity: Extended area, over 10 percent of thickness	----	----	----	X	----	75	X	----	Not shown. Similar to figure 75.	X	----	Not shown. Similar to figure 75.
Extended area, under 10 percent of thickness	----	----	----	----	X	Not shown.	----	X	Not shown.	----	X	Not shown.
Small localized area, reinforcement not exposed	----	----	----	----	X	76	----	X	Not shown. Similar to figure 76.	----	X	Not shown. Similar to figure 76.
Small localized area, under 10 percent of thickness	----	----	----	Not a defect.		Not shown.	Not a defect.		Not shown.	Not a defect.		Not shown.

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Pricker mark, through to reinforcement (note: when specified in procurement document for service requirement, not a defect)	----	----	----	X	----	Not shown. Similar to figure 77.	X	----	77	X	----	Not shown. Similar to figure 77.
Pricker mark or air tracks, small area, not through to reinforcement	----	----	----	----	X	Not shown. Similar to figure 78.	----	X	78	----	X	Not shown. Similar to figure 78.
Ridge: Lamination or separation	----	----	----	X	----	79	X	----	Not shown. Similar to figure 79.	----	----	Not shown. Similar to figure 79.
Large	----	----	----	X	----	80	X	----	Not shown. Similar to figure 80.	X	----	Not shown. Similar to figure 80.
Small	----	----	----	----	X	81 and 82	----	X	Not shown. Similar to figures 81 and 82.	----	X	Not shown. Similar to figures 81 and 82.
Note: Edges of depressions due to wrapper wrinkles are usually sharply defined and regularly patterned thus distinguishing them from surface irregularities caused by buckled carcass which have rounded edges. Air Force procurement only—more than 6 percent of hose length	----	----	----	X	----	81 and 82	X	----	Not shown. Similar to figures 81 and 82.	----	X	Not shown. Similar to figures 81 and 82.
Rough	----	----	----	X	----	Not shown.	X	----	83	X	----	Not shown. Similar to figure 83.
Slit				See Lead, slit								
Split	----	----	----	X	----	Not shown. Similar to figures 84 and 85.	X	----	84 and 85	X	----	Not shown. Similar to figures 84 and 85.
Stripper cut				See Lead, stripper cut								
Thickness, over, not mentioned in procurement document	----	----	----	Not a defect.		Not shown. Similar to figure 86.	Not a defect.		86	Not a defect.		Not shown. Similar to figure 86.
Thickness, under, below tolerance light. (Also see light (gage))	----	----	----	X	----	87	X	----	87	X	----	87
Thickness, eccentric (lopsided), outside tolerance	----	----	----	X	----	Not shown. Similar to figure 88.	X	----	88	X	----	Not shown. Similar to figure 88.
Thickness, eccentric (lopsided) tolerance met	----	----	----	Not a defect.		Not shown. Similar to figure 88.	Not a defect.		88	Not a defect.		Not shown. Similar to figure 88.
Wrinkles	----	----	----	----	----	----	----	----	----	X	----	89
Wrinkles, wrapper, medium	----	----	----	X	----	90	----	----	----	----	----	----
Wrinkles, wrapper, small	----	----	----	----	X	91	----	----	----	----	----	----
Wrinkles, wrapper, fine	----	----	----	Not a defect.		92 and 93	----	----	----	----	----	----

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TABLE I. Classification of defects, cont'd.

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Ends:												
Capped:												
Fabric exposed	----	----	----	X	----	94	----	----	----	----	----	----
Flash or rind, excessive	----	----	----	X	----	95	----	----	----	----	----	----
Flash or rind, small	----	----	----	Not a defect.		95	----	----	----	----	----	----
Loose	----	----	----	X	----	96	----	----	----	----	----	----
Missing, where specified	----	----	----	X	----	Not shown.	----	----	----	----	----	----
Cut:						Not shown.						
Angle or diagonal, specified length	X	----	97	X	----	Similar to	X	----	98	X	----	98
Angle or diagonal, random length	Not a defect.		97	Not a defect.		figures 97 and 98.	Not a defect.		----	Not a defect.		----
Ends:			Not shown.									
Ragged, specified length	X	----	Similar to	X	----	99	X	----	100	X	----	100
Ragged, random length	Not a defect.		figures 99 and 100.	Not a defect.		99	Not a defect.		100	Not a defect.		100
Untrimmed:						Not shown.			Not shown.			
Specified length	X	----	101	X	----	Similar to	X	----	Similar to	X	----	101
Random length	Not a defect.		101	Not a defect.		figure 101.	Not a defect.		figure 101.	Not a defect.		101
Reinforced: (also see ends capped)												
Lamination (separation)	----	----	----	X	----	102 and 103	----	----	----	X	----	Not shown. Similar to figures 102 and 103.
Void	----	----	----	X	----	104, 105, and 106	----	----	----	X	----	Not shown. Similar to figures 105 and 106.
Fittings (couplings, end):												
Connections, flanges:			Not shown.			Not shown.						Not shown.
Angle set	X	----	Similar to figure 107.	X	----	Similar to figure 107.	X	----	107	X	----	Similar to figure 107.
Bands, clamps, ferrules or nipples —incorrectly placed	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Bands, clamps, ferrules or nipples —wrong size	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Bond, physical or vulcanized to hose, loose	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Bond, static wire or electrical to coupling where specified missing loose (does not make good electrical contact)	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Cracked or split metal	X	----	108 and 109	X	----	108 and 109	X	----	108 and 109	X	----	108 and 109
Ferrule, not crimped properly	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.

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Gasket (washer) incorrect size or missing	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Gasket, tail for expansion ring coupling, incorrect size or missing	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Gasket, tail, or binder washer for expansion ring coupling, not properly placed in recess	X	----	110	X	----	Not shown. Similar to figure 110.	X	----	Not shown. Similar to figure 110.	X	----	Not shown. Similar to figure 110.
Hole, bolt, flange connection missing outside tolerance improperly placed	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Holes, bolt, flange connection, not aligned less than 10 foot length	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Holes, bolt, flange connection, not aligned, 10 feet and over in length	Not a defect.		Not shown.	Not a defect.		Not shown.	Not a defect.		Not shown.	Not a defect.		Not shown.
Insertion distance wrong	X	----	Not shown. Similar to figure 111.	X	----	Not shown. Similar to figure 111.	X	----	Not shown.	X	----	Not shown. Similar to figure 111.
Nipple, loose	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Nipple not expanded (internal expanded type only)	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Expansion ring:				See figure 112								
Loose or not expanded	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Placement:												
In waterway, too much gap	X	----	112	X	----	Similar to figure 112.	X	----	Similar to figure 112.	X	----	Similar to figure 112.
Sleeve, not tightened properly	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Swivel, female, tight or frozen	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
General:												
Cement in tube	----	X	113 and 114	----	X	113 and 114	----	X	113 and 114	----	X	113 and 114
Clamp mark	X	----	115 and 116	----	----	----	X	----	117	----	----	----
Crushed, wire reinforced	----	----	----	X	----	118	X	----	118	----	----	----
Fabric pulled out from coupling	X	----	119	X	----	Not shown. Similar to figure 119.	X	----	Not shown. Similar to figure 119.	----	----	----
Kinked, wire reinforced	----	----	----	X	----	120	X	----	Not shown. Similar to figure 120.	----	----	----
Rough, surface	See Cover: light (gage) figures 60 and 61											
Skips, corrugated surface	----	----	----	----	----	----	Not a defect.		121	----	----	----
Smooth, corrugated surface	----	----	----	----	----	----	Not a defect.		122	----	----	----
Twisted, nonwire reinforced	----	----	----	----	----	----	X	----	123	----	----	----
Printing or branding:												
Color, wrong	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Illegible	X	----	Not shown. Similar to figure 124.	X	----	124	X	----	125	X	----	Not shown. Similar to figure 125.
Information, wrong	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.

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TABLE I. Classification of defects, cont'd.

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Printing or branding: cont'd. Location, improper	----	X	126	----	X	Not shown. Similar to figure 126.	----	X	Similar to figure 126.	----	X	Not shown. Similar to figure 126.
Loose, large area	----	----	----	X	----	127	X	----	127	----	----	----
Loose, small area	----	----	----	----	X	Not shown.	----	X	Not shown.	----	----	----
Missing, when specified	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Slipped brand, legible	----	----	----	----	X	Not shown. Similar to figure 128.	----	X	128	----	----	----
Slipped legible	----	----	----	Not a defect.		129	Not a defect.		Not shown. Similar to figure 129.	----	----	----
Reinforcement: Braided: Ball-up	X	----	Not shown. Similar to figures 130 and 131.	X	----	Not shown. Similar to figures 130 and 131.	X	----	130 and 131	----	----	----
Broken strand, more than two per plait or two in 2 inches	----	----	----	----	----	----	X	----	132	----	----	----
Broken strand, less than two per plait or less than two every 2 inches	----	----	----	----	----	----	----	X	Not shown.	----	----	----
Crossed strand, first braid	----	----	----	----	----	----	X	----	133	----	----	----
Crossed strand, other than first braid	----	----	----	----	----	----	Not a defect.		134	----	----	----
Dough (filler) where specified, absent or missing	----	----	----	X	----	135	X	----	135	----	----	----
Dough (filler) where specified, excessive	----	----	----	X	----	136	X	----	Not shown. Similar to figure 136.	----	----	----
Flare, wire at end 1/8 inch greater than braid O.D.	----	----	----	----	----	----	X	----	137	----	----	----
Friction between braids where specified, missing	----	----	----	X	----	Not shown. Similar to figure 138.	X	----	138	----	----	----
Knot in yarn	----	----	----	X	----	Not shown. Similar to figure 139.	X	----	139	----	----	----
Looped strand, first braid, class A	----	----	----	X	----	Not shown.	X	----	140	----	----	----
Classes B and C	----	----	----	----	X	Similar to figure 140.	----	X	140	----	----	----
Other than first braid	----	----	----	----	X	Not shown. Similar to figure 140.	----	X	140	----	----	----

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Looped wires, more than two per plait (carrier) or two in 2 inches	----	----	----	X	----	Not shown. Similar to figure 141.	X	----	141	----	----	----
Looped wires, less than two per plait (carrier) or less than two in 2 inches	----	----	----	----	X	142	----	X	142	----	----	----
Loose braid more than two plaits (carriers) in one direction	----	----	----	----	X	----	X	----	Not shown.	----	----	----
Loose braid, more than 2 plaits (carriers) in either direction	----	----	----	----	X	----	----	X	143	----	----	----
Loose braid, one plait (carrier) in either direction	----	----	----	----	----	----	Not a defect.		144 and 145	----	----	----
Pitch, out of tolerance	X	----	146	X	----	147	----	----	----	----	----	----
Missing braid or ply	----	----	----	X	----	Not shown.	X	----	Not shown.	----	----	----
Missing plait (carrier or strand)-more than one	----	----	----	X	----	Not shown.	X	----	Not shown.	----	----	----
One only, class A	----	----	----	X	----	148	X	----	148	----	----	----
One only, classes B and C	----	----	----	----	X	148	----	X	148	----	----	----
Rusty wire	----	----	----	X	----	Not shown.	X	----	Not shown.	----	----	----
Wavy braid	----	----	----	----	X	Not shown.	----	X	149	----	----	----
Wire through cover	----	----	----	X	----	Not shown. Similar to figure 152.	X	----	150 and 151	X	----	152
Wire through tube	----	----	----	X	----	153	X	----	Not shown. Similar to figure 153.	X	----	153
Knitted, skip	----	----	----	X	----	154	X	----	154	X	----	154
Woven jacket, rubber covered (also see cover: woven jacket, nonrubber covered): ball-up	----	----	----	X	----	155	X	----	155	----	----	----
Broken or missing warp yarns: exceeding 4	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	----	----	----
Four or less	----	X	Not shown.	----	X	Not shown.	----	X	Not shown.	----	----	----
Yarn imbedded between jackets: More than 2 feet long	X	----	156	X	----	Not shown.	X	----	Not shown.	----	----	----
Less than 6 inches from end	X	----	156	X	----	Similar to figure 156.	X	----	Similar to figure 156.	----	----	----
Less than 2 feet long	----	X	156	----	X		----	X		----	----	----
6 inches or more from end	----	X	156	----	X		----	X		----	----	----
Wrapped fabric: Buckled or wrinkled	----	----	----	X	----	157, 158, and 159	----	----	----	X	----	157, 158, and 159
Incorrect number of plies or open lap	----	----	----	X	----	160	----	----	----	X	----	160
Lap insufficient (½ inch minimum on ½ inch ID and smaller, minimum lap equal to ID from ½ to 1 inch, inclusive, 1 inch minimum lap on ID over 1 inch for radiator hose minimum lap equal ¼ inch for all sizes)	----	----	----	X	----	Not shown.	----	----	----	X	----	Not shown.

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TABLE I. Classification of defects, cont'd.

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Reinforcement: cont'd. Ply turnback at leading edge—improper ply distribution	----	----	----	X	----	161	----	----	----	X	----	161
Wrapped fabric, wire reinforced: broken wire	----	----	----	X	----	Not shown.	----	----	----	X	----	Not shown.
Corrugations bad	----	----	----	X	----	162	----	----	----	X	----	162
Filler, insufficient (noncorrugated surface hose)	----	----	----	----	X	163	----	----	----	----	X	163
Filler, insufficient (noncorrugated surface hose (Air Force procurement only)	----	----	----	X	----	163	----	----	----	X	----	163
Pitch-incorrect	----	----	----	X	----	164	----	----	----	X	----	164
Pitch-uneven	----	----	----	----	X	165	----	----	----	----	X	165
Wire, rusty or wrong size	----	----	----	X	----	Not shown.	----	----	----	X	----	Not shown.
Wire, not tied down properly at end	----	----	----	X	----	163	----	----	----	X	----	163
Wire, static, bulge: With loose cover	----	----	----	X	----	Not shown. Similar to figure 166.	X	----	166	X	----	Not shown. Similar to figure 166.
Dynamic use (Air Force procurement only)	----	----	----	X	----		X	----	166	X	----	
With adhered cover	----	----	----	Not a defect.			Not a defect.		166	Not a defect.		166
Yarn through cover	----	----	----	X	----	167	X	----	168	X	----	Not shown. Similar to figure 167.
Yarn in cover: Class A	----	----	----	X	----	Not shown. Similar to figure 169.	X	----	169	X	----	Not shown. Similar to figure 169.
Classes B and C	----	----	----	----	X		----	X	169	----	X	
Tube: Blister	(See Cover, rubber covered—blister.)											
Bloom	(See Cover, rubber covered—bloom.)											
Break	(See Cover, rubber covered—break.)											
Buckle	X	----	170	X	----	Not shown. Similar to figure 170.	X	----	Not shown. Similar to figure 170.	X	----	Not shown. Similar to figure 170.
Bulge	(See Tube, buckle.)											
Bunching	(See Tube, buckle.)											
Cement, excess Class A	X	----	171 and 172	X	----	171 and 172	X	----	171 and 172	X	----	171 and 172
Classes B and C	----	X		----	X		----	X		----	X	
Crack	(See Cover, rubber covered—crack.)											
Dent	(See Cover, rubber covered—dent.)											

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Discolored: not a defect unless specified in procurement document	(See Cover, rubber-covered—discolored.)											
Fold	X	----	173	X	----	Not shown. Similar to figure 173.	X	----	Not shown. Similar to figure 173.	X	----	Not shown. Similar to figure 173.
Hole	X	----	Not shown. Similar to figures 174, 175, and 176.	X	----	174	X	----	Not shown. Similar to figures 175 and 176.	X	----	Not shown. Similar to figures 174, 175, and 176.
Lap, open or loose	X	----	Not shown. Similar to figure 177.	X	----	177	X	----	Not shown. Similar to figure 177.	X	----	Not shown. Similar to figure 177.
Loose (separation)	X	----	178, 179, and 180	X	----	182 and 181	X	----	Not shown. Similar to figures 180, 181, and 182.	X	----	Not shown. Similar to figures 180, 181, and 182.
Lubricant, excess or loose:												
Class A	X	----	183	X	----	Not shown. Similar to figure 183.	X	----	Not shown. Similar to figure 183.	X	----	Not shown. Similar to figure 183.
Classes B and C	----	X	183	----	X		----	X		----	X	
Lubricant, normal amount	Not a defect.		184	Not a defect.		184	Not a defect.		184	Not a defect.		184
Lumps	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Missing	X	----	Not shown. Similar to figures 185 and 186.	X	----	Not shown. Similar to figures 185 and 186.	X	----	Not shown. Similar to figures 185 and 186.	X	----	Not shown. Similar to figures 185 and 186.
Off-center:												
Tube-thickness below tolerance	X	----	Not shown. Similar to figure 187.	X	----	Not shown. Similar to figure 187.	X	----	187	X	----	Not shown. Similar to figure 187.
Tube thickness meets tolerance	Not a defect.			Not a defect.			Not a defect.		187	Not a defect.		
Pits, deep	X	----	188 and 189	X	----	Not shown. Similar to figures 188 and 189.	X	----	Not shown. Similar to figures 188 and 189.	X	----	Not shown. Similar to figures 188 and 189.
Pits, small, less than 15 percent of tube thickness:												
Class A	X	----	Not shown. Similar to figure 190.	X	----	190	X	----	Not shown. Similar to figure 190.	X	----	Not shown. Similar to figure 190.
Classes B and C	----	X		----	X	190	----	X		----	X	
Pits, small irregularities	Not a defect.		Not shown. Similar to figure 191.	Not a defect.		Not shown. Similar to figure 191.	Not a defect.		191	Not a defect.		Not shown. Similar to figure 191.
Porosity, more than 15 percent of tube thickness	X	----	Not shown. Similar to figure 192.	X	----	192	X	----	Not shown. Similar to figure 192.	X	----	Not shown. Similar to figure 192.
Porosity-15 percent or less of tube thickness												
Class A	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.	X	----	Not shown.
Classes B and C	----	X		----	X		----	X		----	X	

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TABLE I. Classification of defects, cont'd.

Defect	Fibre jacket			Wrap cured			Press cured			Miscellaneous		
	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number	Major	Minor	Figure number
Tube: cont'd.												
Ridge				(See Cover, rubber-covered—ridge.)								
Ring												
Class A	X	----	Not shown. Similar to figures 193 and 194.	X	----	193 and 194	X	----	Not shown. Similar to figures 193 and 194.	X	----	Not shown. Similar to figures 193 and 194.
Classes B and C	Not a defect.			Not a defect.			Not a defect.			Not a defect.		
Size larger than size shown on figures 189 and 190	----	X		----	X		----	X		----	X	
Ridge, over 1/64 inch or over 10 percent of tube thickness, whichever is greater	X	----	Not shown. Similar to figure 195.	X	----	Not shown. Similar to figures 195 and 196.	X	----	195	X	----	Not shown. Similar to figure 195.
Ridge, less than 1/64 inch or less than 10 percent of tube thickness, whichever is greater												
Class A	----	X	Not shown. Similar to figures 195 and 196.	----	X	Not shown. Similar to figures 195 and 196.	----	X	196	----	X	Not shown. Similar to figures 195 and 196.
Classes B and C	Not a defect.			Not a defect.			Not a defect.			Not a defect.		
Rough, over 1/64 inch or over 10 percent of tube thickness, whichever is greater	X	----	Not shown. Similar to figure 197.	X	----	Not shown. Similar to figure 197.	X	----	Not shown. Similar to figure 197.	X	----	197
Rough, less than 1/64 inch or less than 10 percent of tube thickness, whichever is greater	X	----	198	X	----	Not shown. Similar to figure 198.	X	----	Not shown. Similar to figure 198.	X	----	Not shown. Similar to figure 198.
Scored (slit)	X	----	Not shown. Similar to figure 199.	X	----	199	X	----	Not shown. Similar to figure 199.	X	----	Not shown. Similar to figure 199.
Surface irregularity: when assembled with swaged fittings	X	----	200	X	----	Not shown. Similar to figure 200.	X	----	Not shown. Similar to figure 200.	X	----	Not shown. Similar to figure 200.
All others	----	X	201 and 202	----	X	201 and 202	----	X	Not shown. Similar to figures 201 and 202.	----	X	Not shown. Similar to figures 201 and 202.
Wire through tube	X	----	Not shown. Similar to figure 203.	X	----	203	X	----	Not shown. Similar to figure 203.	X	----	Not shown. Similar to figure 203.
Yarn through tube	X	----	Not shown. Similar to figure 204.	X	----	204	X	----	Not shown. Similar to figure 204.	X	----	Not shown. Similar to figure 204.

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MILITARY INTEREST:

Custodians:

Army—MO
Navy—SH
Air Force—69

Review activities:

Army—MO
Navy—SH, YD
Air Force—69

User activities:

Navy—WP, MC

Preparing activity:

Navy—SH

CIVIL AGENCY INTEREST:

GSA–FSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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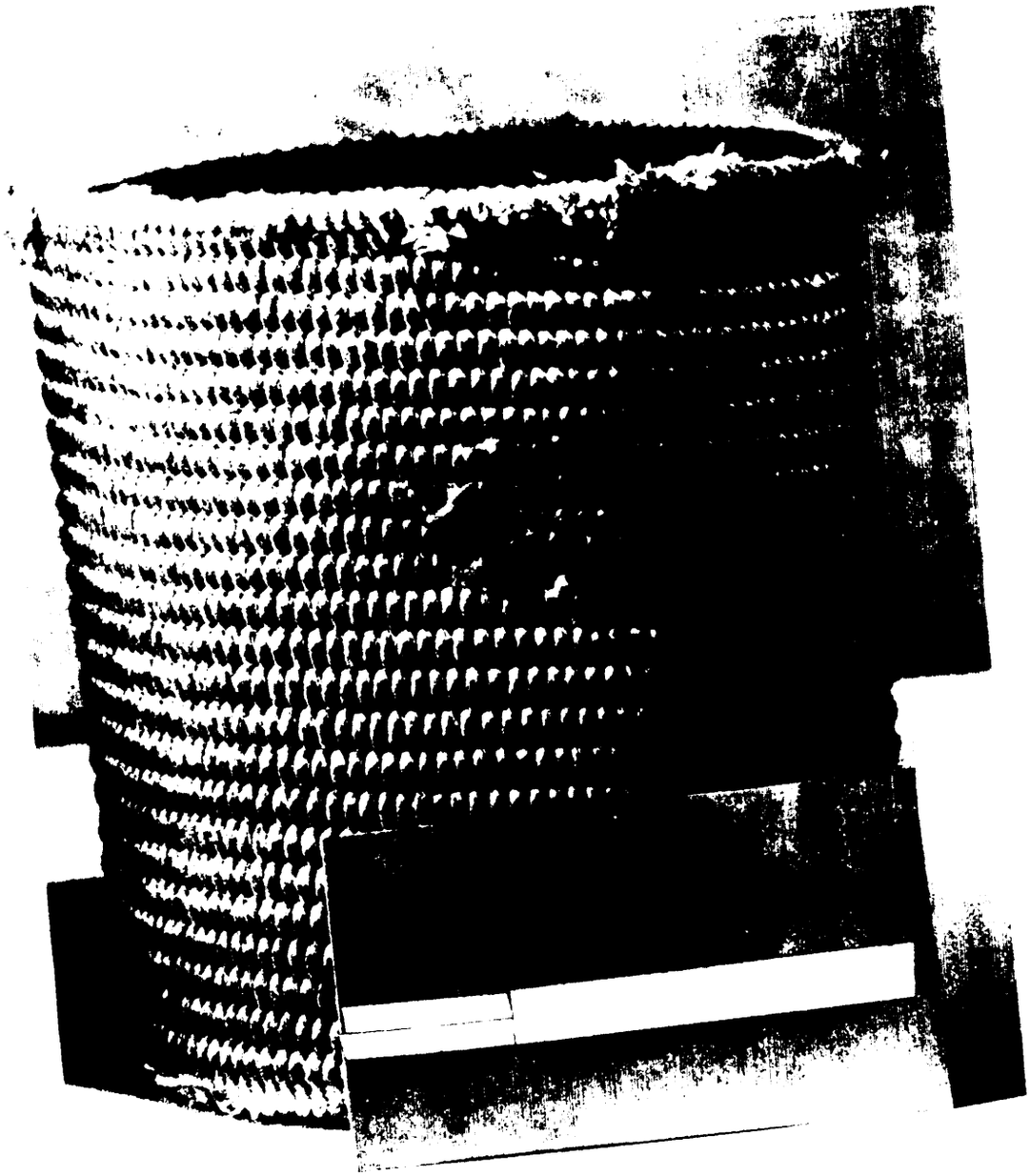


FIGURE 1. Abrasion—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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FIGURE 2. Friction strike through—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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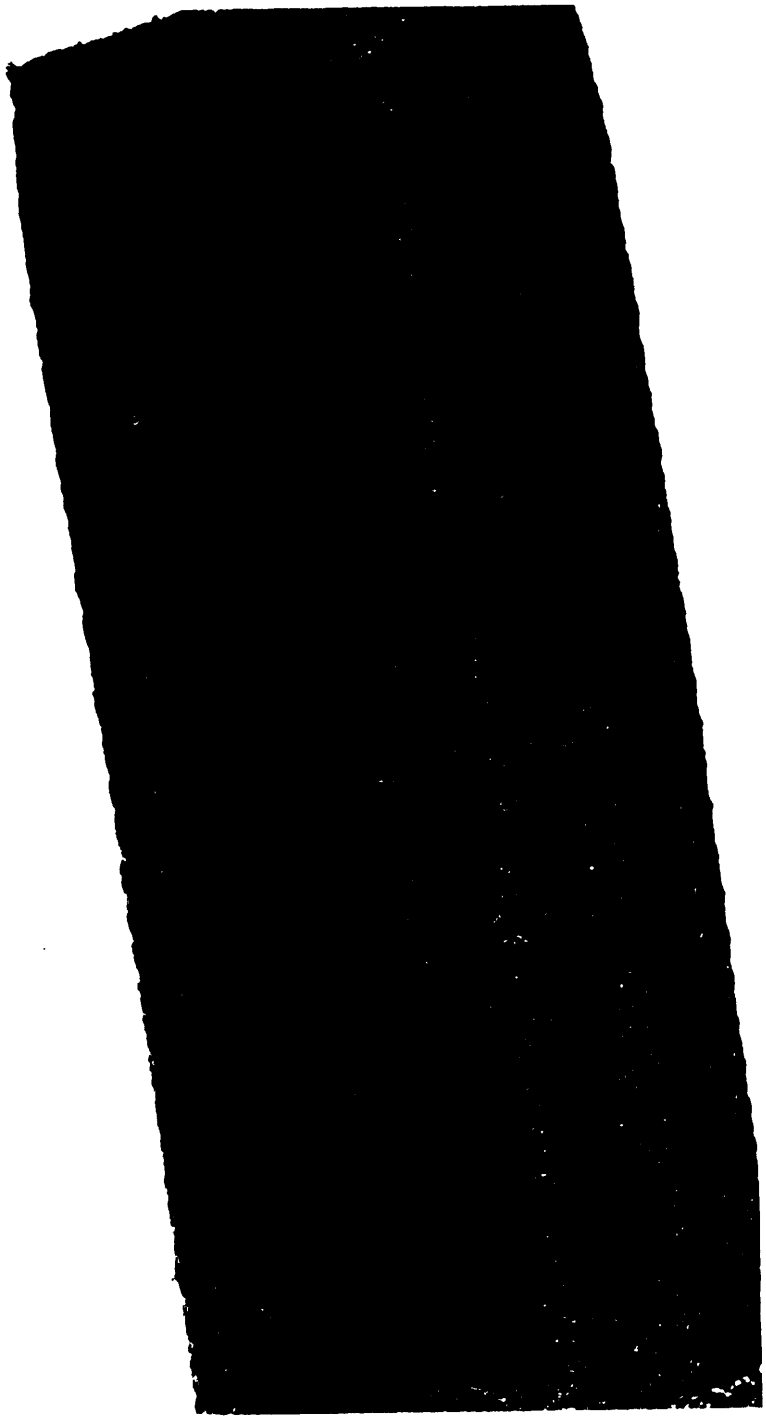


FIGURE 3. Friction strike through—minor.

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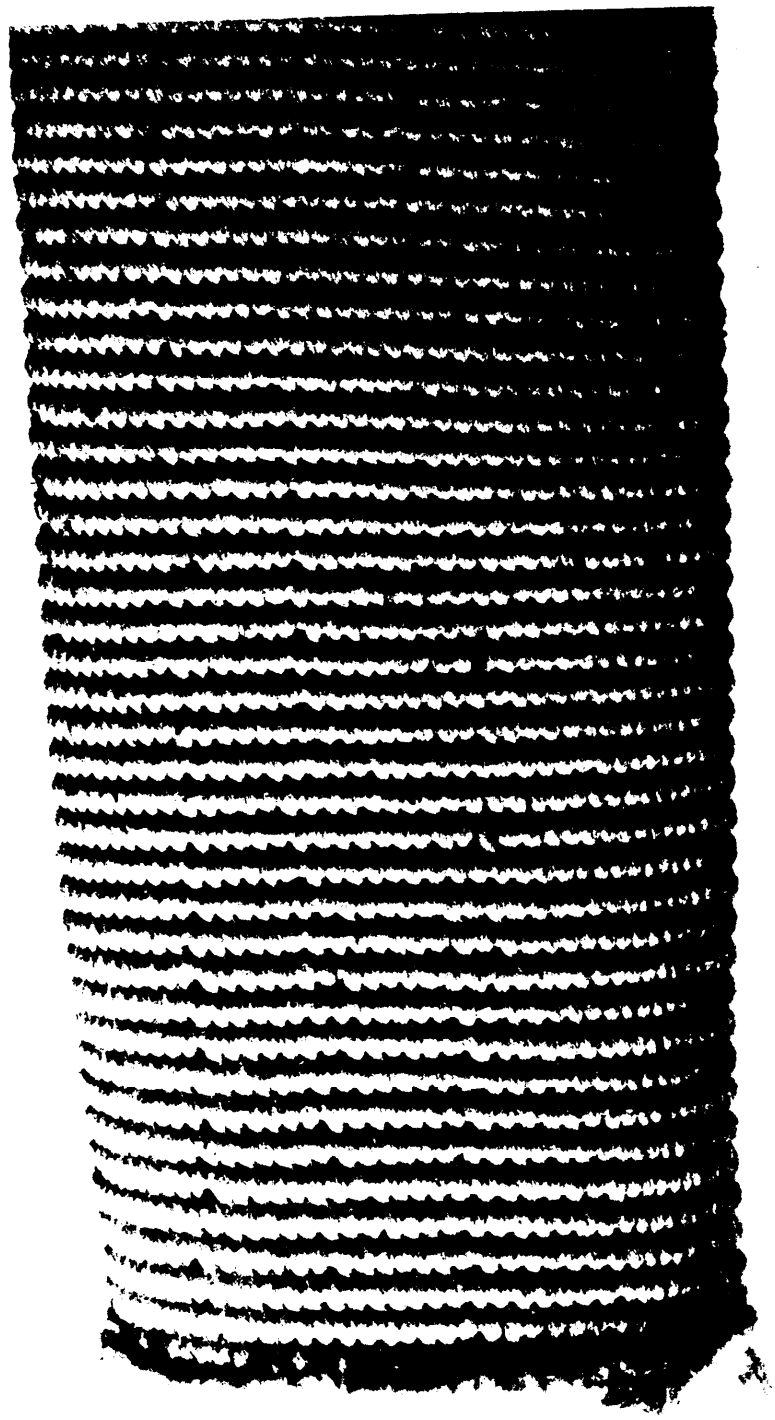


FIGURE 4. Jacket insufficient—major.

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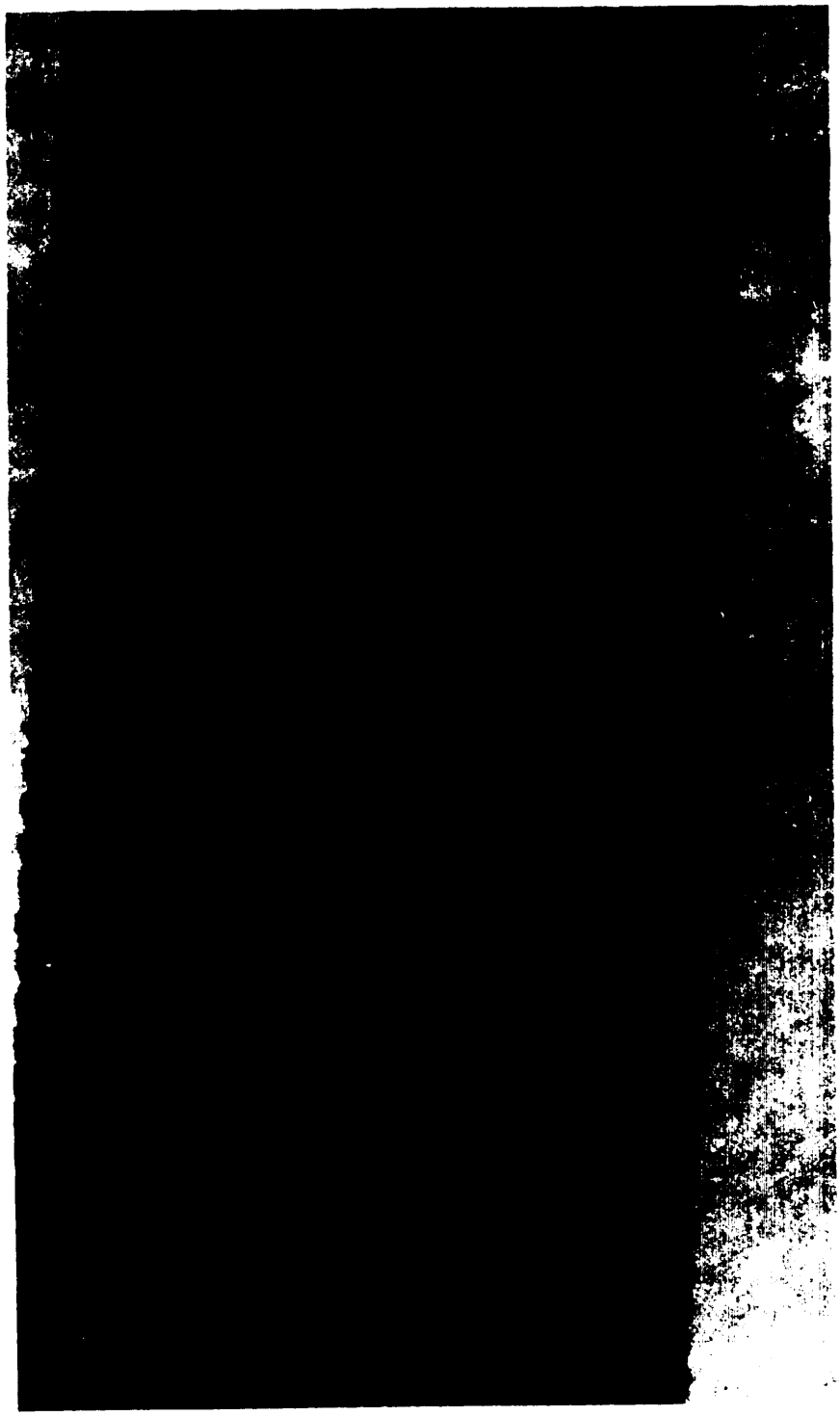


FIGURE 5. Jacket insufficient—minor.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T12:54Z
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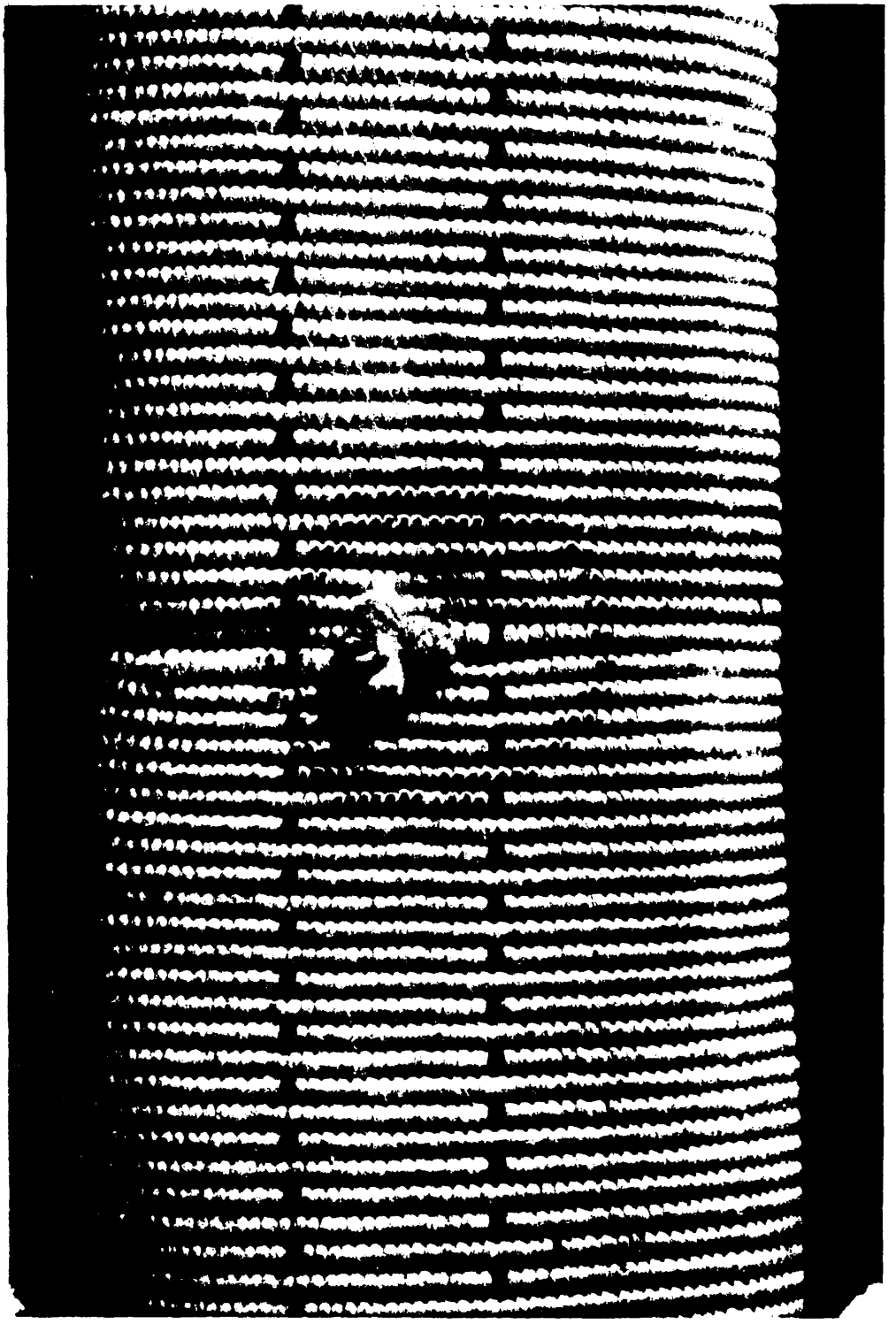


FIGURE 6. Knot filler—major.

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FIGURE 7. Knot filler—not a defect.

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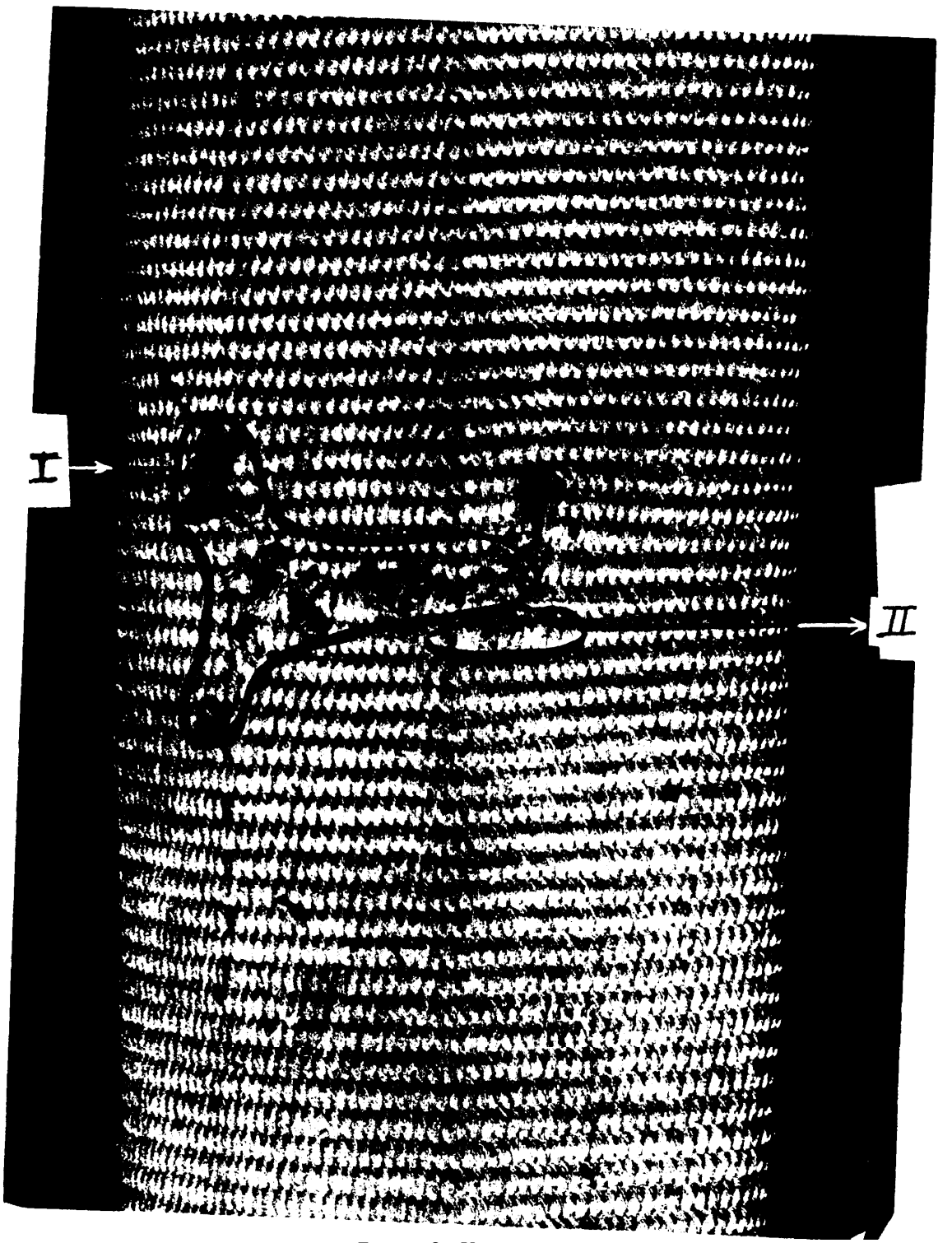


FIGURE 8. Yarn:

1—Knotted—major.
II—Irregularity—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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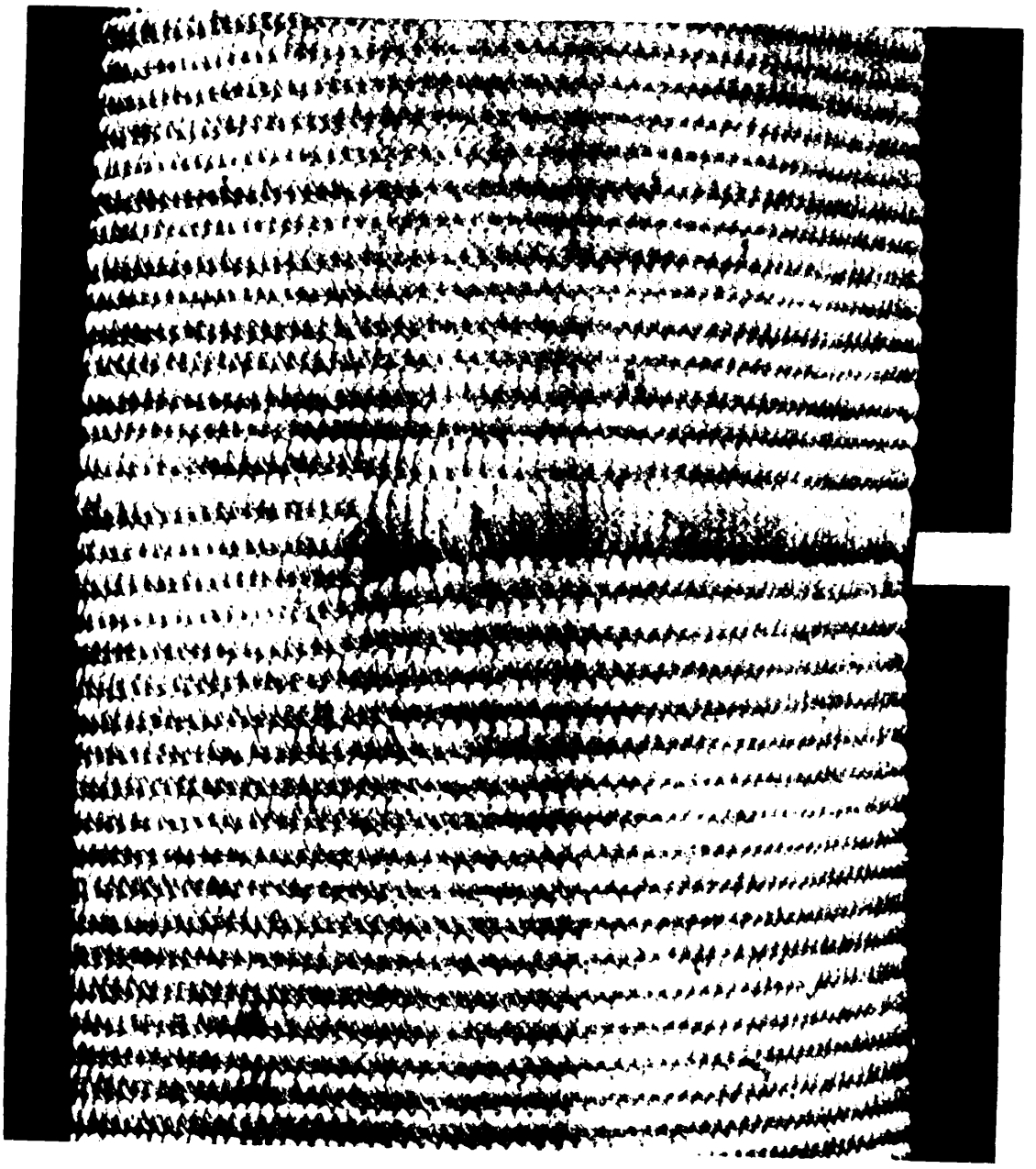


FIGURE 9. Pick missing or broken—major.

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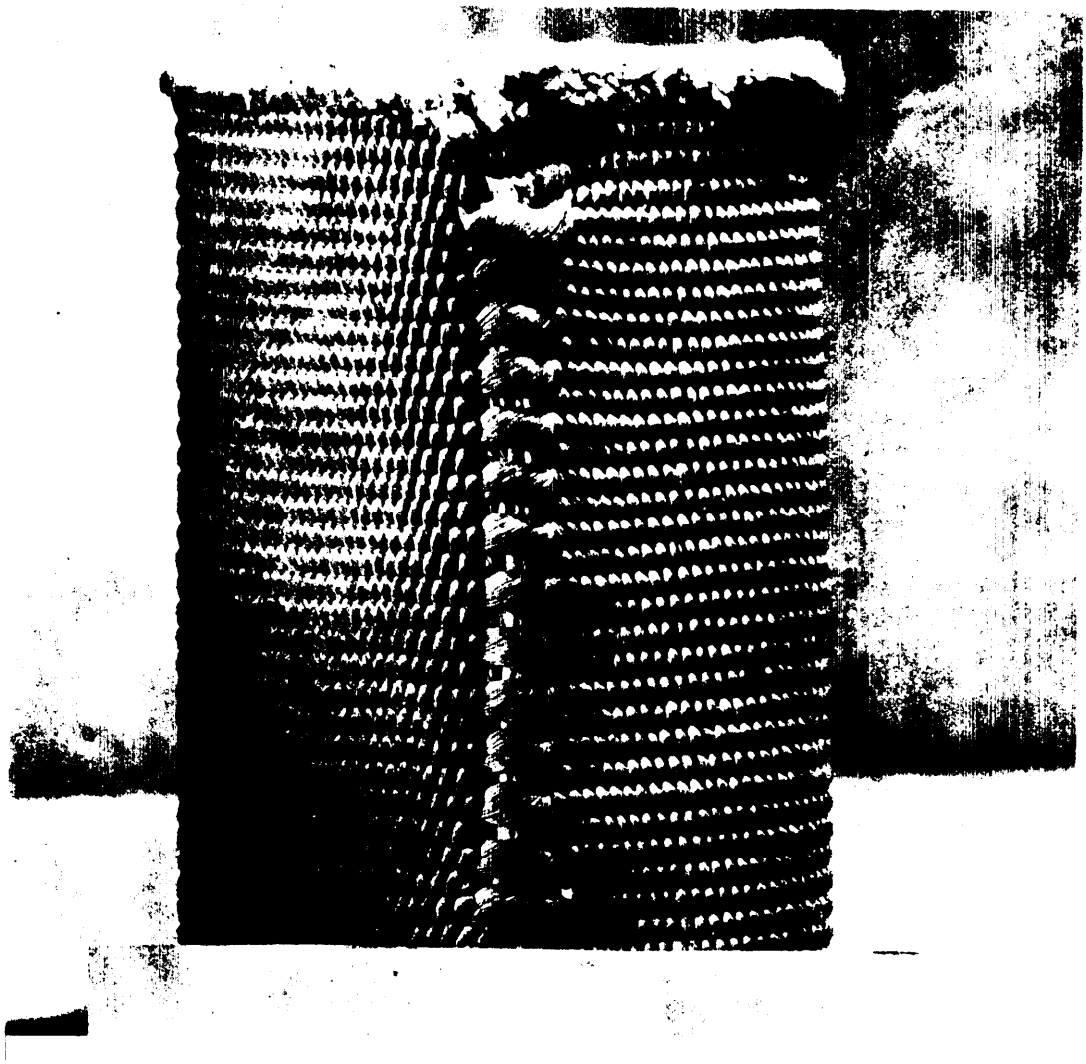


FIGURE 10. Skips, warp—major.

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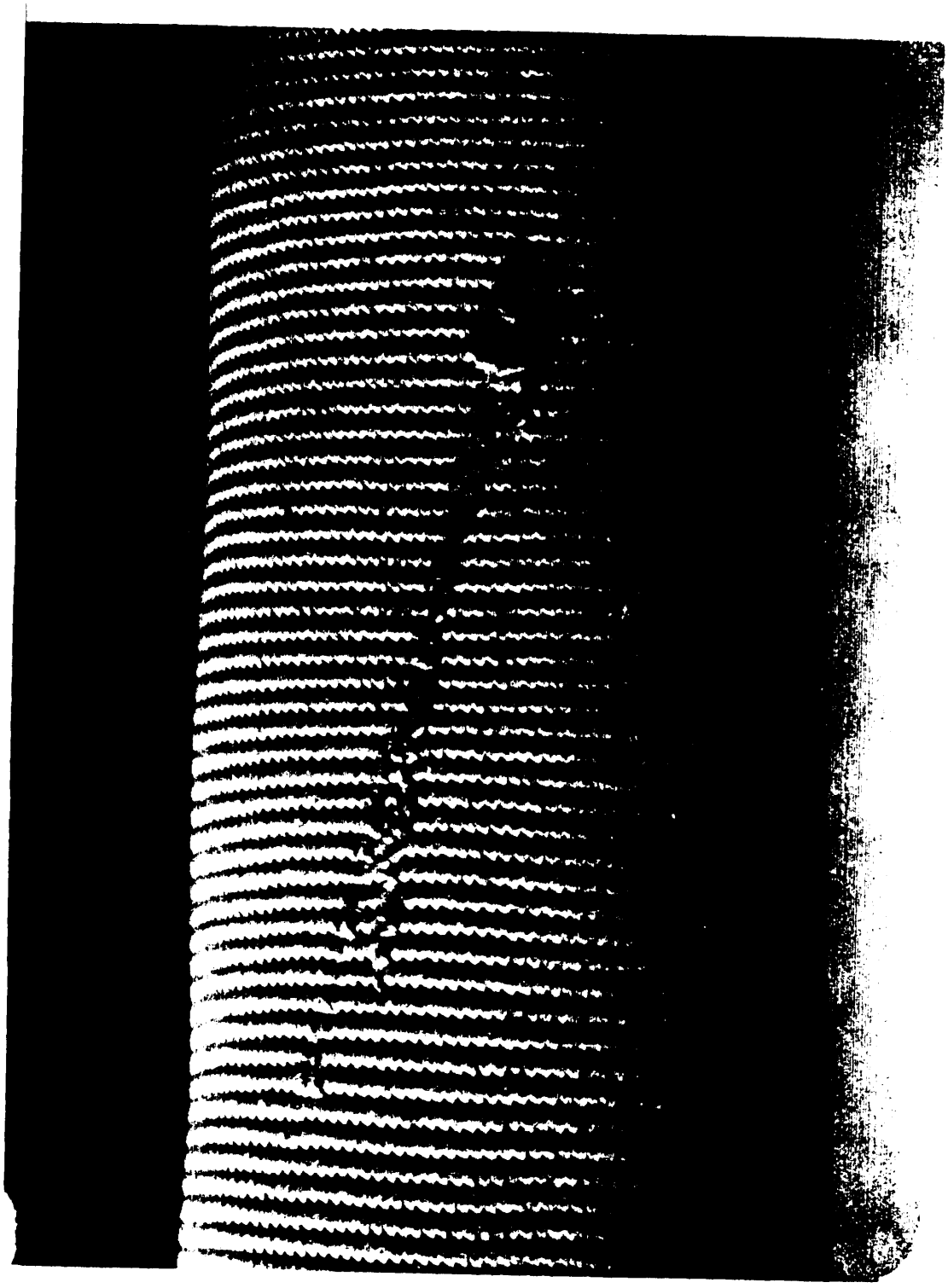


FIGURE 11. Snag—major.

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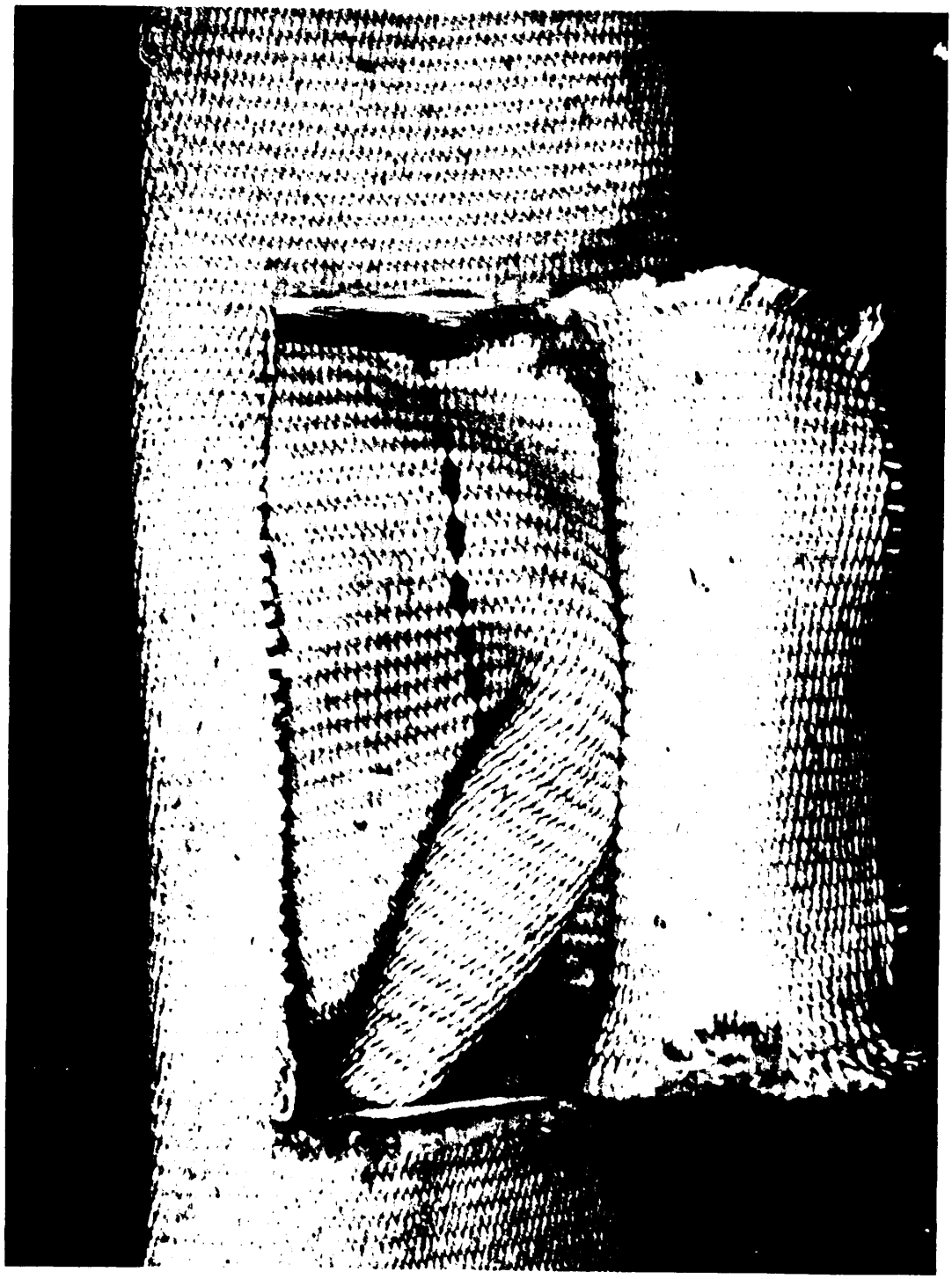


FIGURE 12. Twist, inner jacket—major.

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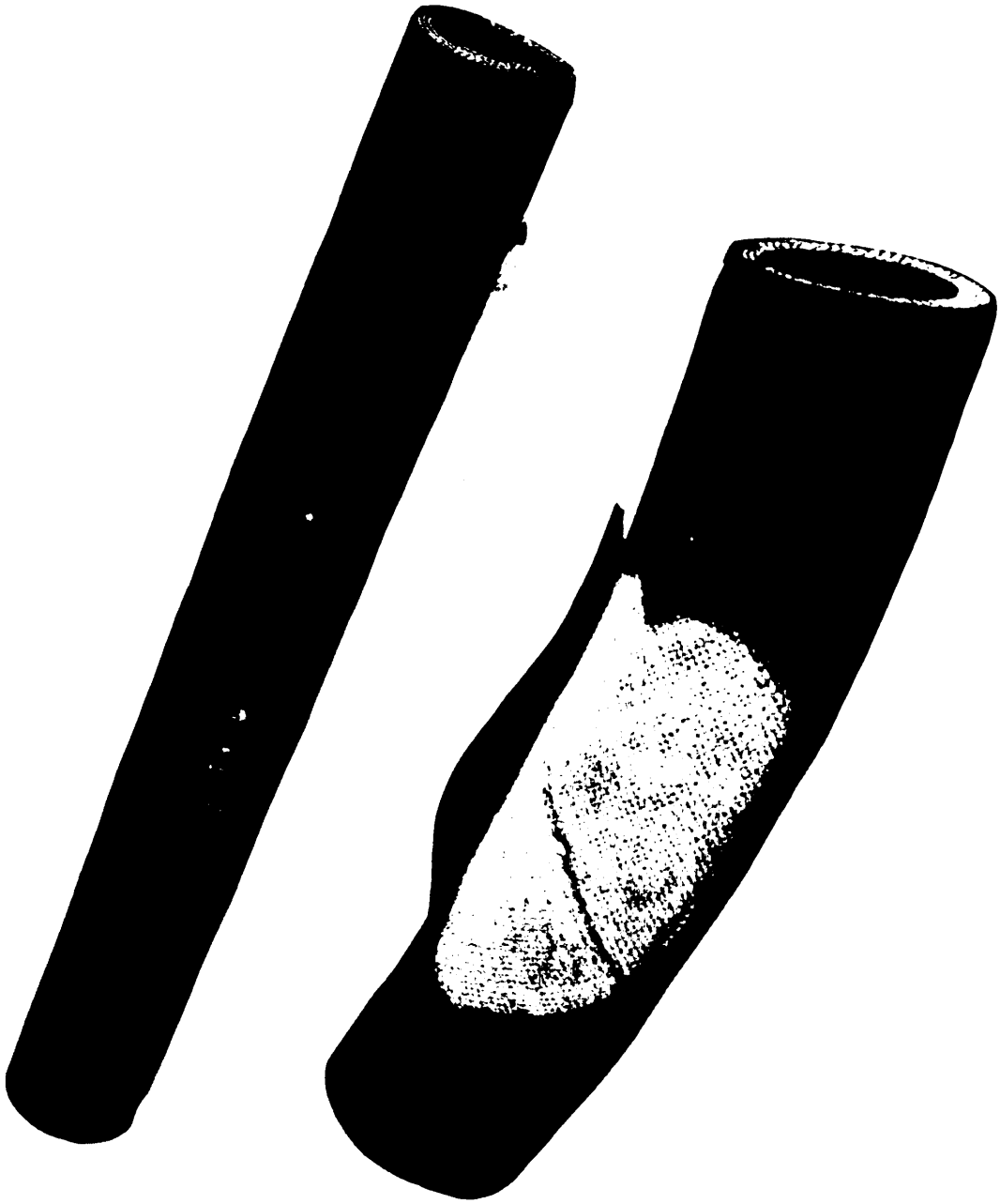


FIGURE 13. Blister under cover, large—major (cover removed for clarity).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 14. Blister under cover, large—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

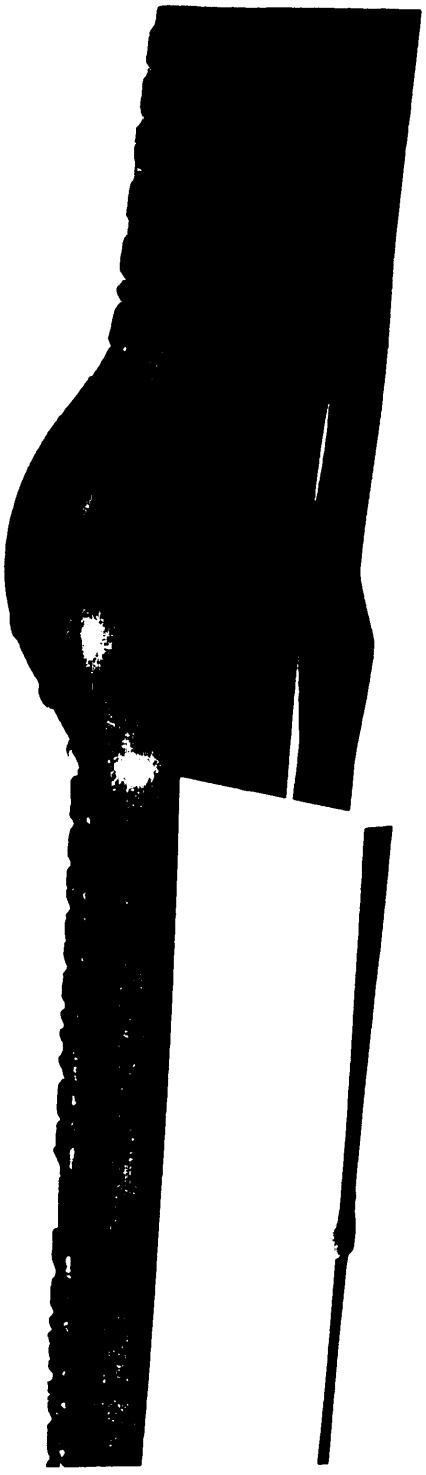


FIGURE 15. Blister under cover, large—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

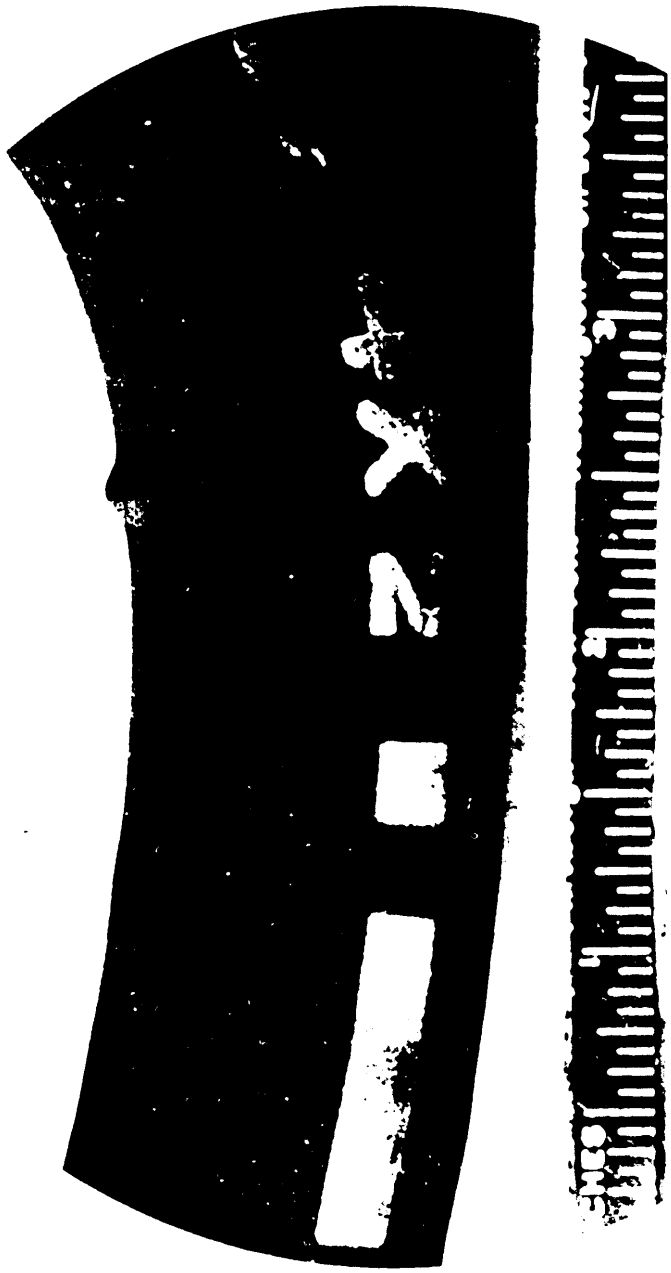


FIGURE 16. Blister under cover, large—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

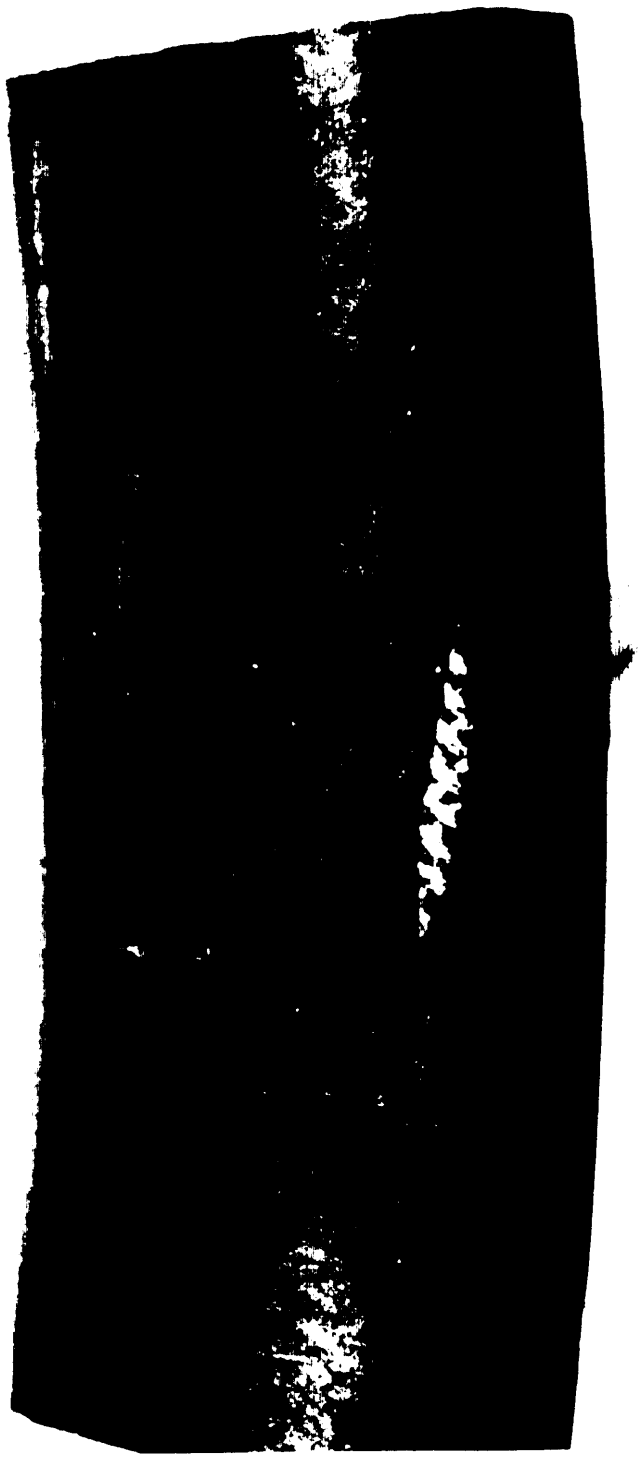


FIGURE 17. Break—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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FIGURE 18. Buckle (because interior contour will be affected)—major.

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FIGURE 19. Bulge, large: dynamic use (Air Force procurement only), and class A —major: classes B and C—minor.

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Check the source to verify that this is the current version before use.

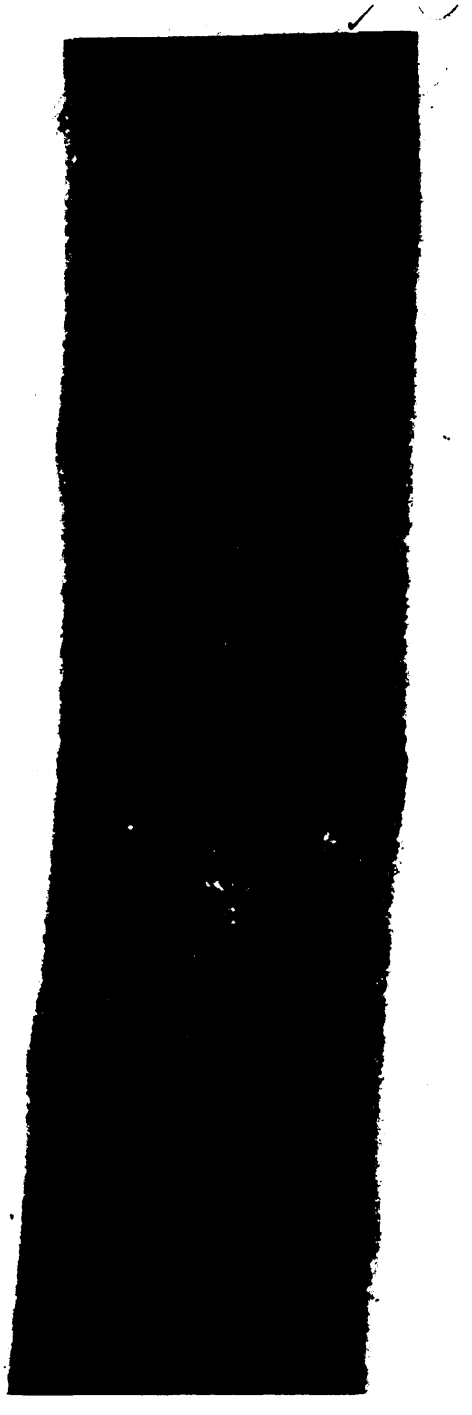


FIGURE 20. Bulge, small: dynamic use—service abrasion (Air Force procurement only)—major; other use—not a defect.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 21. Bump: in contour—minor; in nipple area—major.

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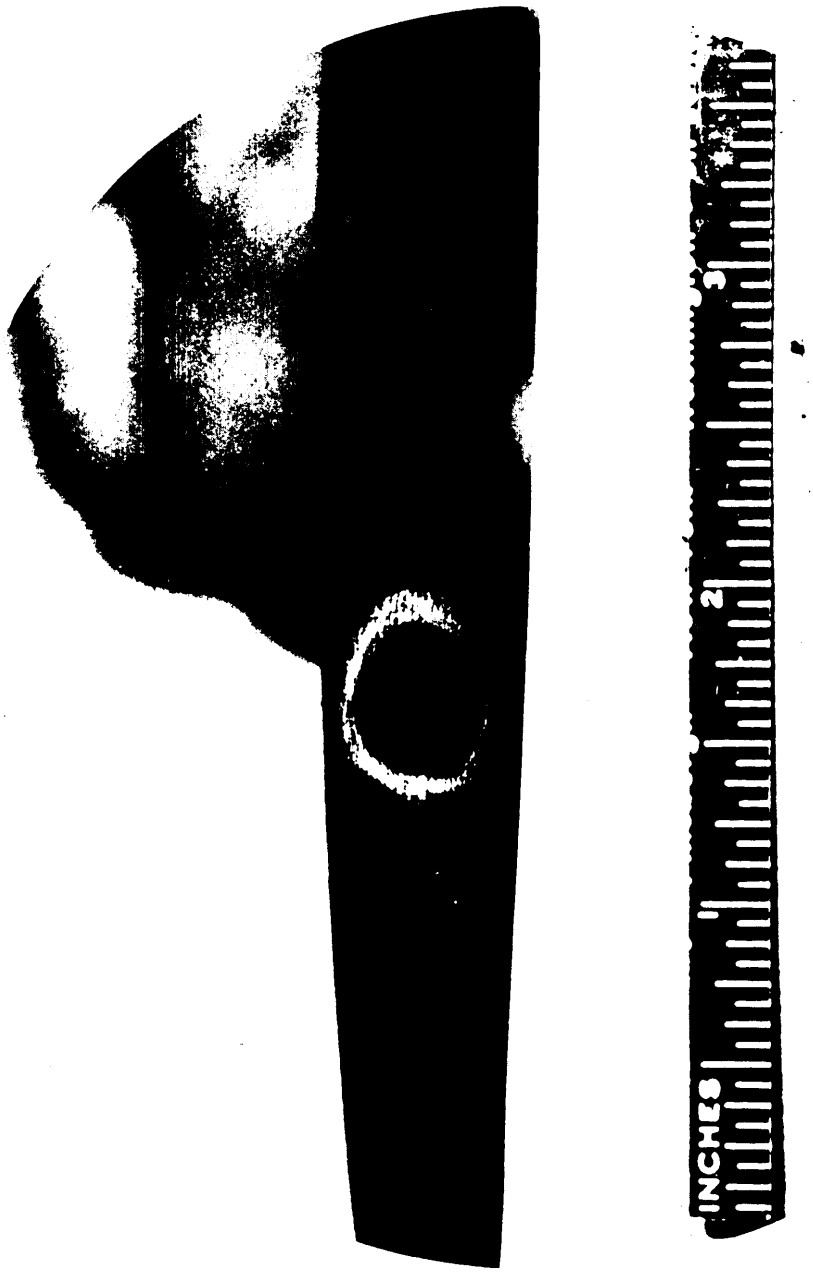


FIGURE 22. Crack—major.

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FIGURE 23. Dent—major,

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Check the source to verify that this is the current version before use.



FIGURE 24. Dent—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

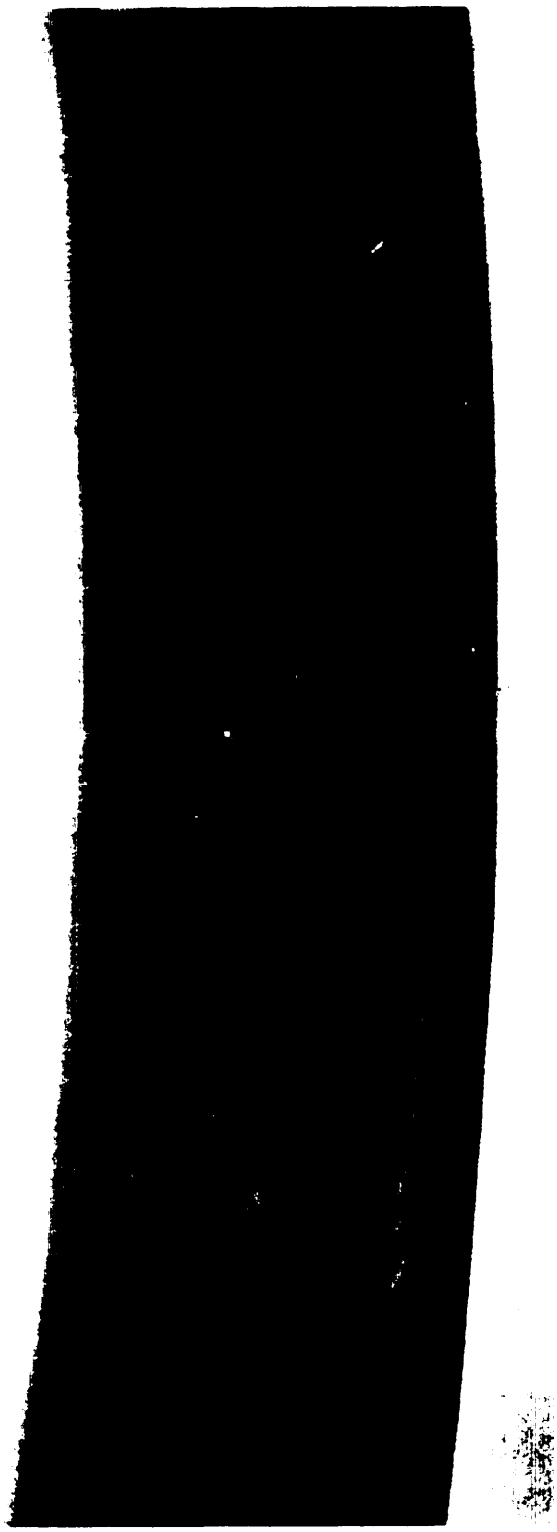


FIGURE 25. Depression, large area—major.

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Check the source to verify that this is the current version before use.

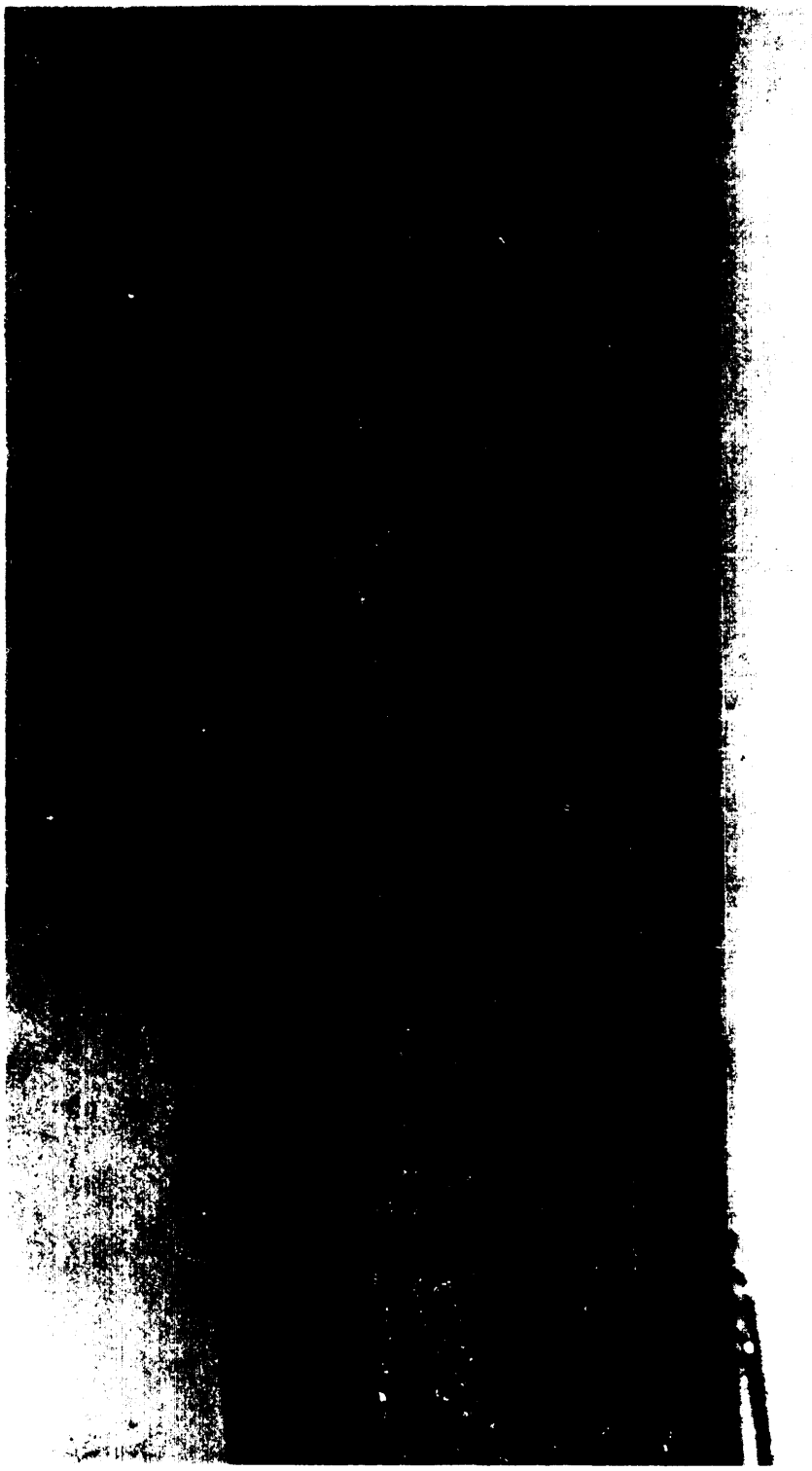


FIGURE 26. Depression, large area—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

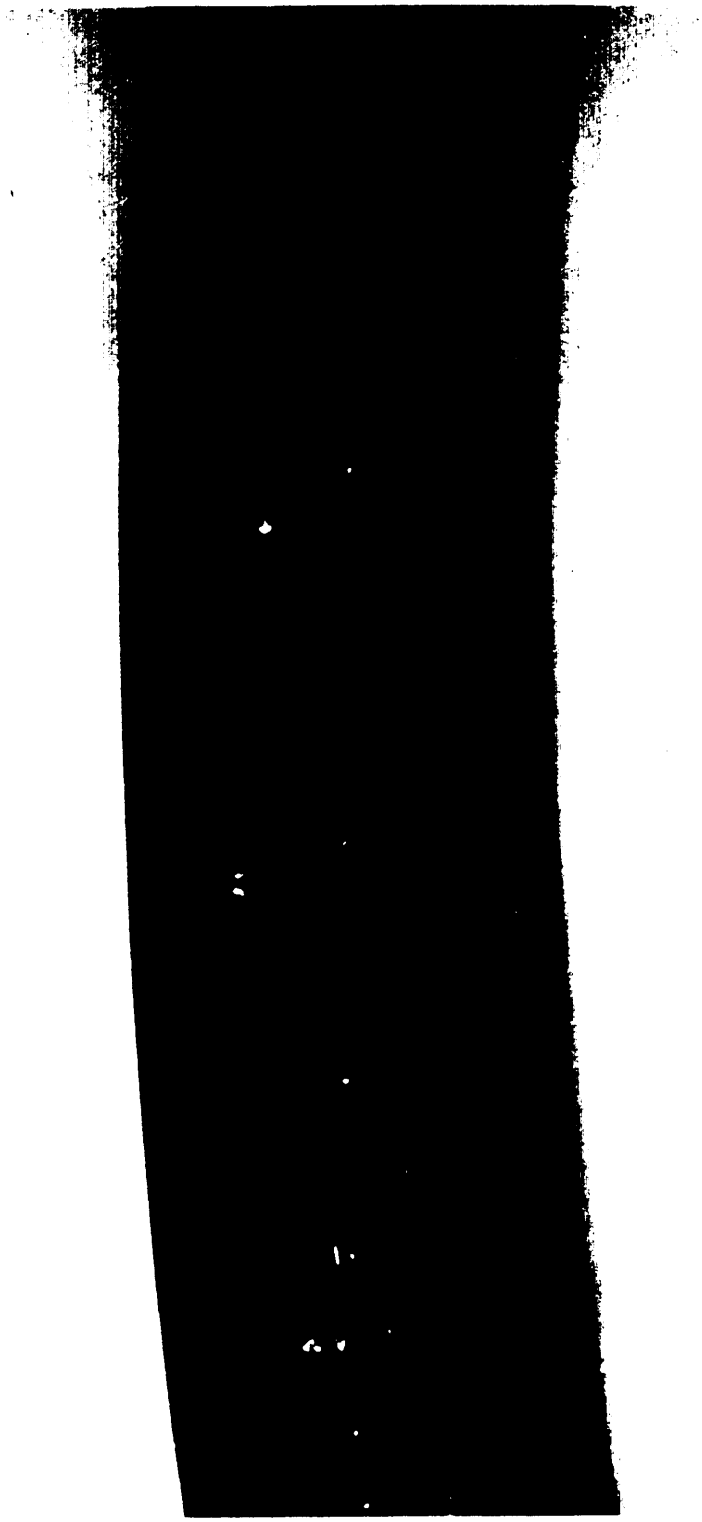


FIGURE 27. Depression. deep—major.

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Check the source to verify that this is the current version before use.

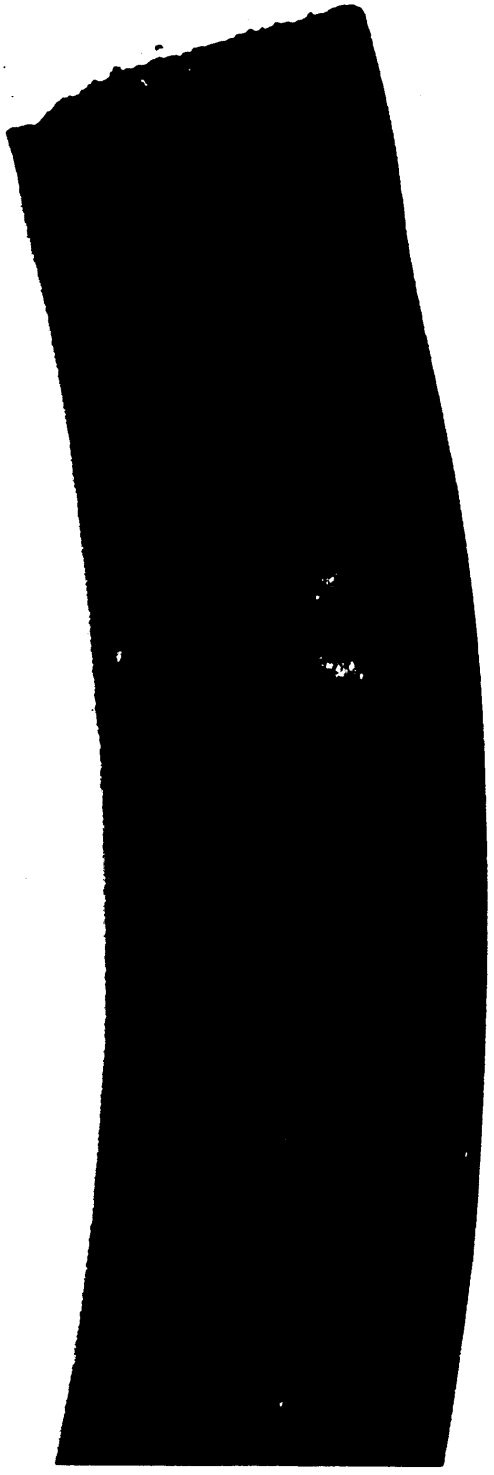


FIGURE 28. Depression, deep—major.

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Check the source to verify that this is the current version before use.

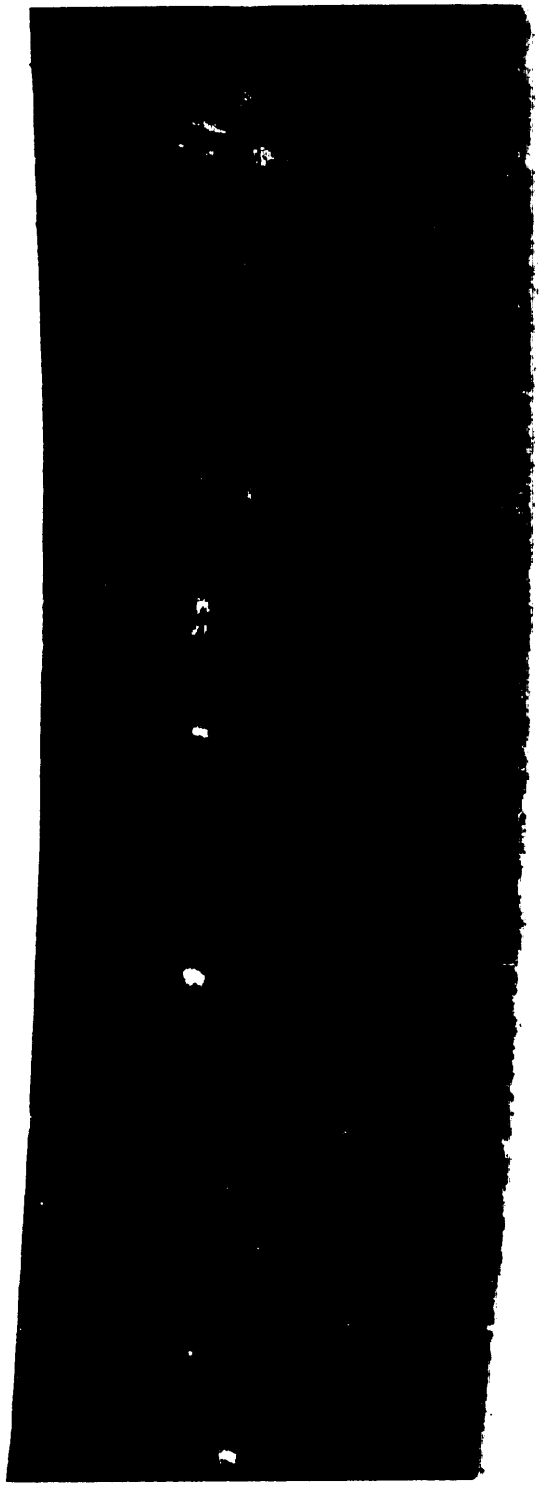


FIGURE 29. Depression, small and light: class A—major; classes B and C—minor.

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Check the source to verify that this is the current version before use.

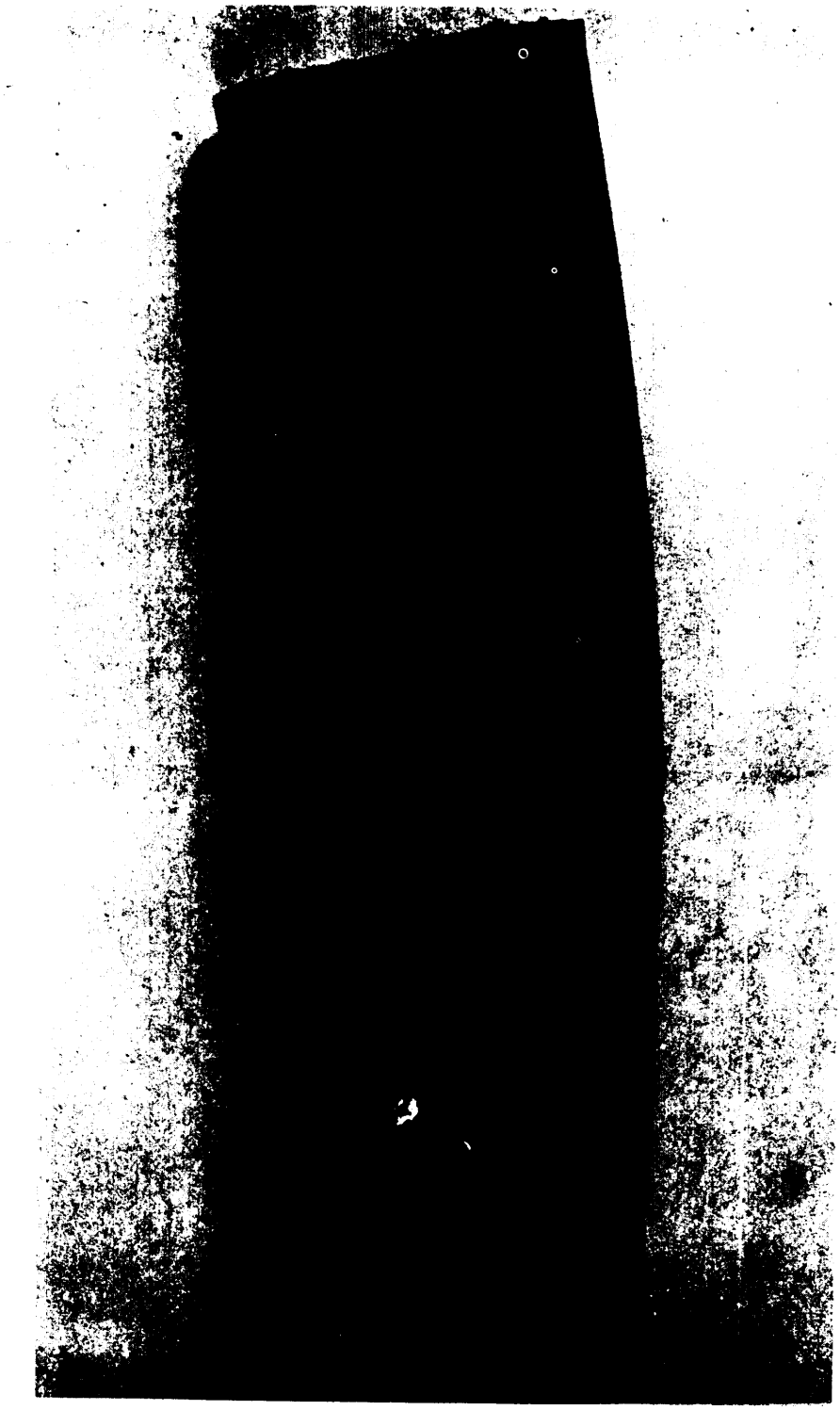


FIGURE 30. Depression, small and light: class A—major; classes B and C—minor.

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Check the source to verify that this is the current version before use.

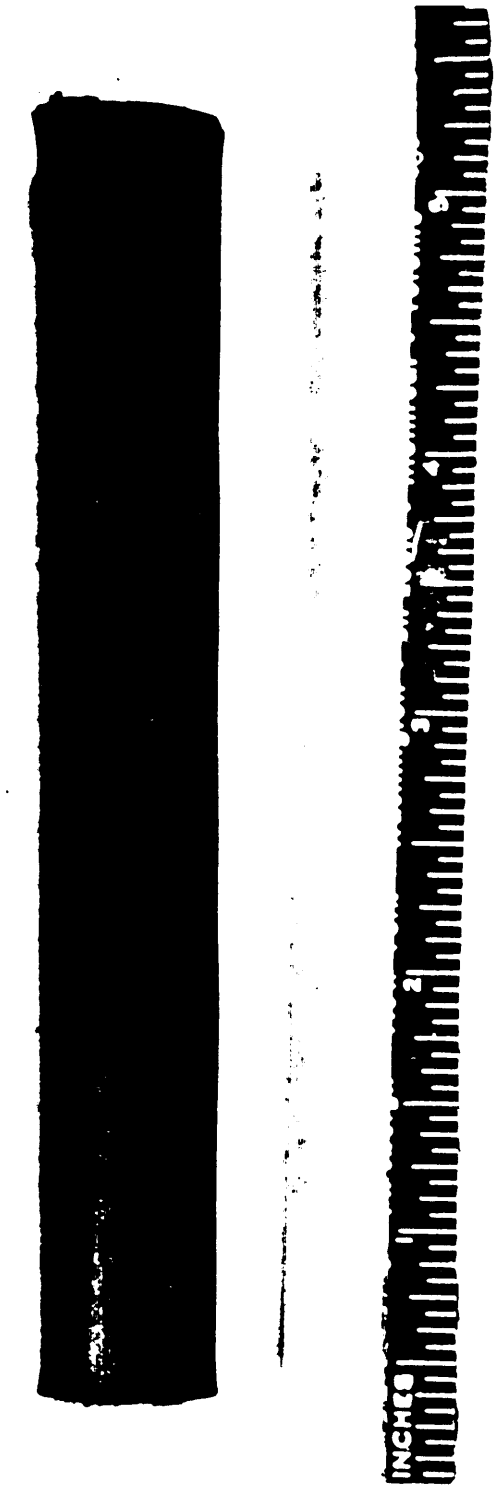


FIGURE 31. Depression. small and light: class A—major: classes B and C—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

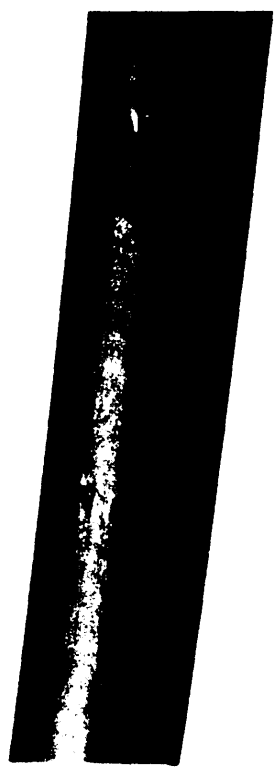


FIGURE 32. Depression, small and light: class A—major: classes B and C—minor.

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Check the source to verify that this is the current version before use.

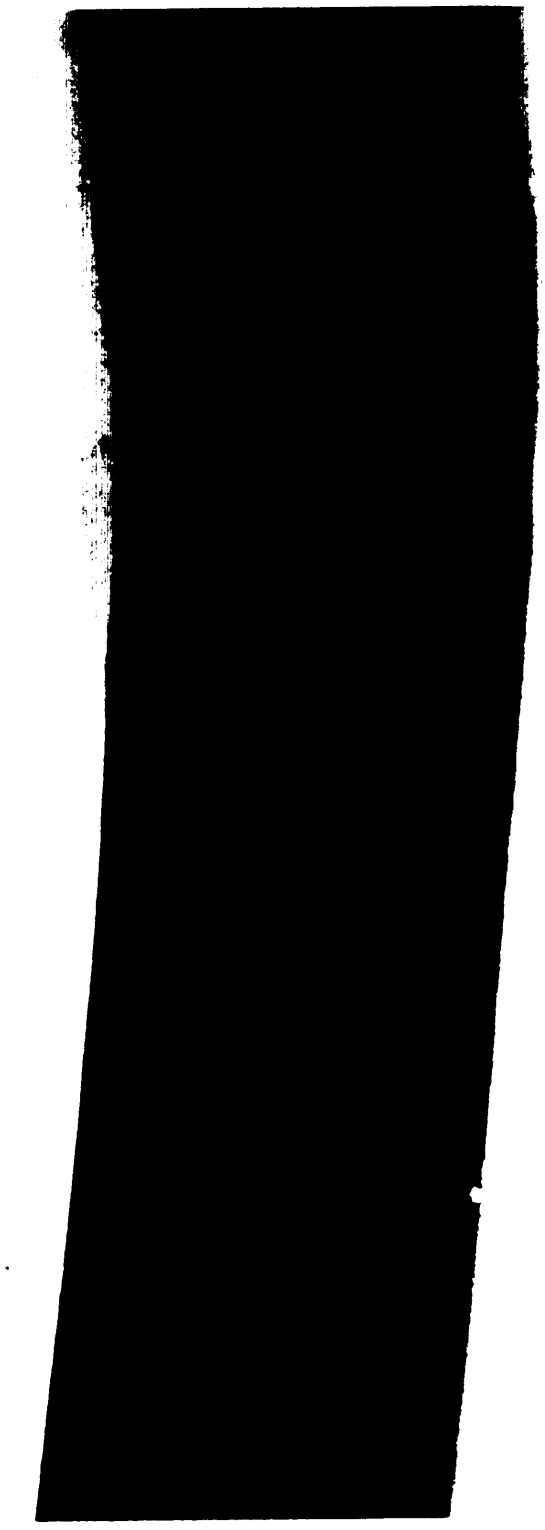


FIGURE 33. Depression, small and light—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 34. Depression, small and light—minor.

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Check the source to verify that this is the current version before use.



FIGURE 35. Discolored—not a defect.

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FIGURE 36. Dog leg—major.

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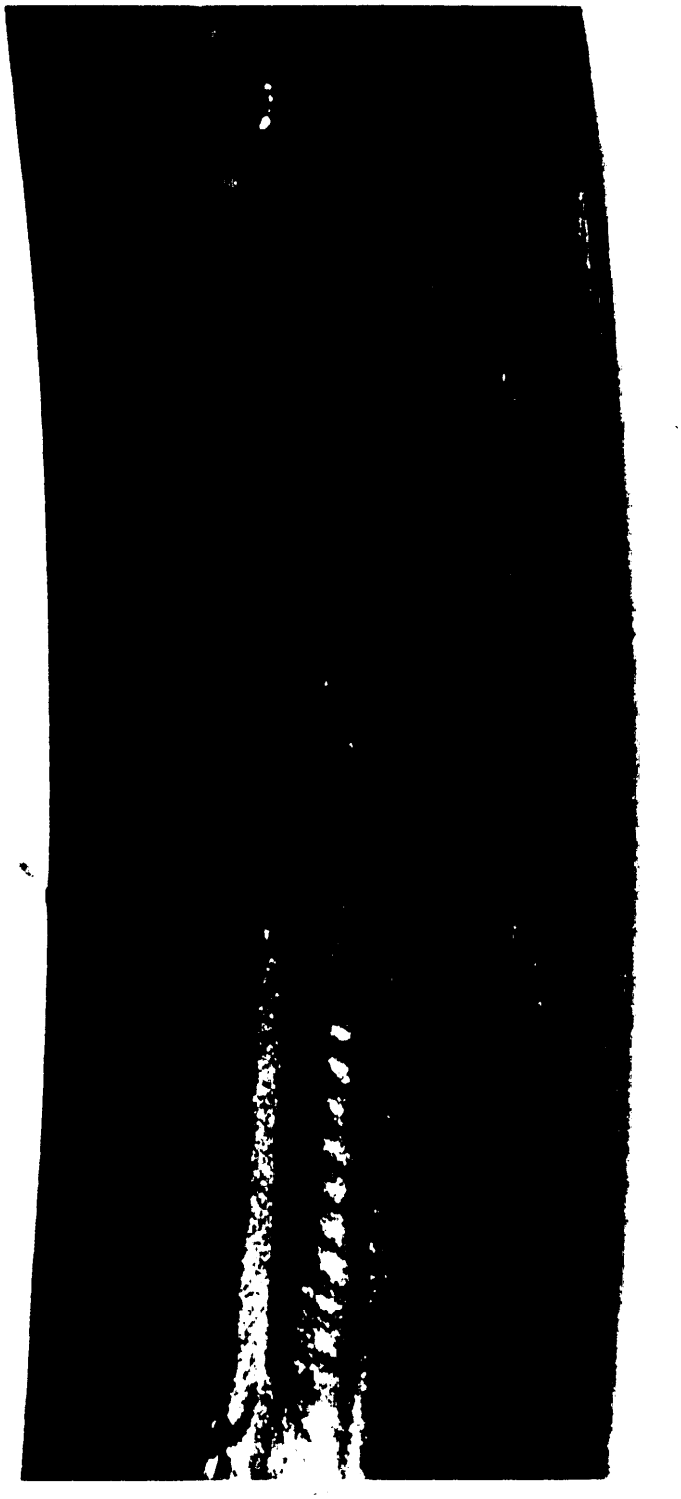


FIGURE 37. Groove, deep—major.

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FIGURE 38. Rupture, large—major.

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FIGURE 39. Hole: reinforcement exposed—major.

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FIGURE 40. Hole: reinforcement exposed—major.

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Check the source to verify that this is the current version before use.

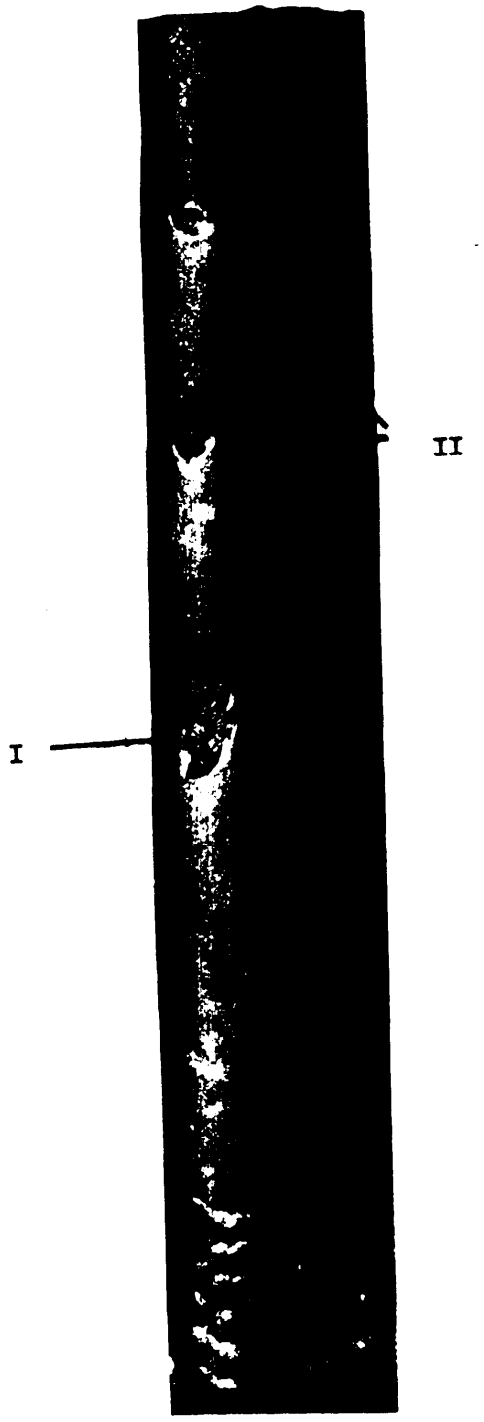


FIGURE 41. Hole, small.

I—Reinforcement exposed—major.
II—Reinforcement not exposed—minor.

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Check the source to verify that this is the current version before use.

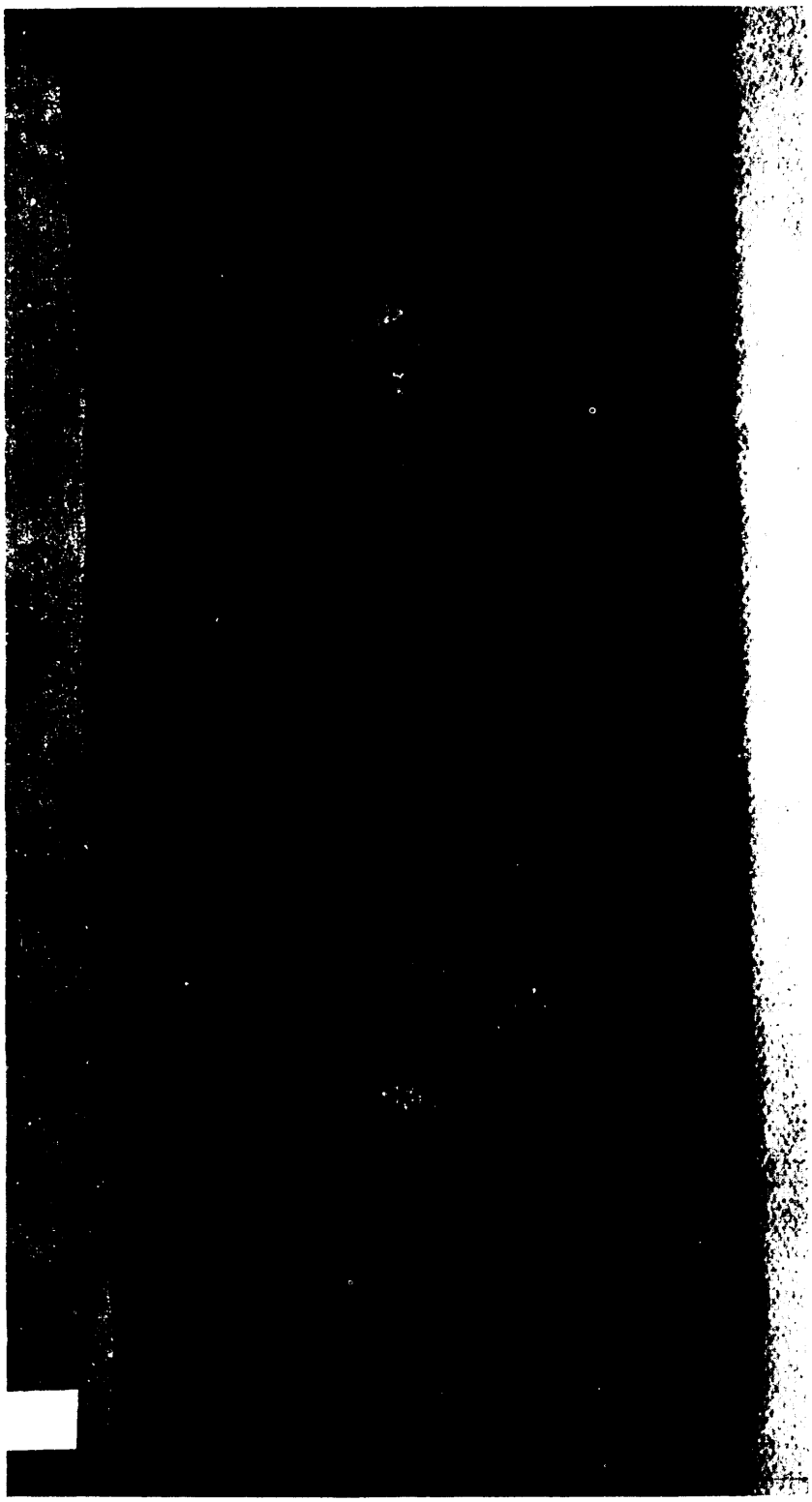


FIGURE 42. Lap: foldover, opening—major.

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Check the source to verify that this is the current version before use.



FIGURE 43. Lap: foldover, opening—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

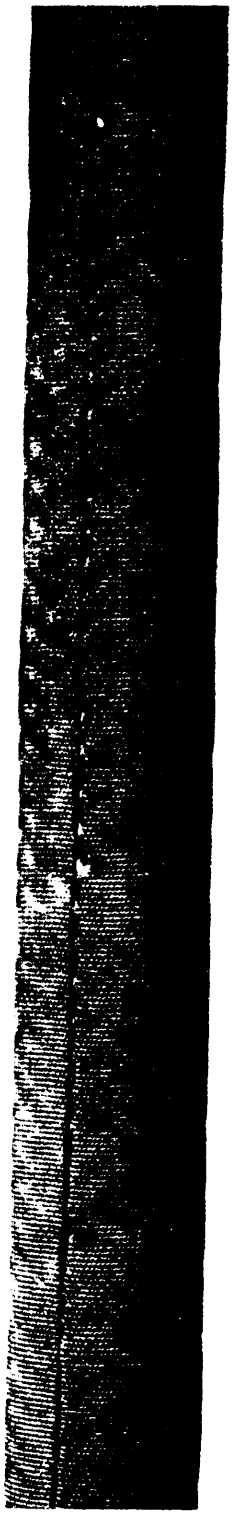


FIGURE 44. Lap: foldover, opening—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 45. Lap, open in fabric—major.

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Check the source to verify that this is the current version before use.

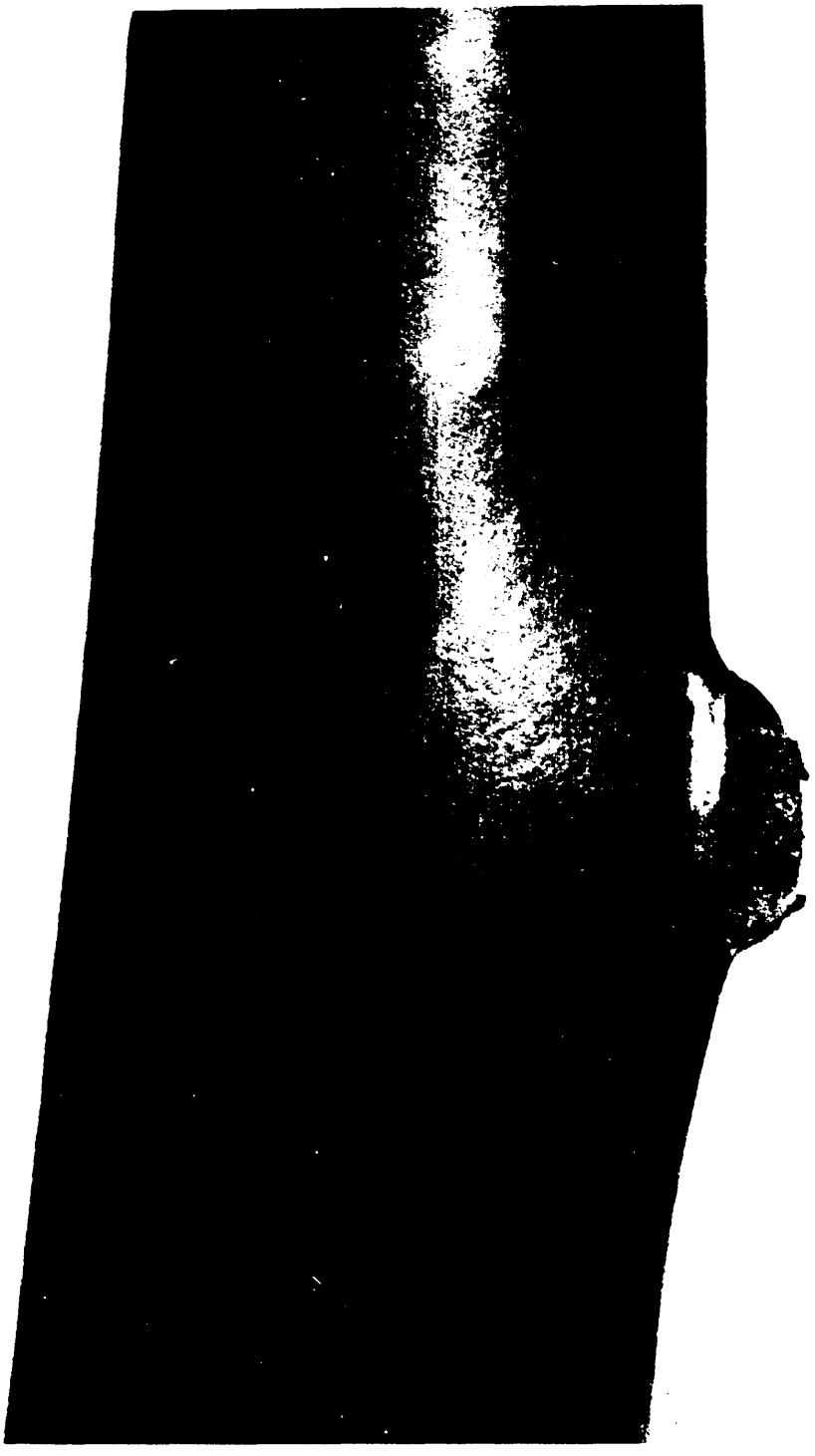


FIGURE 46. Lead, burst—major,

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FIGURE 47. Charge mark—major.

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FIGURE 48. Charge mark:

I —major.
II —minor.
III—Not a defect.

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Check the source to verify that this is the current version before use.

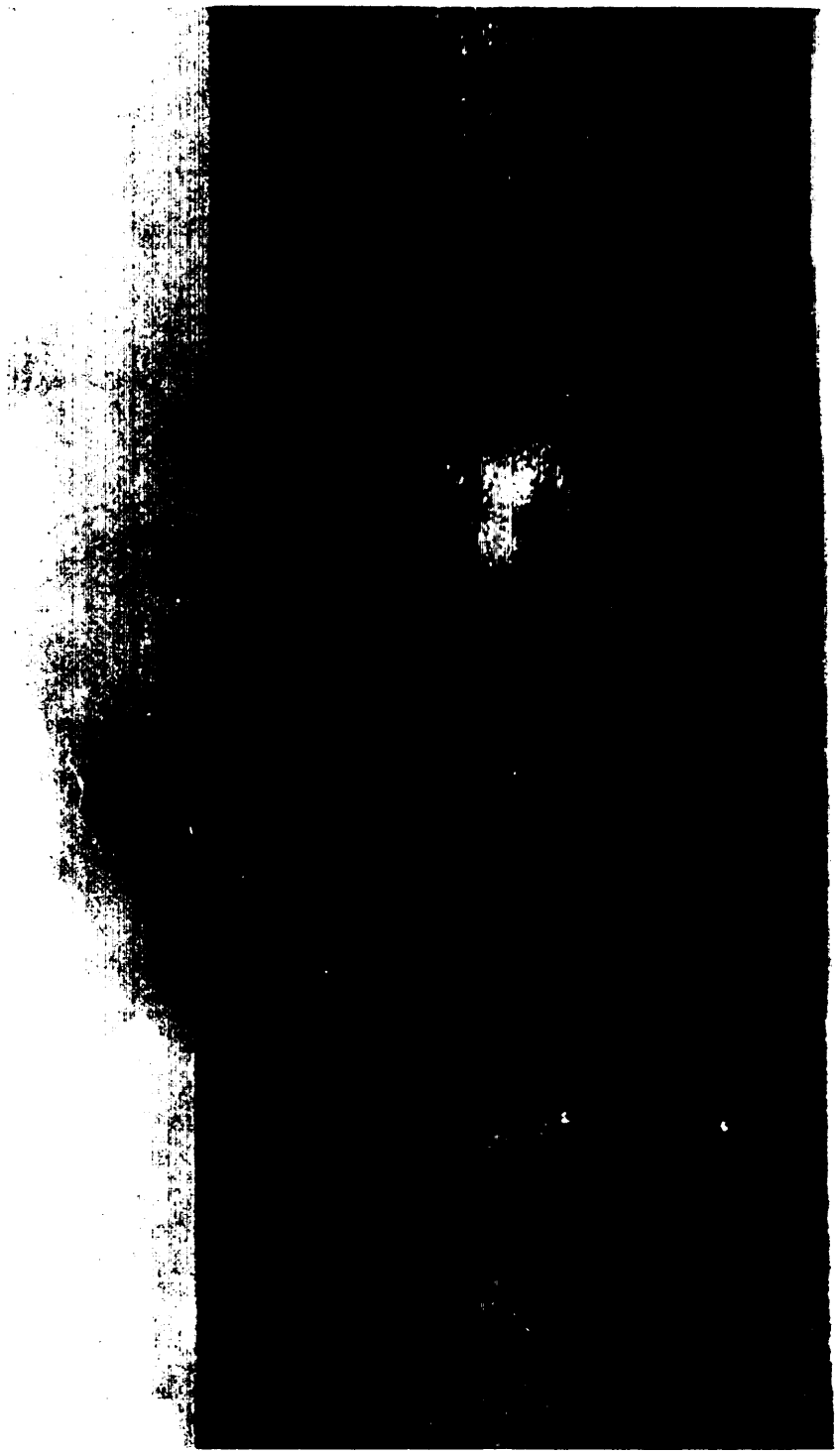


FIGURE 49. Charge mark: class A—minor; classes B and C—not a defect.

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Check the source to verify that this is the current version before use.

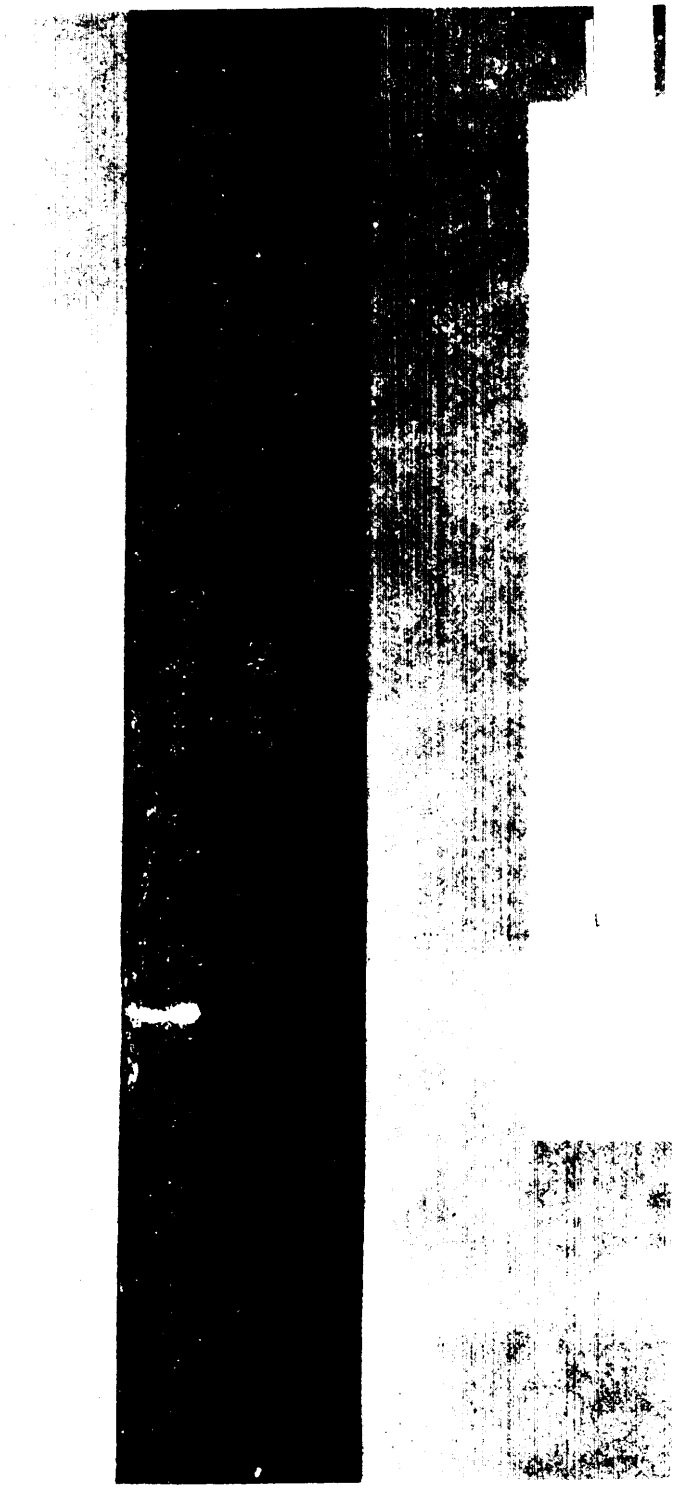


FIGURE 50. Flakes, lead: classes A and B—major; class C—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 51. Flakes, lead: class A—major: class B—minor: class C—not a defect.

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Check the source to verify that this is the current version before use.

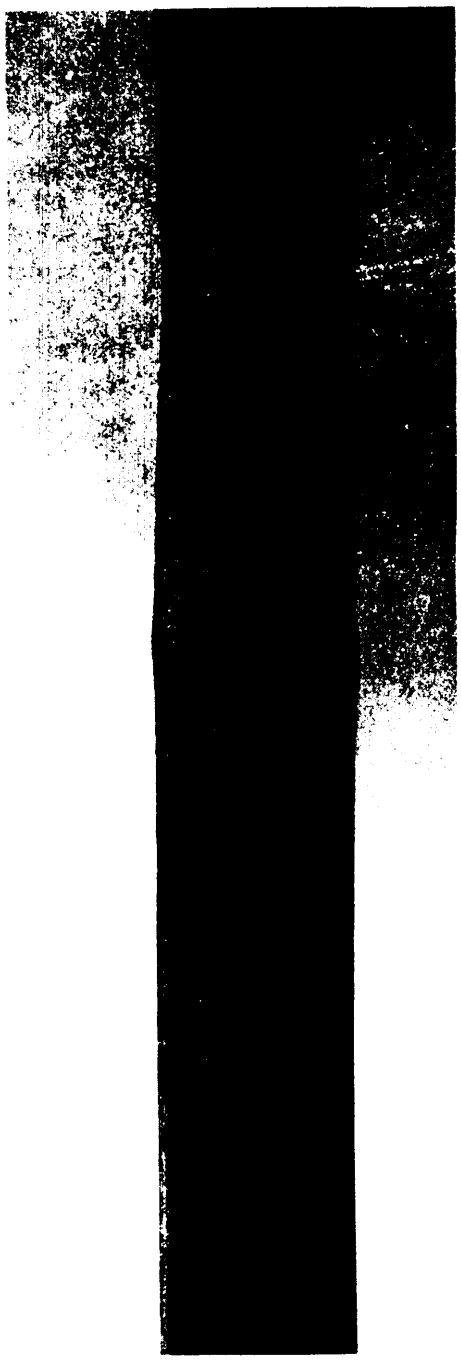


FIGURE 52. Machine stop: class A—major; class B—minor; class C—not a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

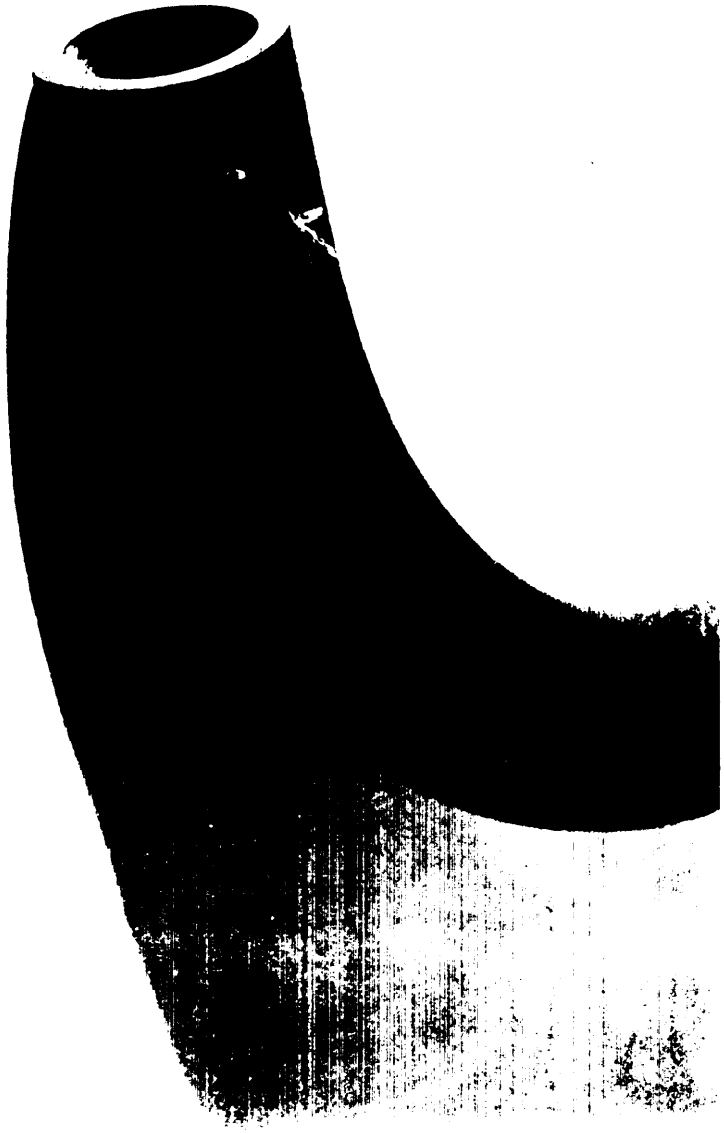


FIGURE 53. Pop, lead, through body—major.

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FIGURE 54. Pop, lead, reinforcement exposed—major.

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Check the source to verify that this is the current version before use.



FIGURE 55. POP, lead, reinforcement not exposed: class A—major; class B—minor; class C—not a defect.

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Check the source to verify that this is the current version before use.

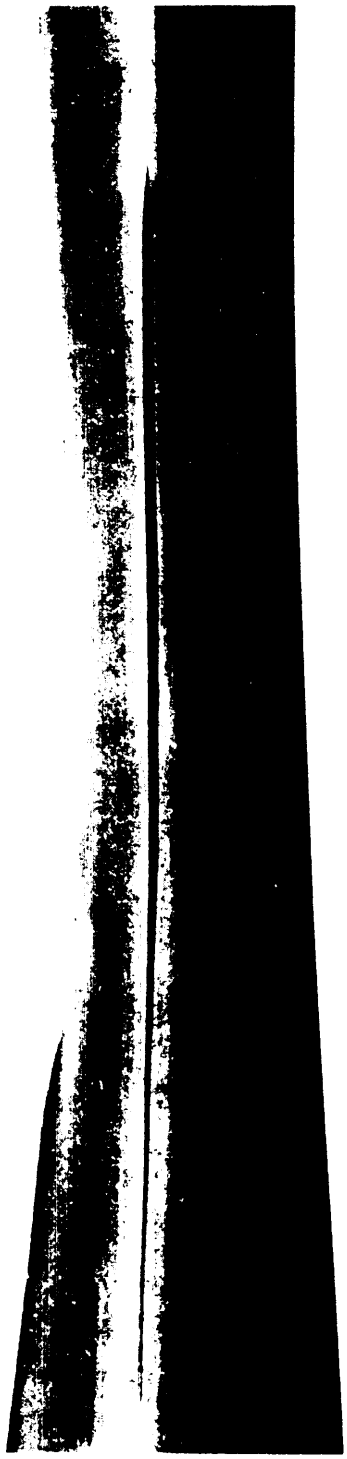


FIGURE 56. Slit or stripper cut—major.

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Check the source to verify that this is the current version before use.

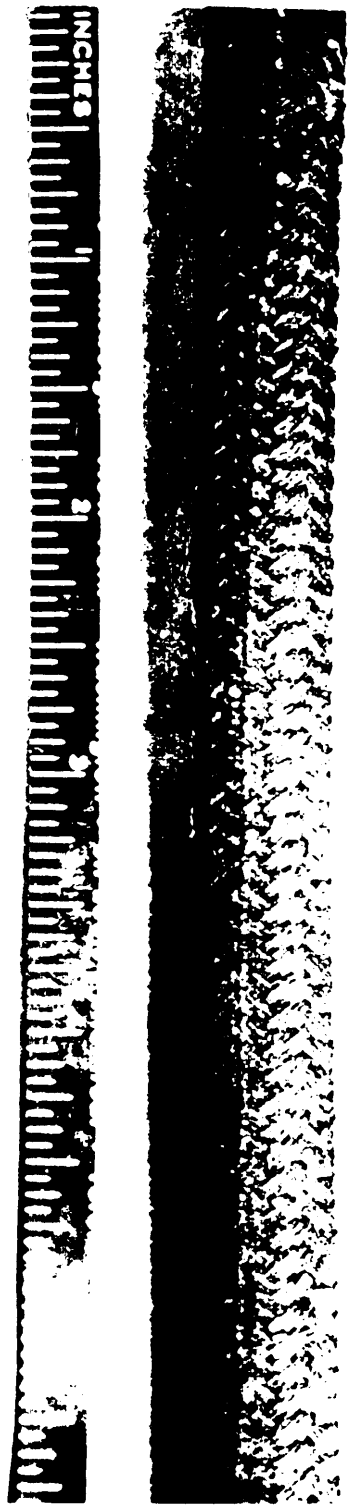


FIGURE 57. Stripper cut (into braided cover)—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

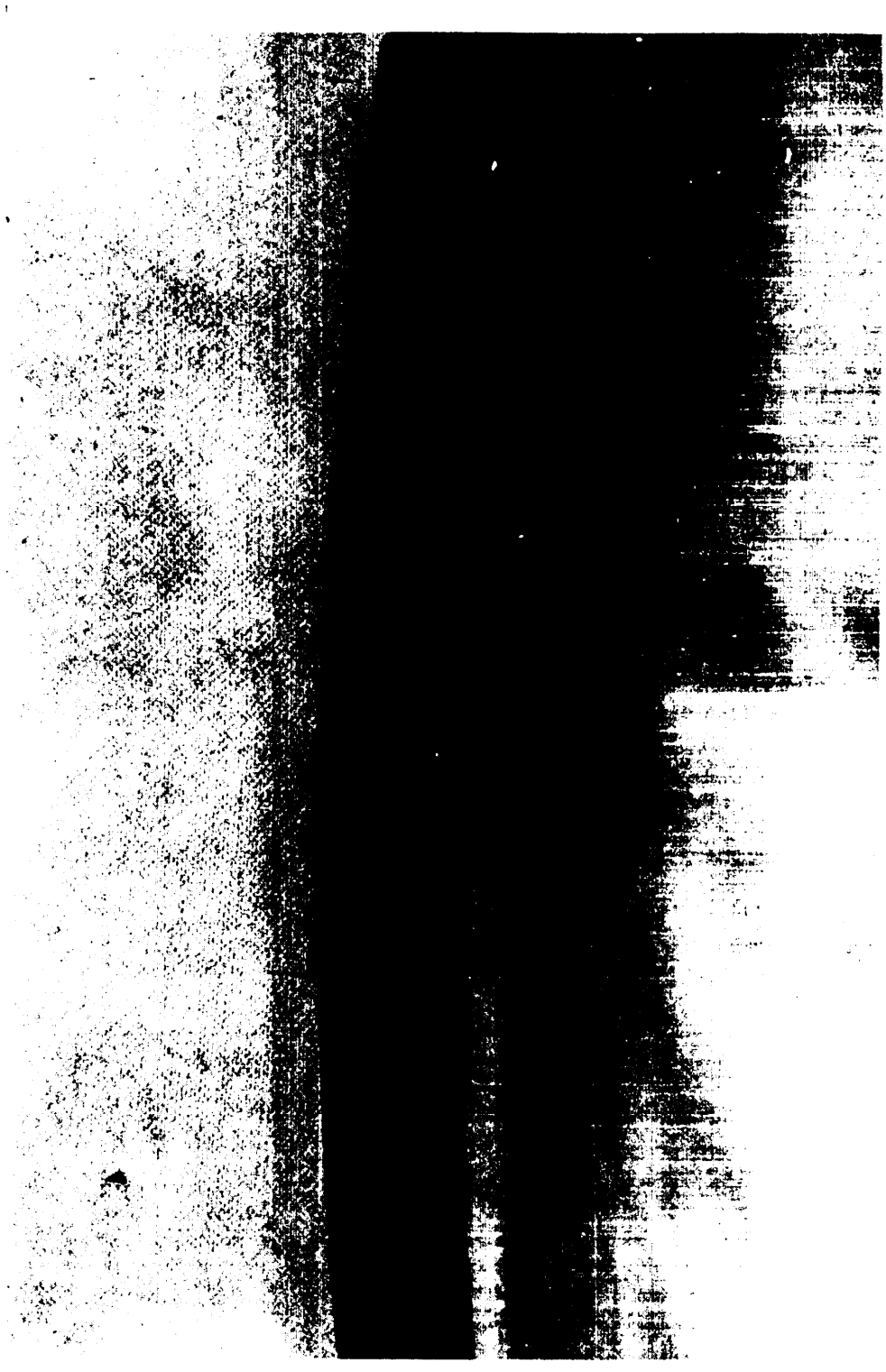


FIGURE 58. Light cover, under gage, or thickness below tolerance limit—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

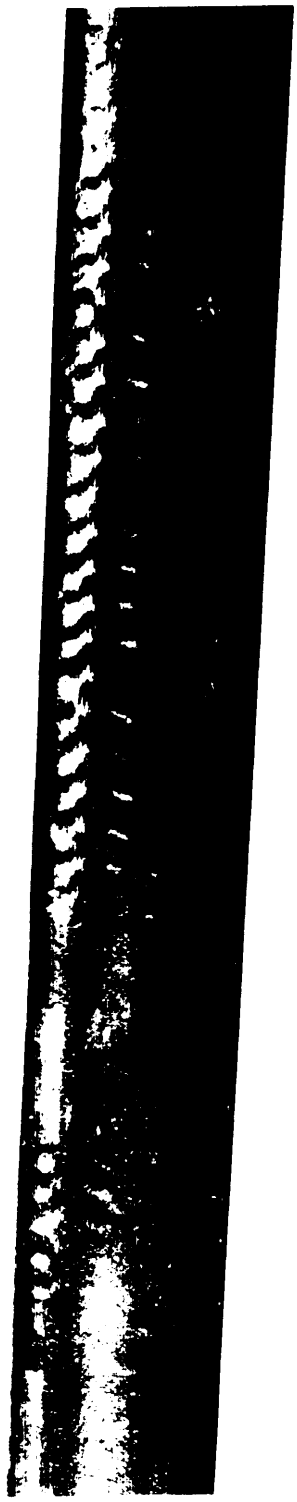


FIGURE 59. Light cover, under gage, or thickness below tolerance limit—major.

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Check the source to verify that this is the current version before use.



FIGURE 60. Light cover (roughness), thickness not below specified or not specified —minor.

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FIGURE 61. Light cover (roughness), thickness not below specified or not specified —minor.

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FIGURE 62. Light cover (roughness), thickness not below specified or not specified —minor.

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FIGURE 63. Light cover (roughness), thickness not below specified or not specified —minor.

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Check the source to verify that this is the current version before use.

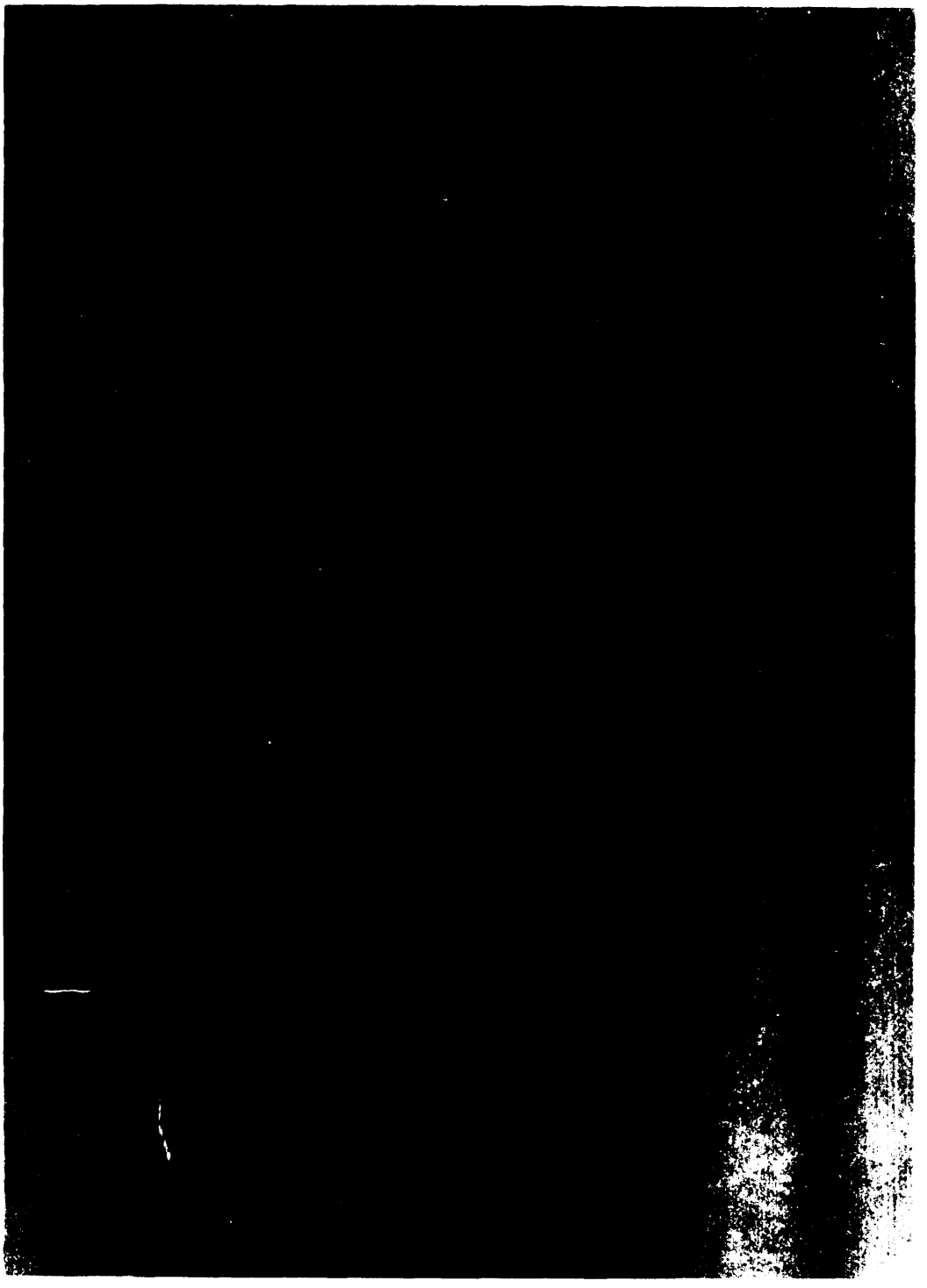


FIGURE 64. Loose cover—major.

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FIGURE 65. Loose cover—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

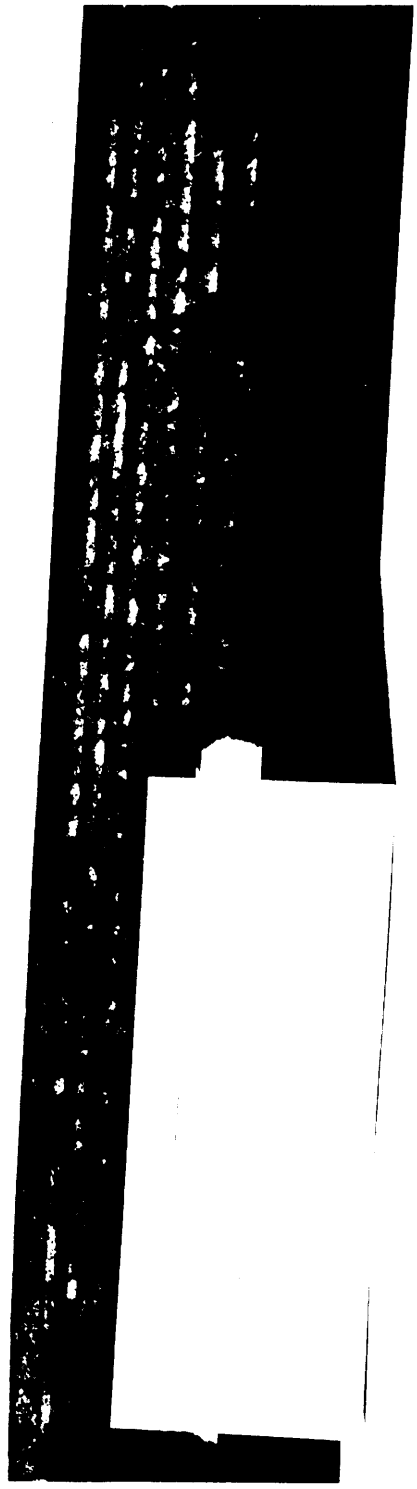


FIGURE 66. Lubricant, excess—not a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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FIGURE 67. LUMP, foreign material inclusion or lamination—major.

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Check the source to verify that this is the current version before use.

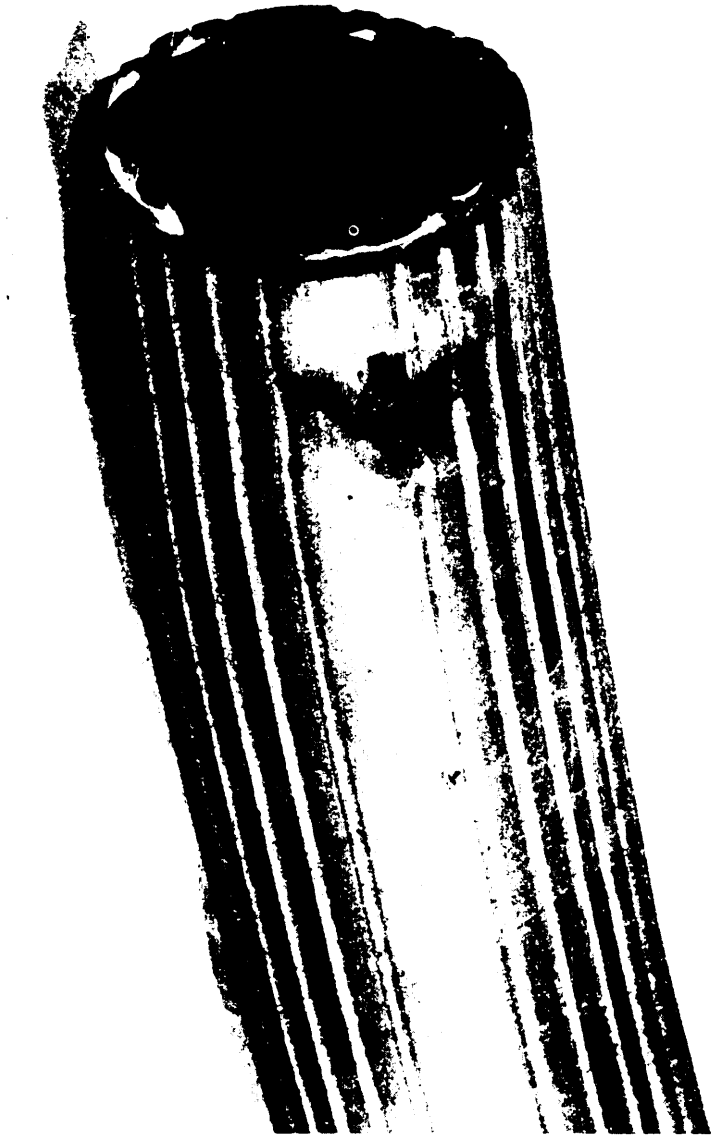


FIGURE 68. Lump, foreign material inclusion or lamination—major.

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Check the source to verify that this is the current version before use.

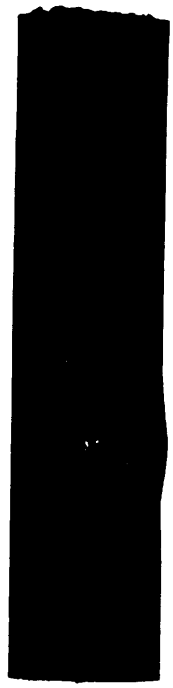


FIGURE 69. Lump, small protrusion, no hole: class A—major: classes B and C—minor.

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Check the source to verify that this is the current version before use.

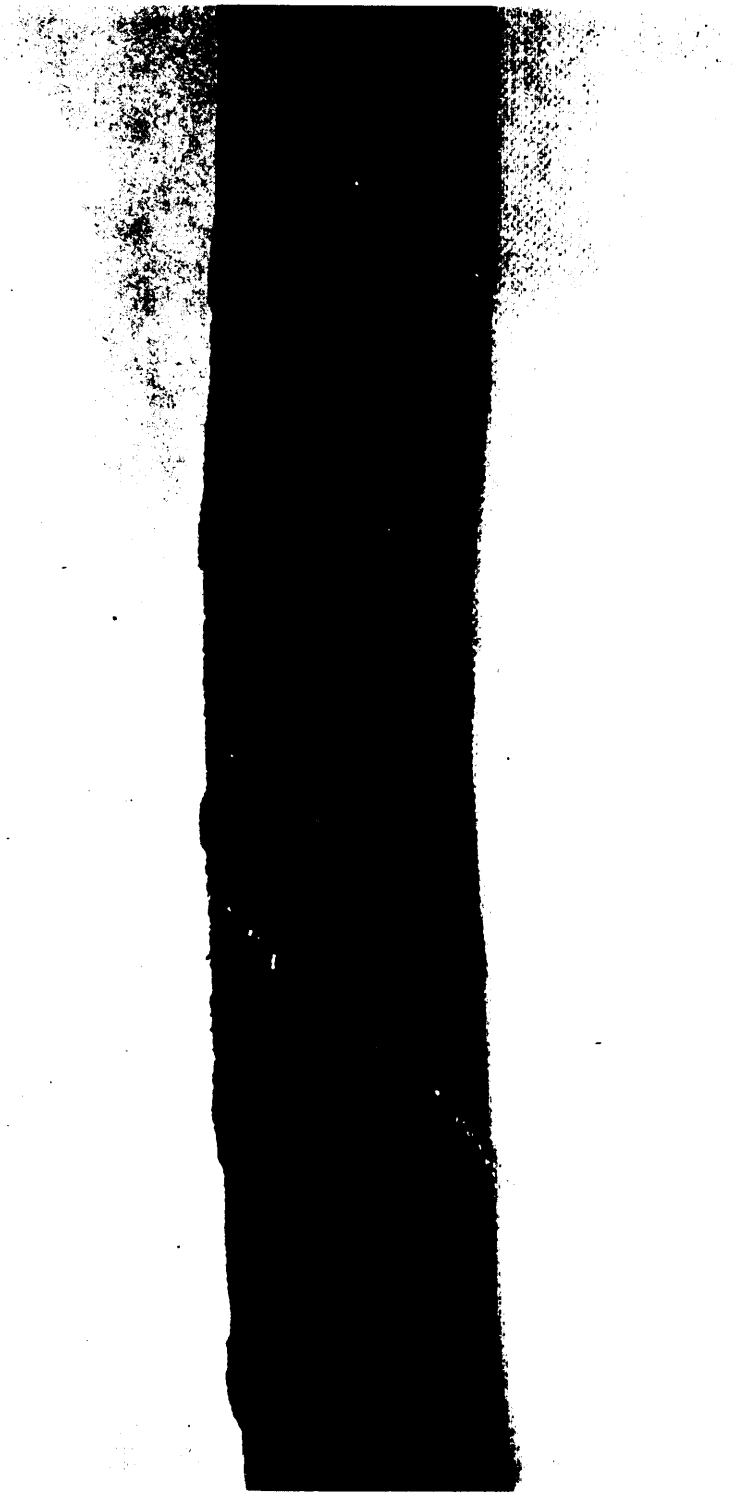


FIGURE 70. Missing cover, reinforcement exposed—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

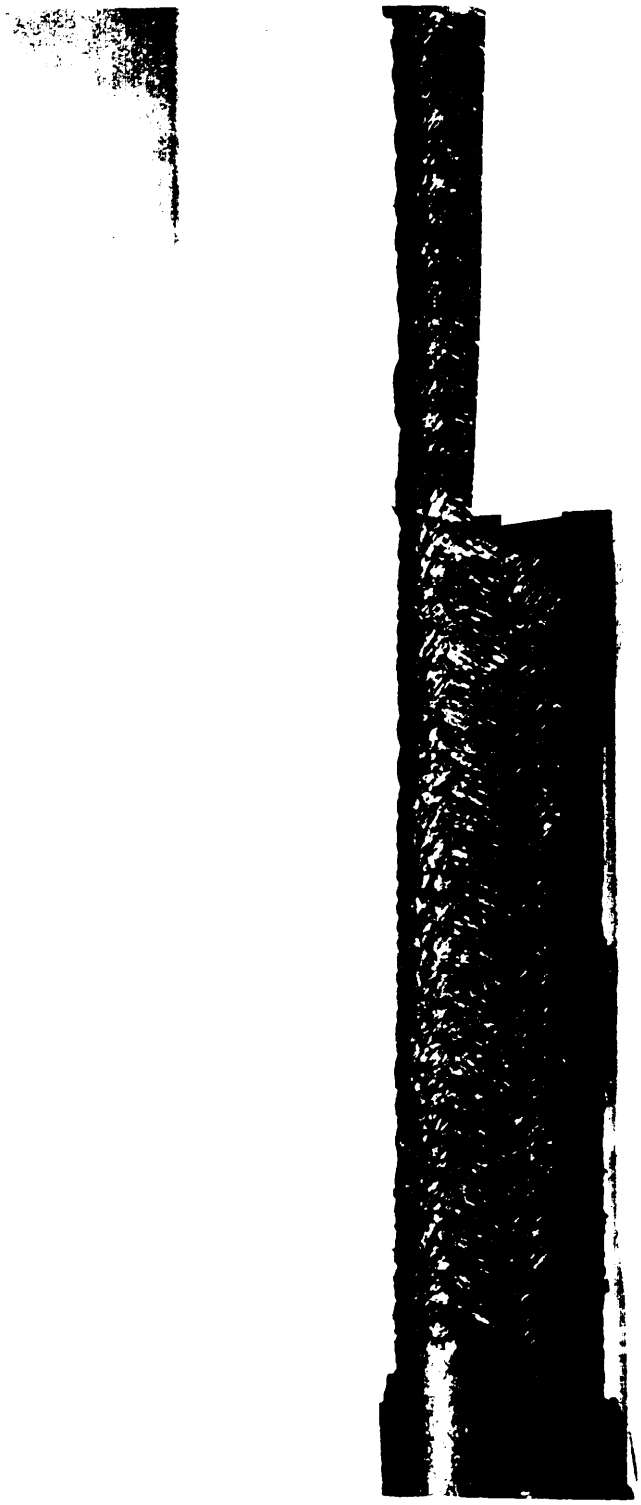


FIGURE 71. Missing cover, reinforcement exposed—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

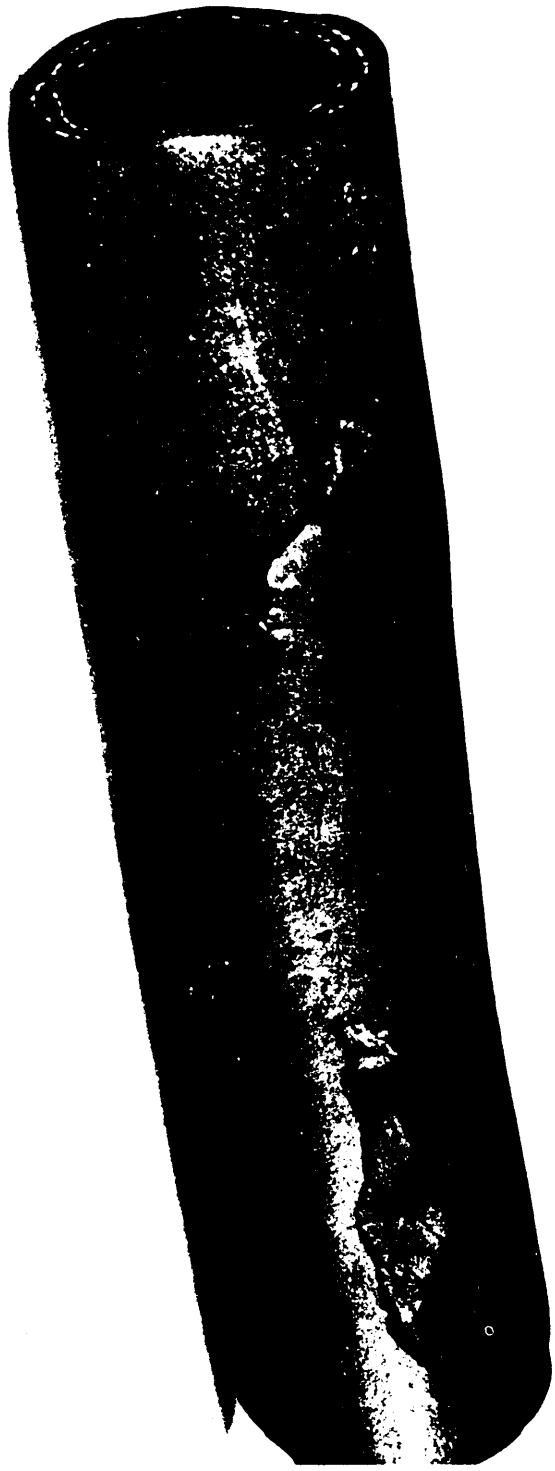


FIGURE 72. Patched cover, open edge—major.

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Check the source to verify that this is the current version before use.

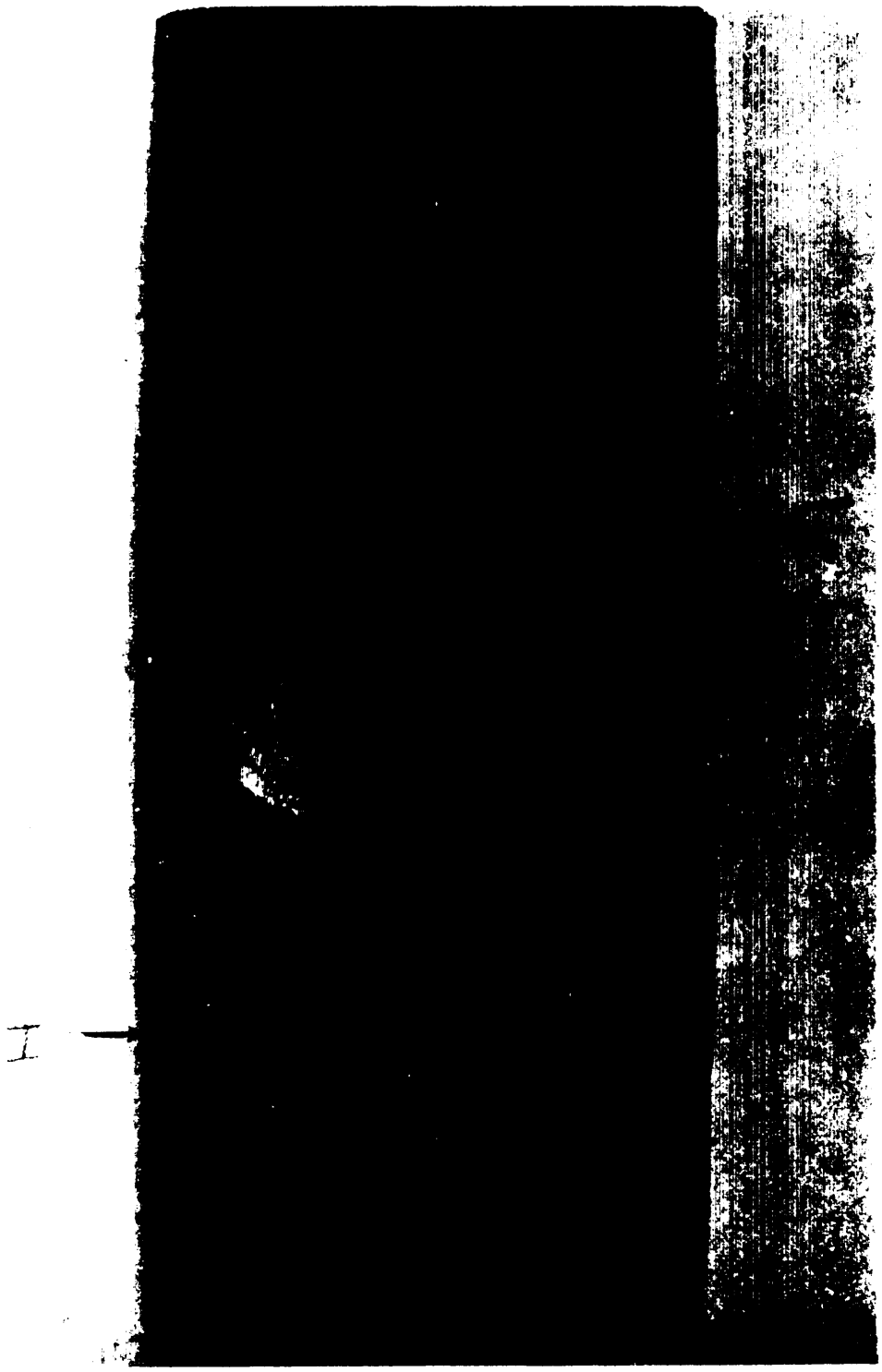


FIGURE 73. Patched cover:

I—Incorrect thickness. high or low—major.
II—Correct thickness and securely cured—not a defect.

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Check the source to verify that this is the current version before use.

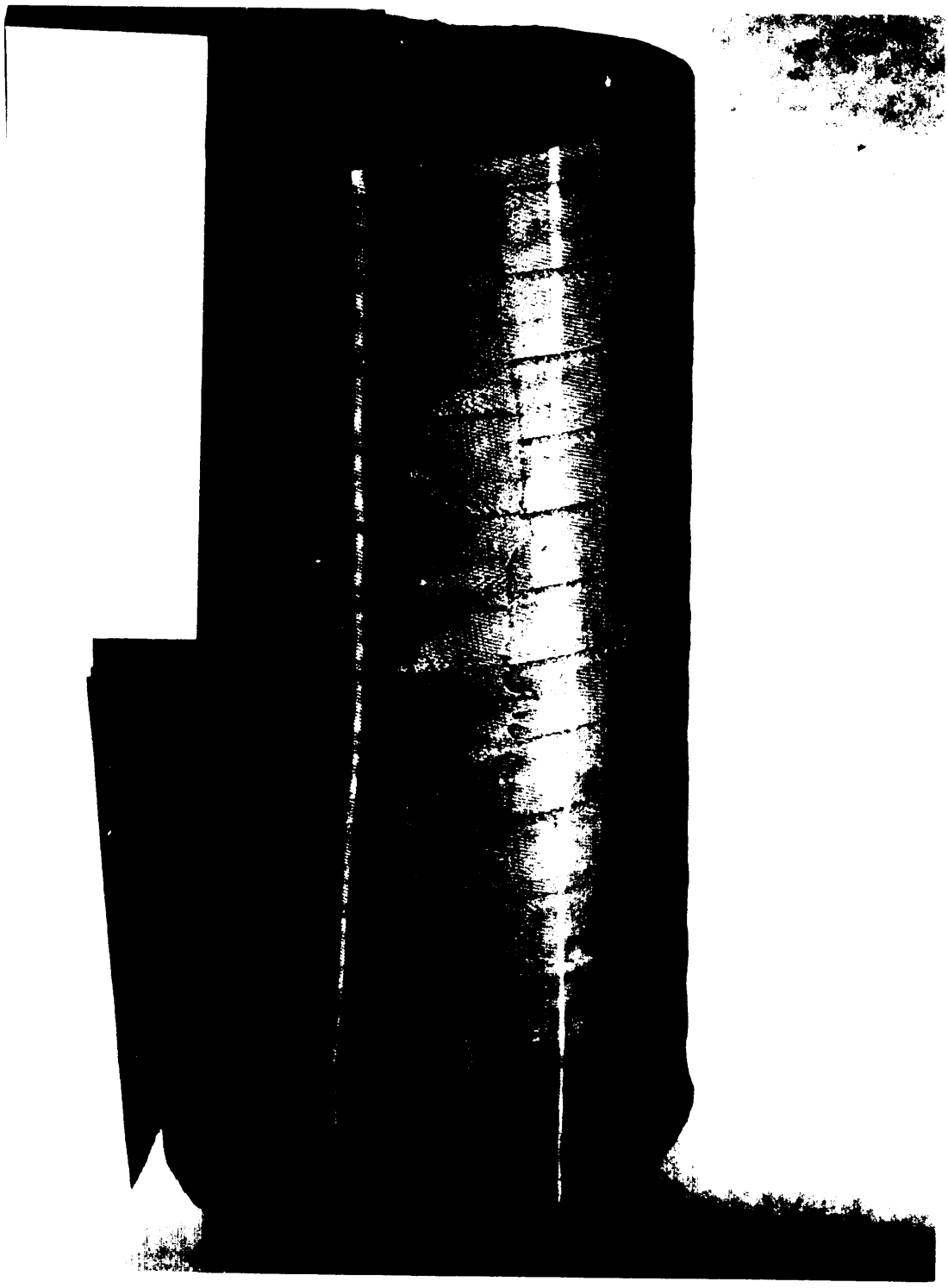


FIGURE 74. Patched cover—not a defect.

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Check the source to verify that this is the current version before use.

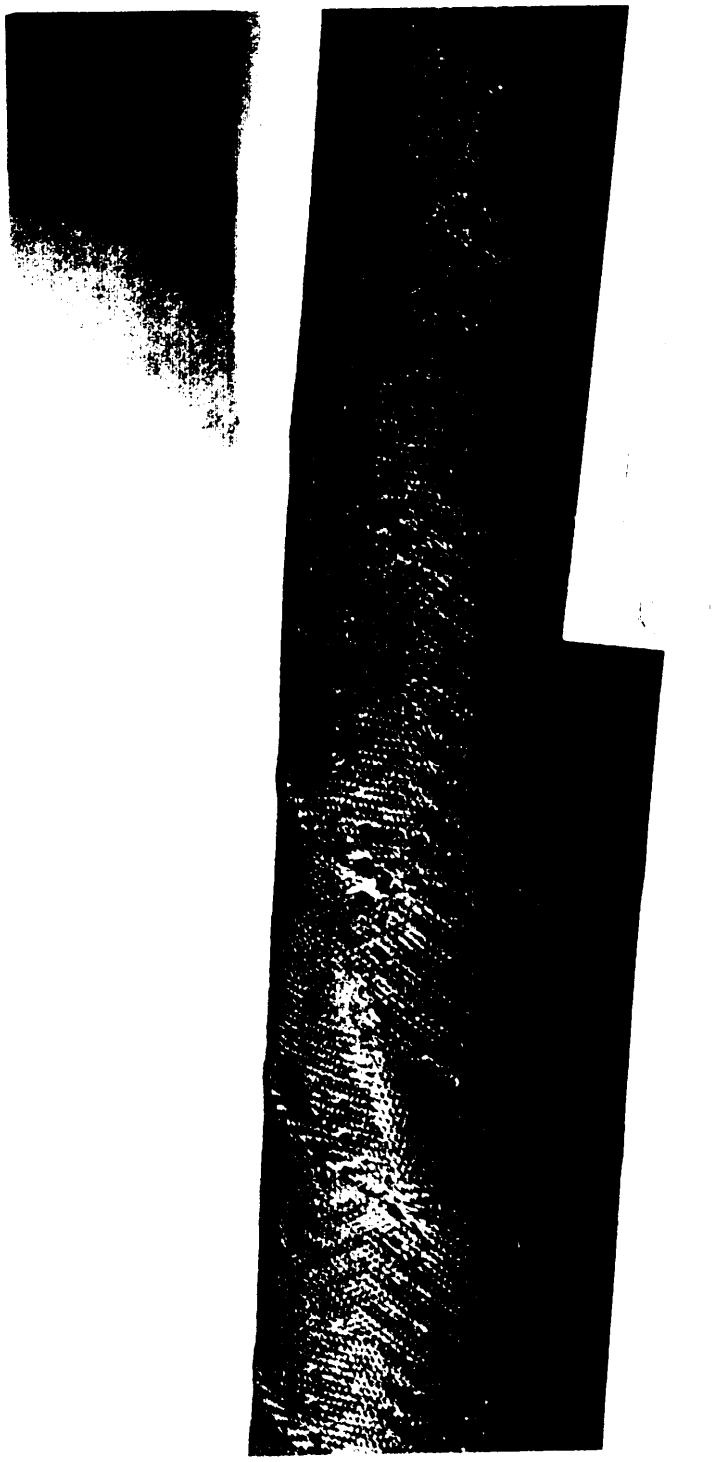


FIGURE 75. Porosity, extended area, over 10 percent of thickness—major.

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Check the source to verify that this is the current version before use.

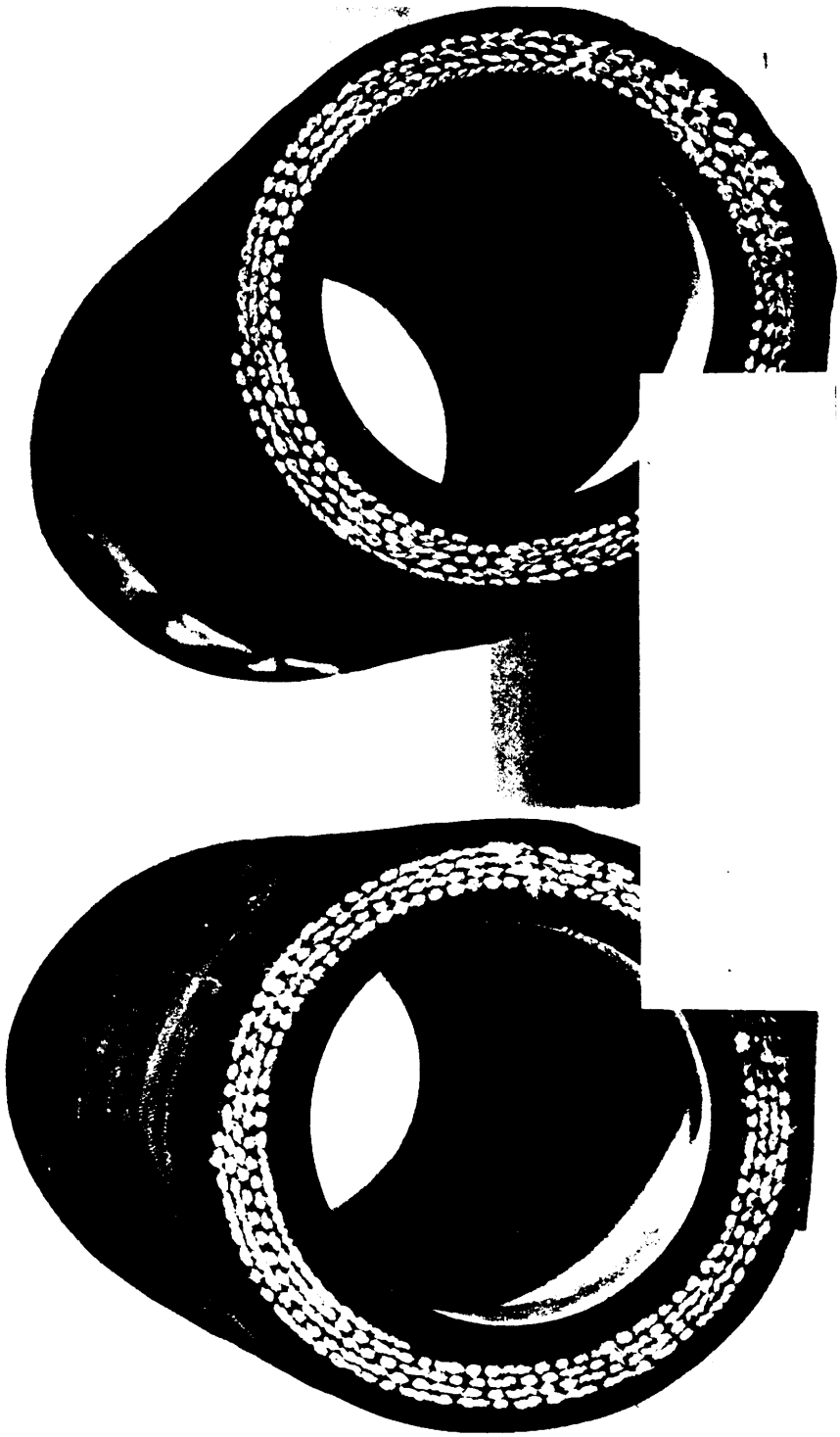


FIGURE 76 Porosity, small localized arrea, reinforcement not exposed—minor.

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FIGURE 77. Pricker marks through to reinforcement—major (see note in table I for other classification).

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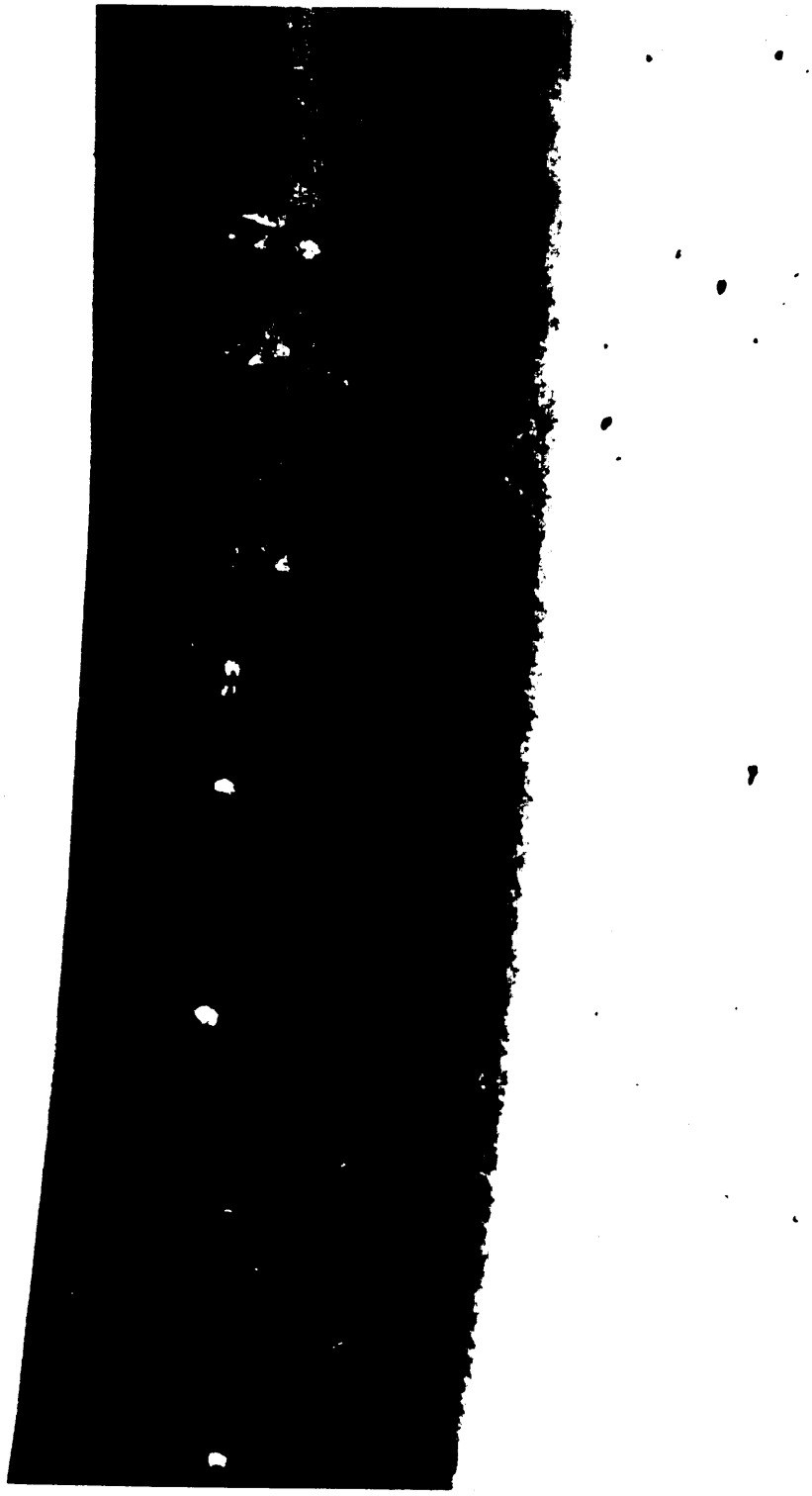


FIGURE 78. Pricker mark or air tracks, small area, not through to reinforcement—minor.

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FIGURE 79. Ridge, lamination or separation—major.

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FIGURE 80. Ridge, large—major.

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FIGURE 81. Ridge. small—minor (see note in table I for other classification).

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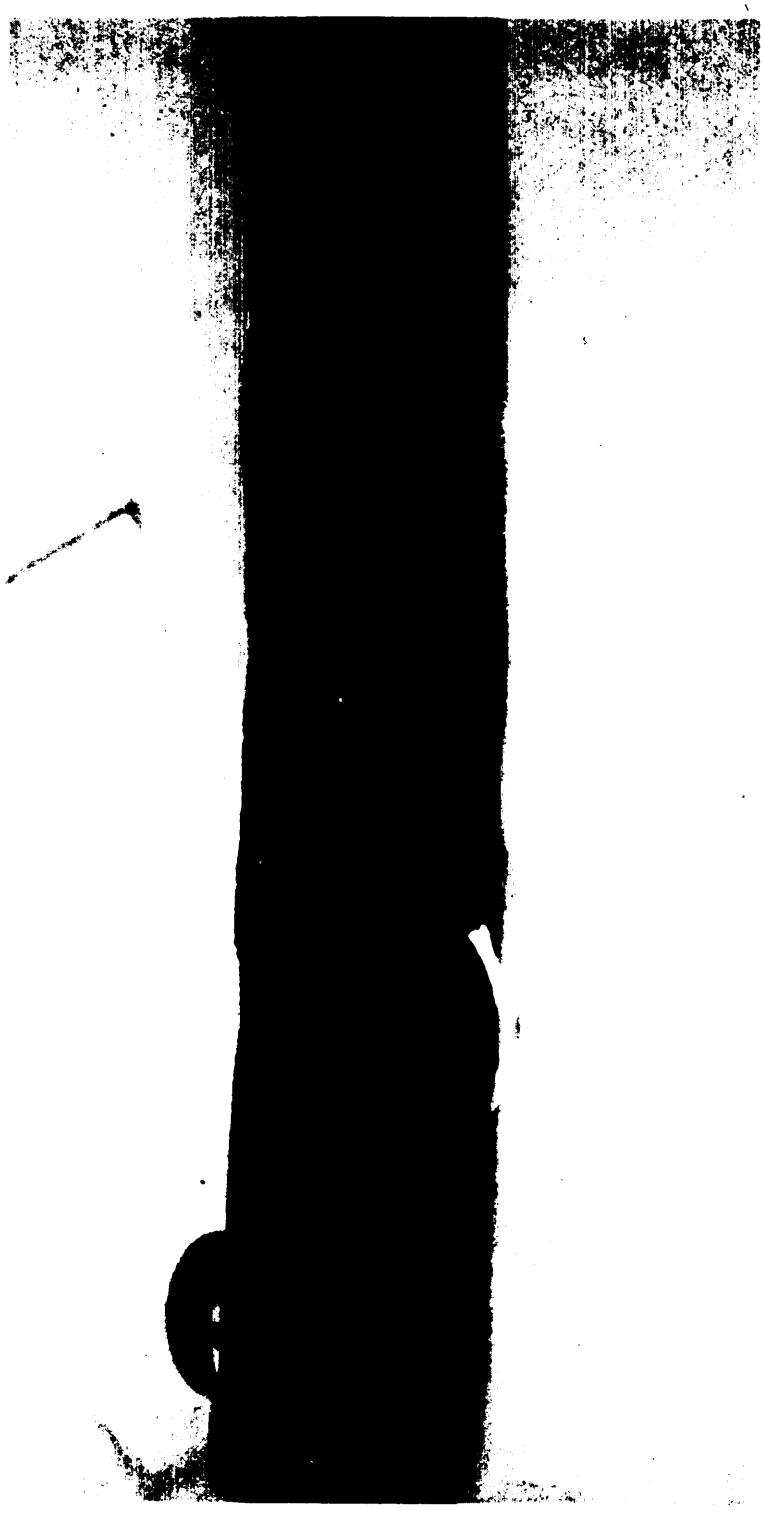


FIGURE 82. Ridge, small—minor (see note in table I for other classification).

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Check the source to verify that this is the current version before use.

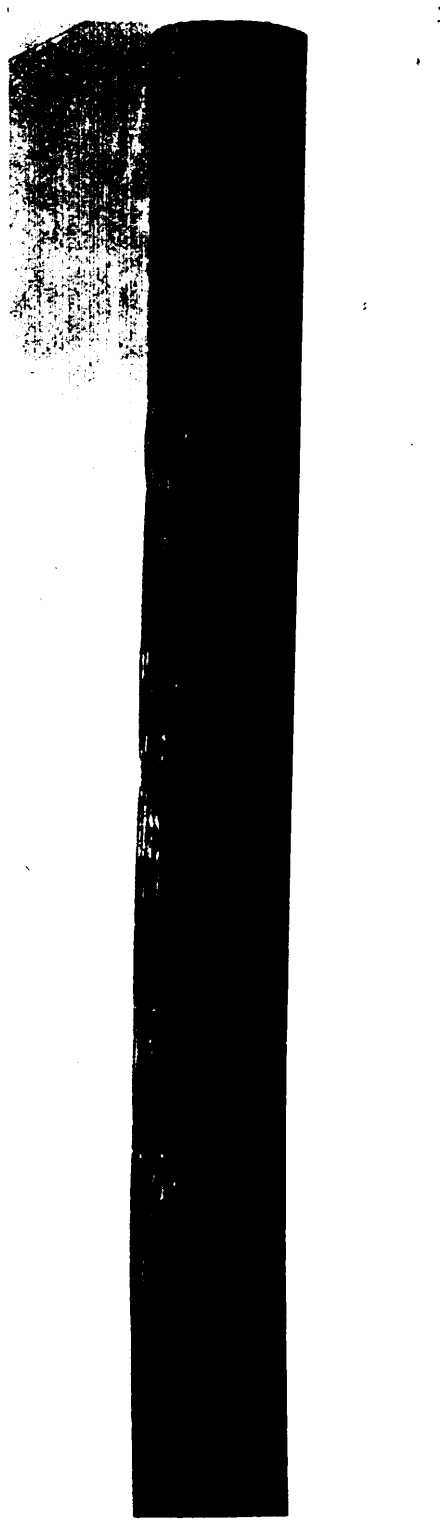


FIGURE 83. Rough cover—major.

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FIGURE 84. Split cover—major.

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FIGURE 85. Split cover—major.

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Check the source to verify that this is the current version before use.

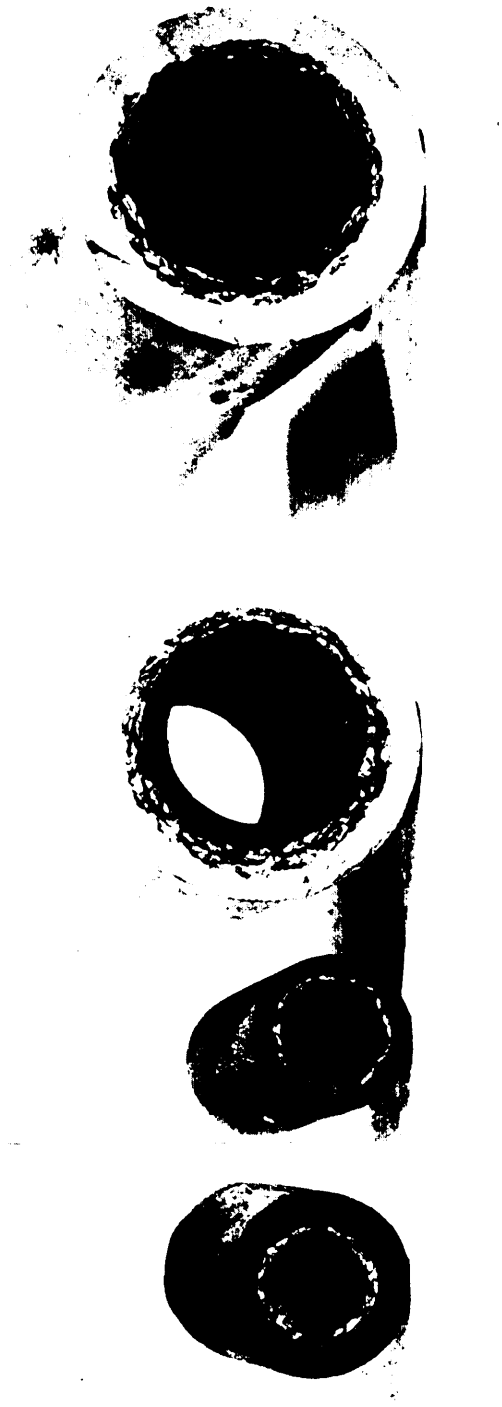
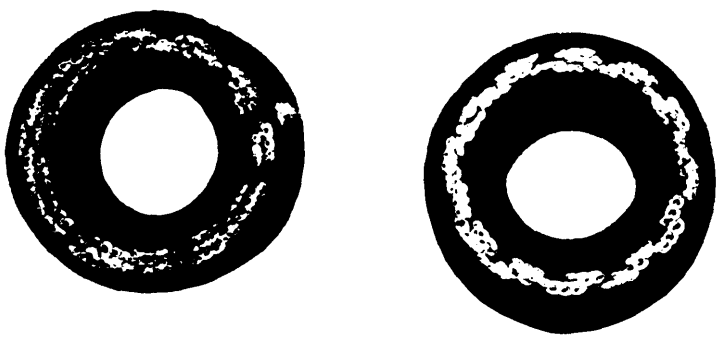


FIGURE 86. Thickness, over (not mentioned in procurement document)—not a defect.

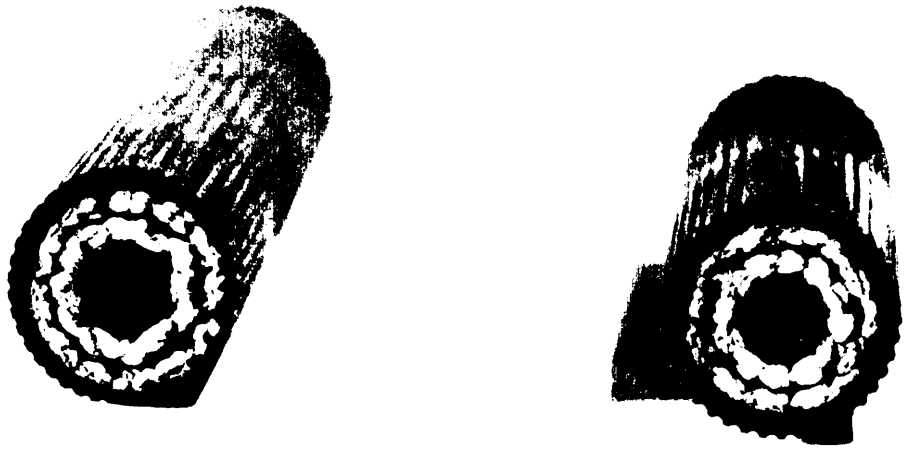
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



I II

FIGURE 87. Thickness, under, below tolerance limit:
I—Thin cover—major.
II—Normal hose.

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I II

FIGURE 88. Thickness:
I—Eccentric (lopsided): outside tolerance—major; tolerance met—not a defect.
II—Normal hose.

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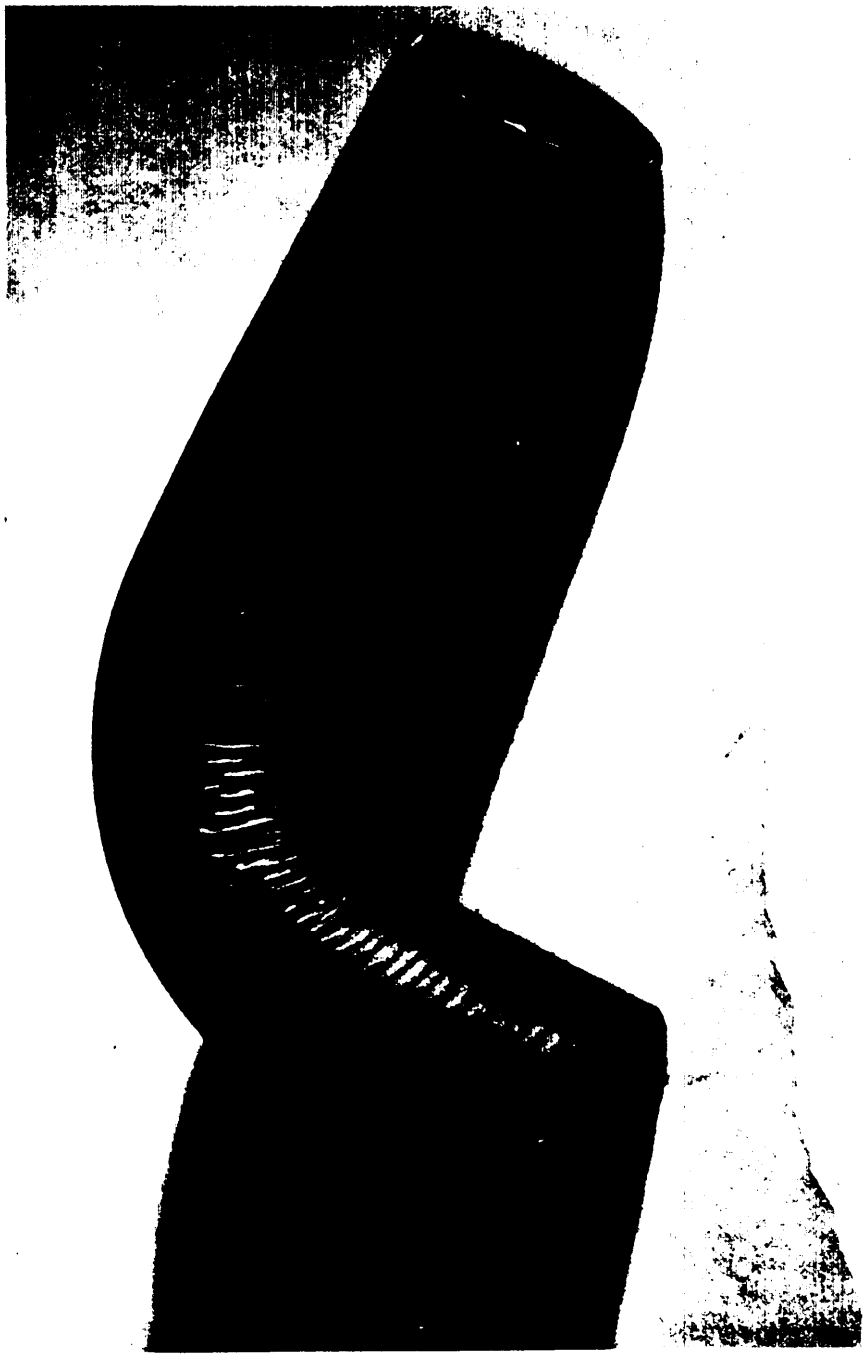


FIGURE 89. Wrinkles—major.

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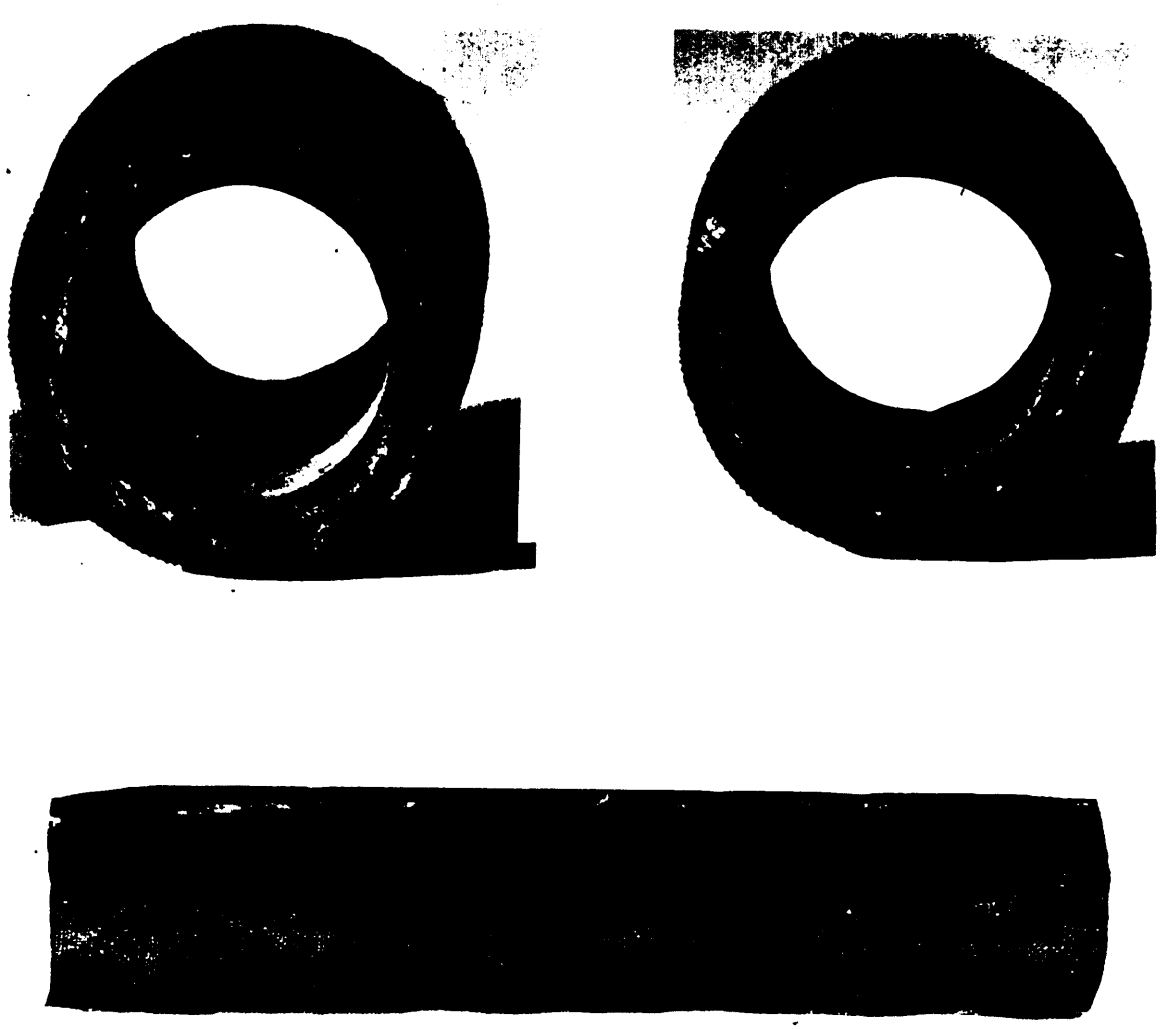


FIGURE 90. Wrinkles, wrapper, medium—major.

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FIGURE 91. Wrinkles, wrapper, small—minor.

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FIGURE 92. Wrinkles, wrapper, fine—not a defect.

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FIGURE 93. Wrinkles, wrapper, fine—not a defect.

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Check the source to verify that this is the current version before use.

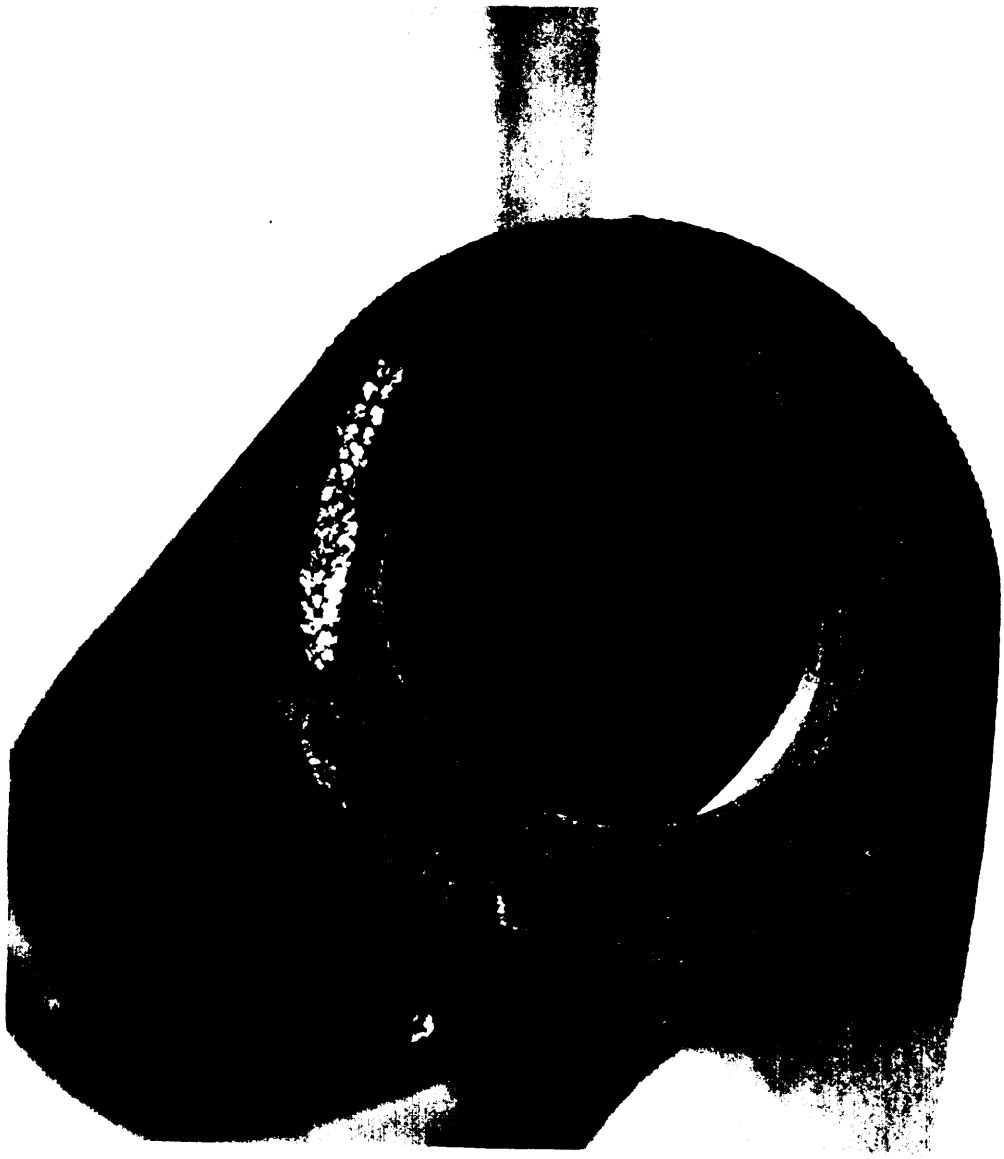
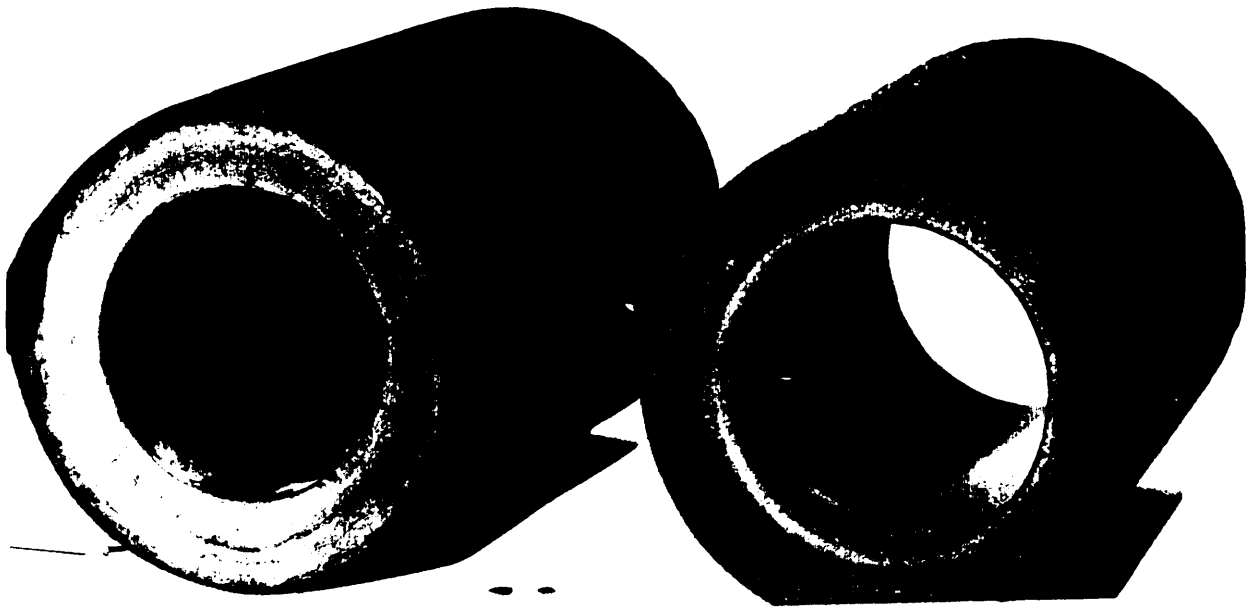


FIGURE 94. Capped, fabric exposed—major.

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I—Excessive—major.

II—Small—not a defect.

FIGURE 95. Flash or rind:

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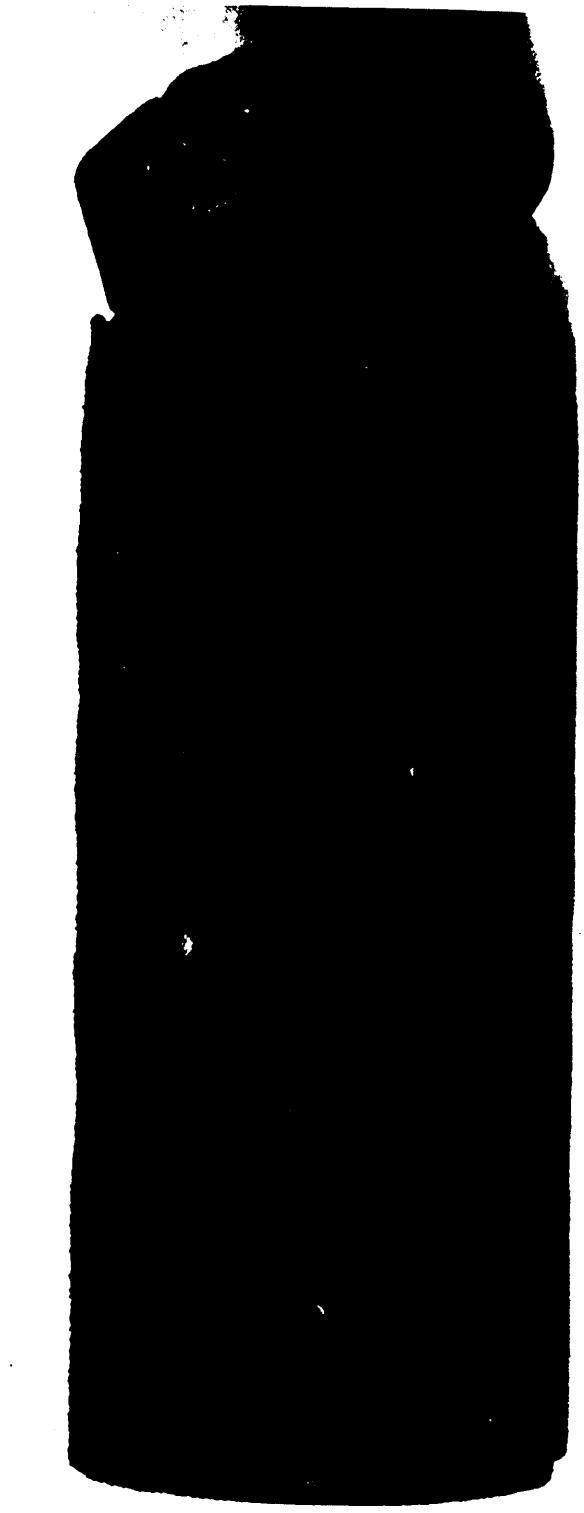


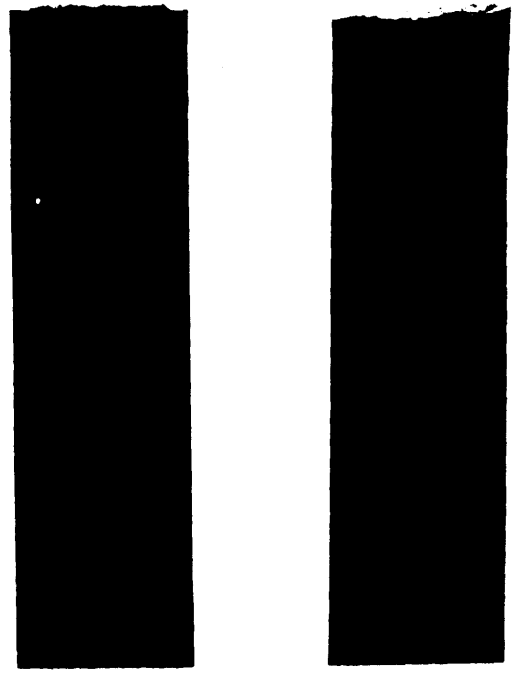
FIGURE 96. Loose ends—major.

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FIGURE 97. Cut, angle or diagonal: specified lengh—major; random length—not a defect.

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I II

I—Normal hose. II—Angle end cut.

FIGURE 98. Cut, angle or diagonal: specified length—major; random length—not a defect.

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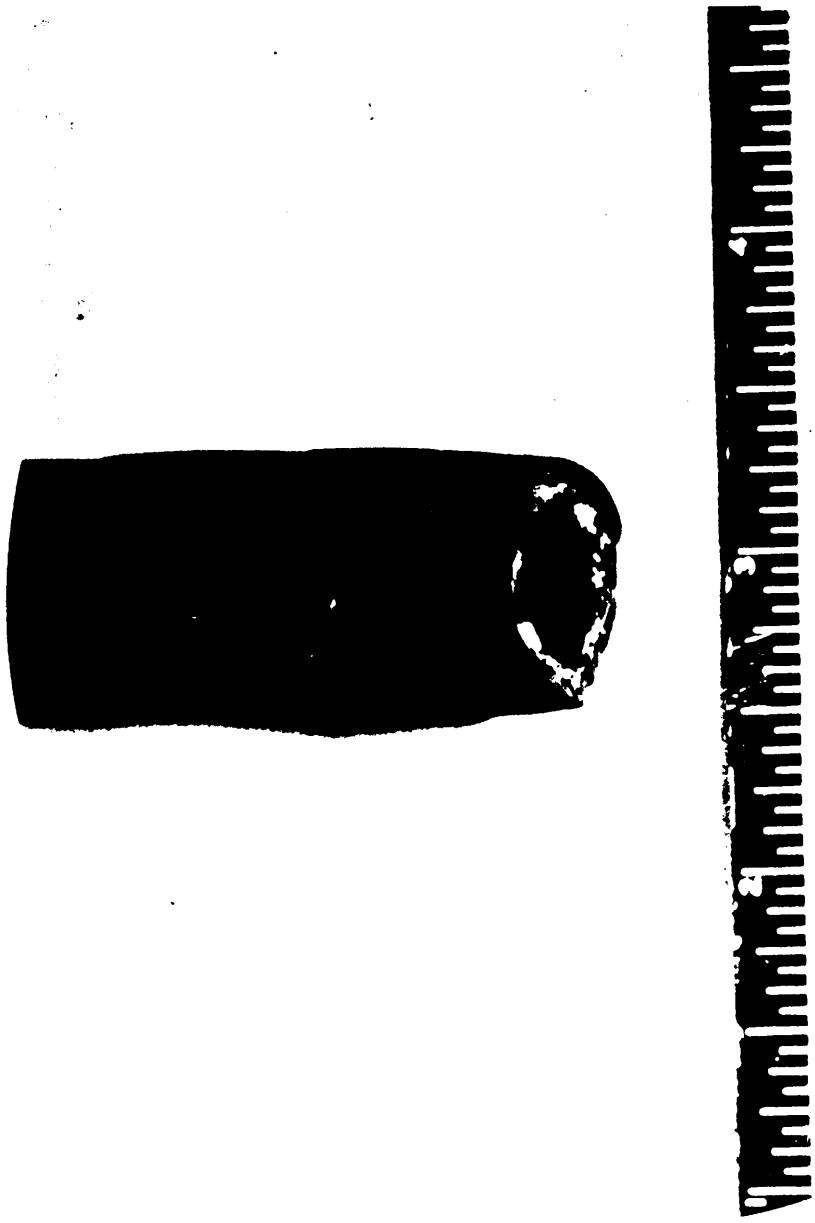


FIGURE 99. Ragged ends: specified length—major: random length—not a defect.

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FIGURE 100. Ragged ends: specified length—major: random length—not a defect.

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FIGURE 101. Untrimmed ends: specified length—major; random length—not a defect.

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Check the source to verify that this is the current version before use.

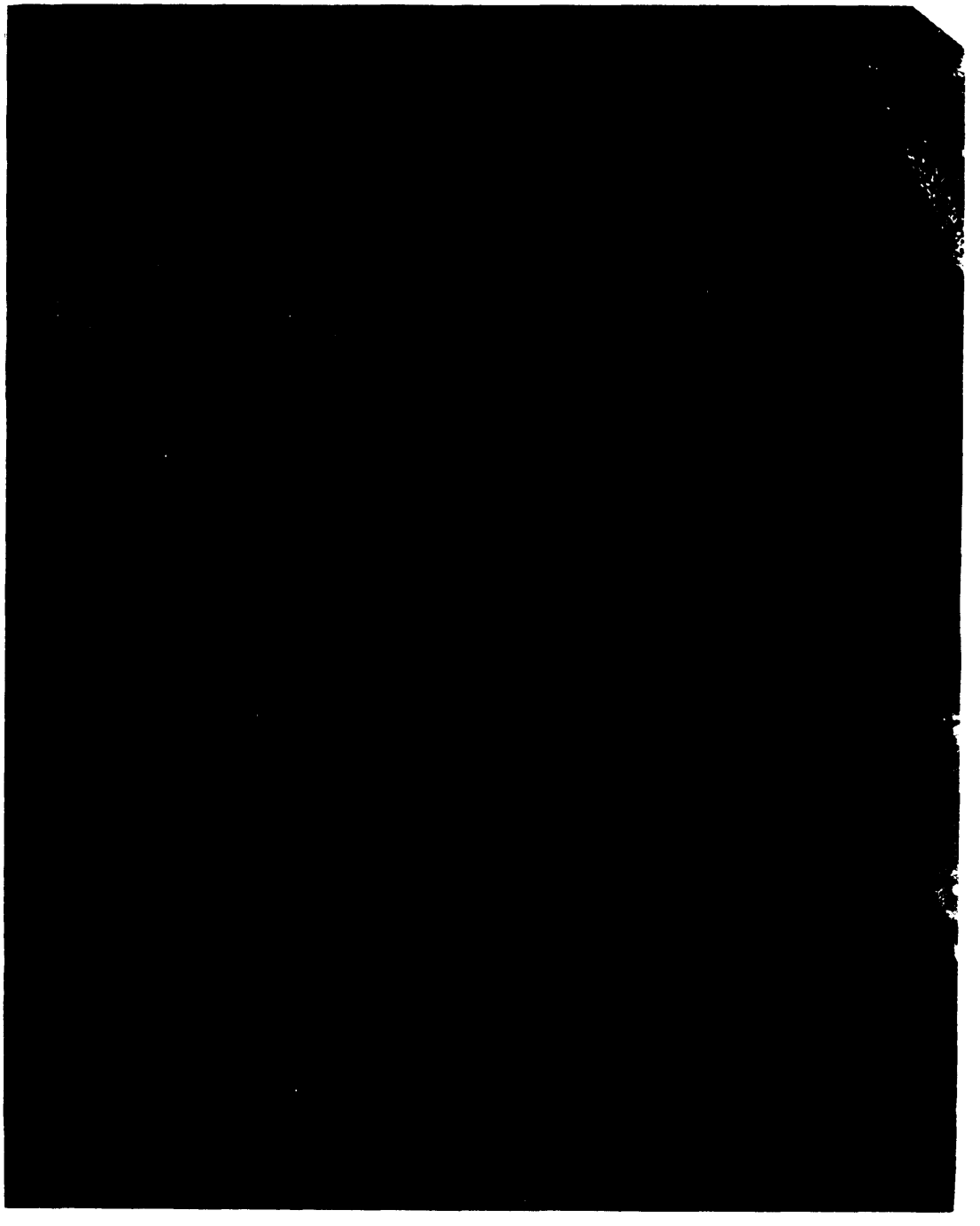


FIGURE 102. Reinforced ends, lamination (separation)—major.

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Check the source to verify that this is the current version before use.

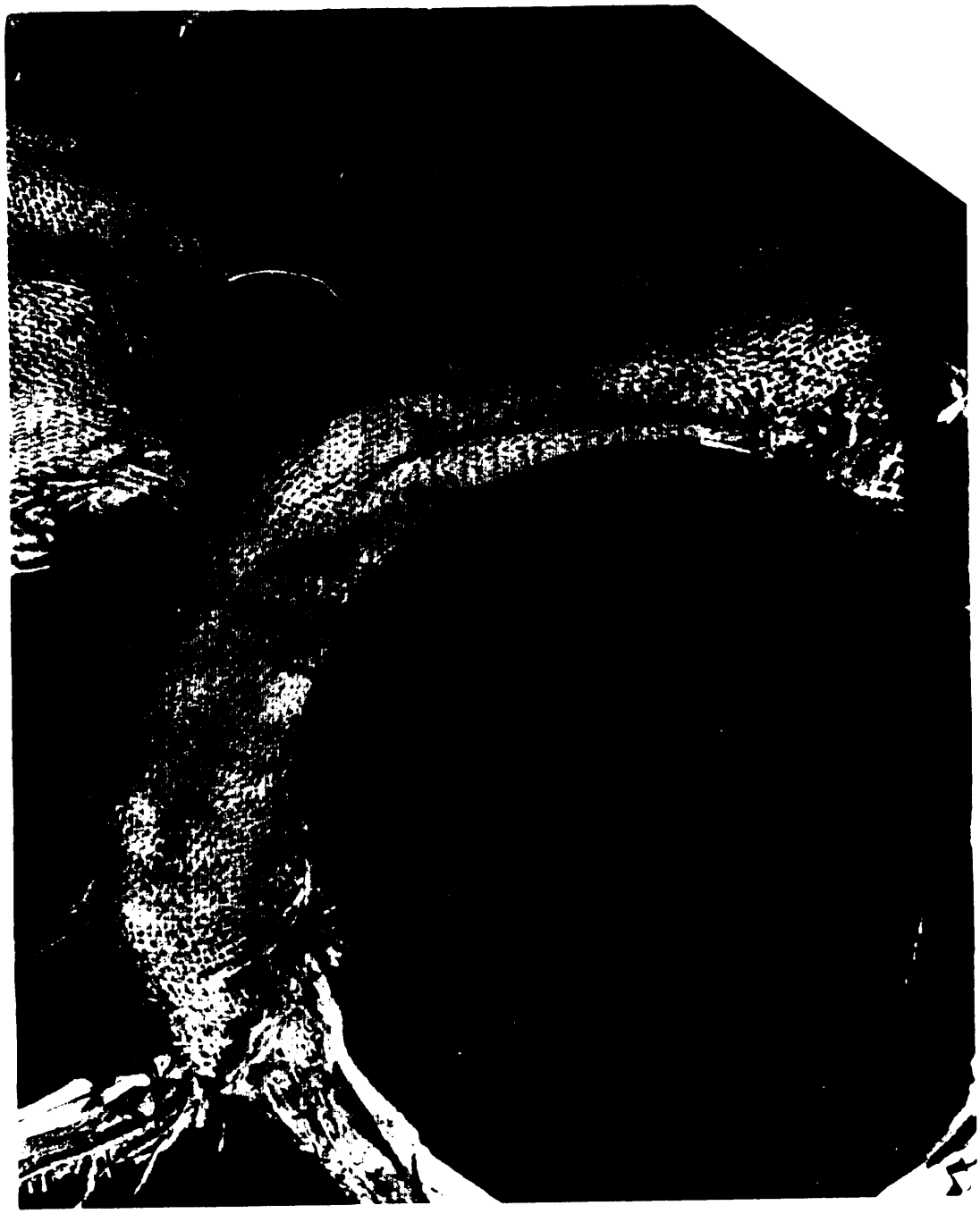


FIGURE 103. Reinforced ends, lamination (separation)—major.

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FIGURE 104. Reinforced ends, void—major.

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Check the source to verify that this is the current version before use.

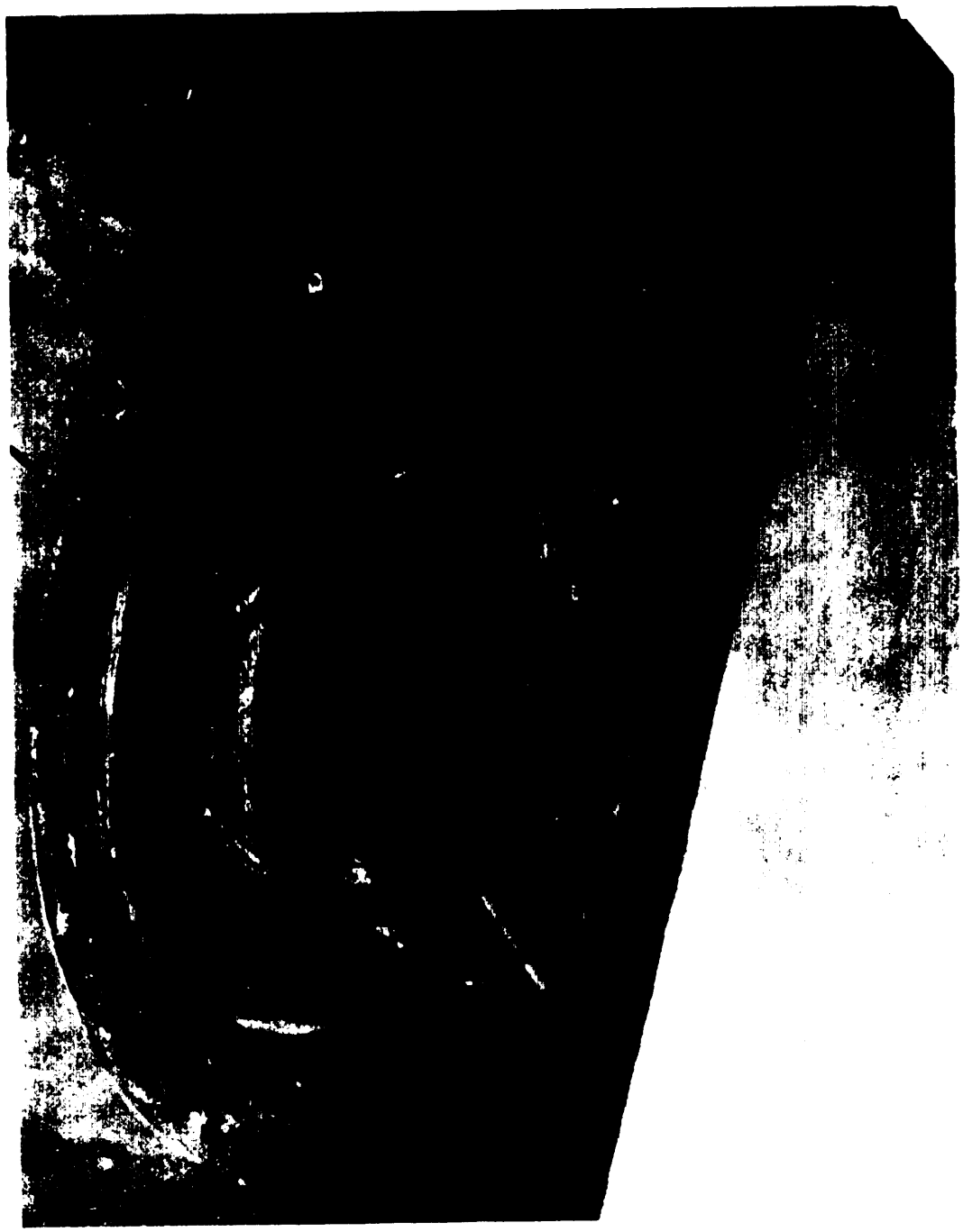


FIGURE 105. Reinforced ends, void—major.

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FIGURE 106. Reinforced ends, void—major.

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FIGURE 107. Connections, flanges, angle set—major.

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Check the source to verify that this is the current version before use.

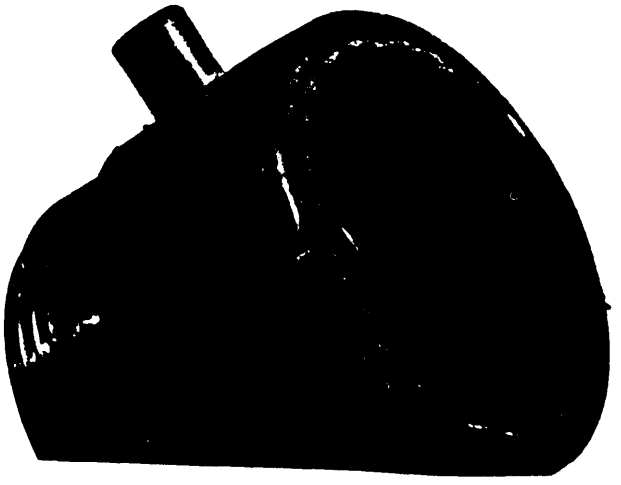


FIGURE 108. Cracked or split metal couplings—major.

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Check the source to verify that this is the current version before use.

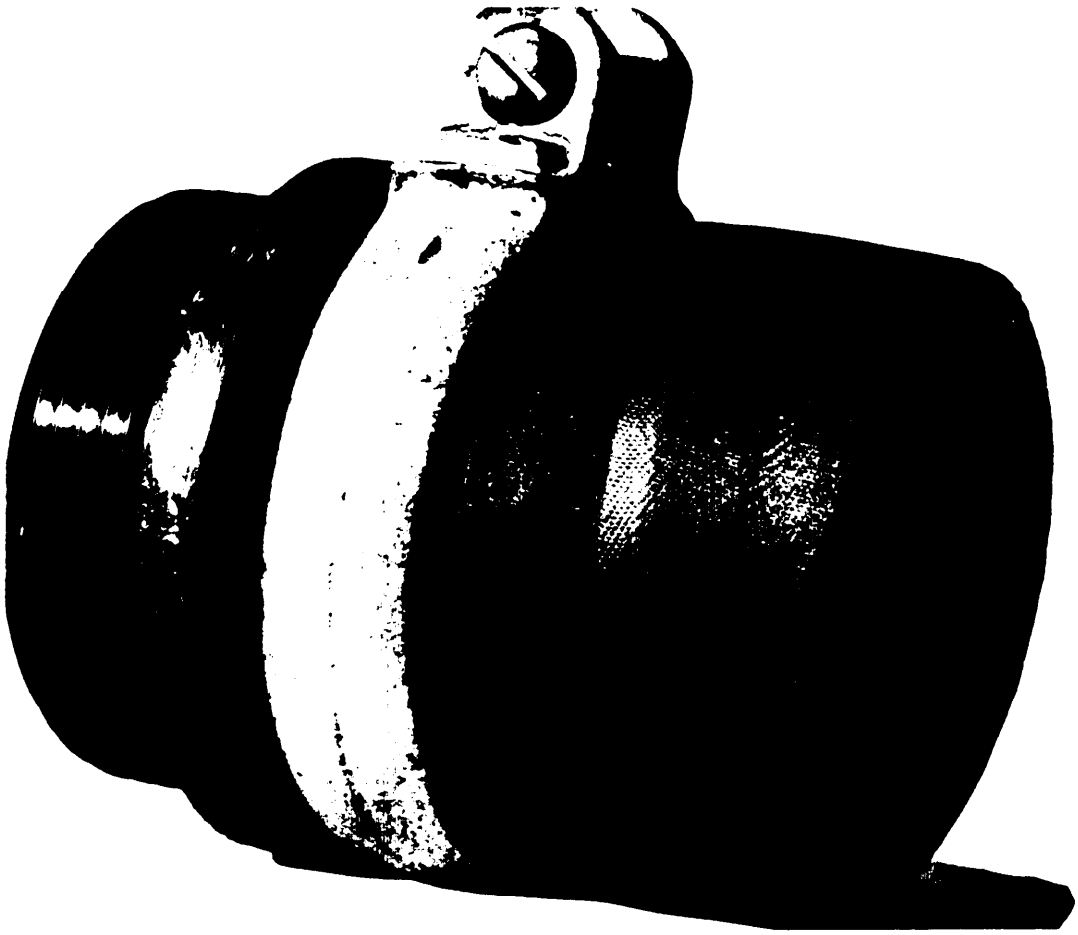


FIGURE 109. Cracked or split metal clamp—major.

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Check the source to verify that this is the current version before use.

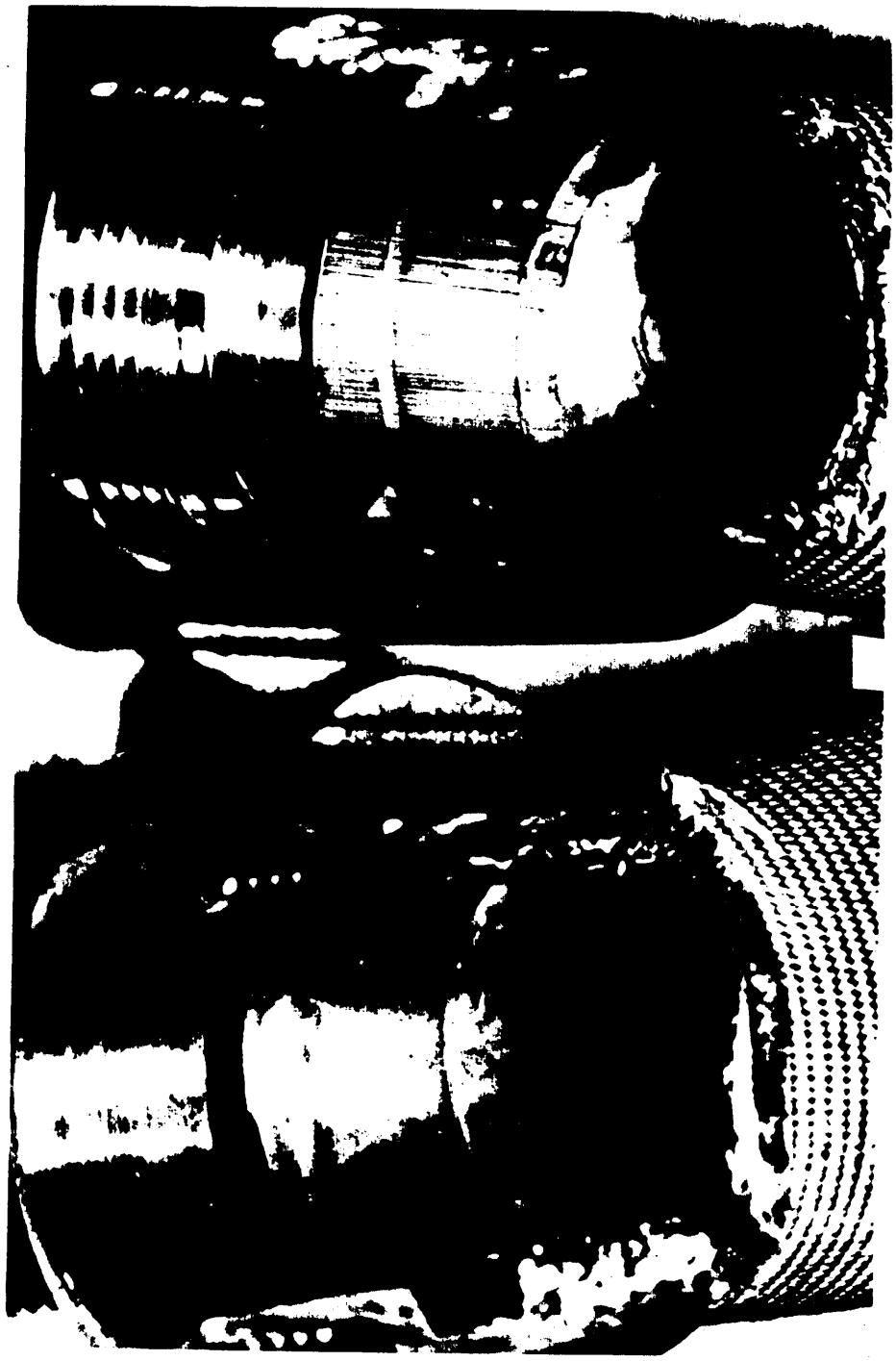


FIGURE 110. Gasket, tail or binder washer for expansion ring coupling:
I—Not properly placed—major.
II—Correct placement.

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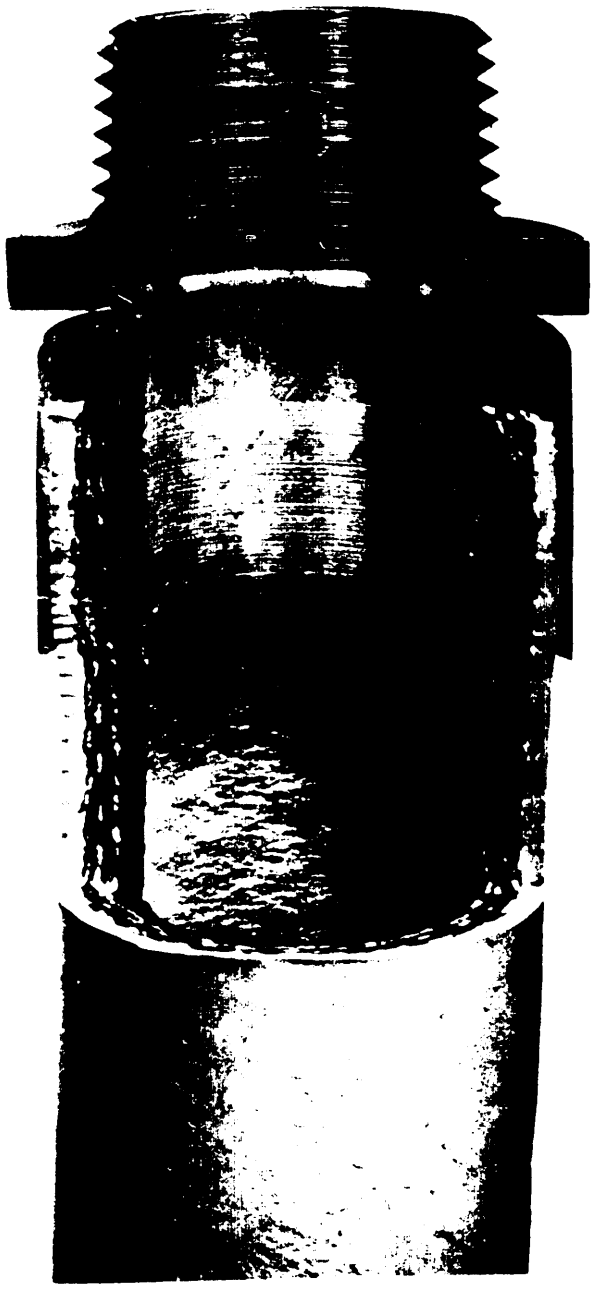


FIGURE III. Insertion distance wrong—major.

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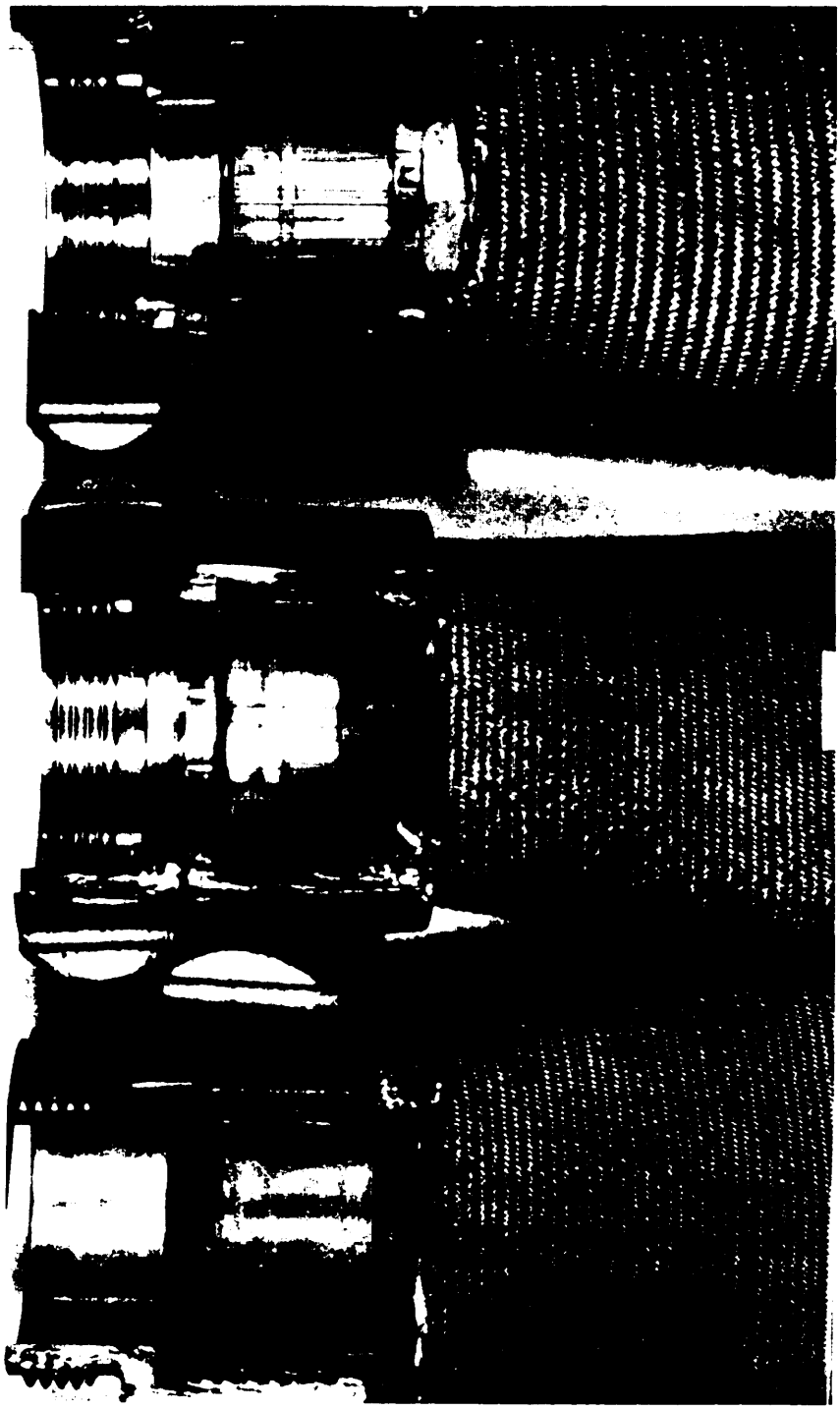


FIGURE 112. Ring, expansion, improperly placed:
I—Correct.
II—Ring in waterway—major.
III—Too much gap—major.

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FIGURE 113. Cement in tube—minor.

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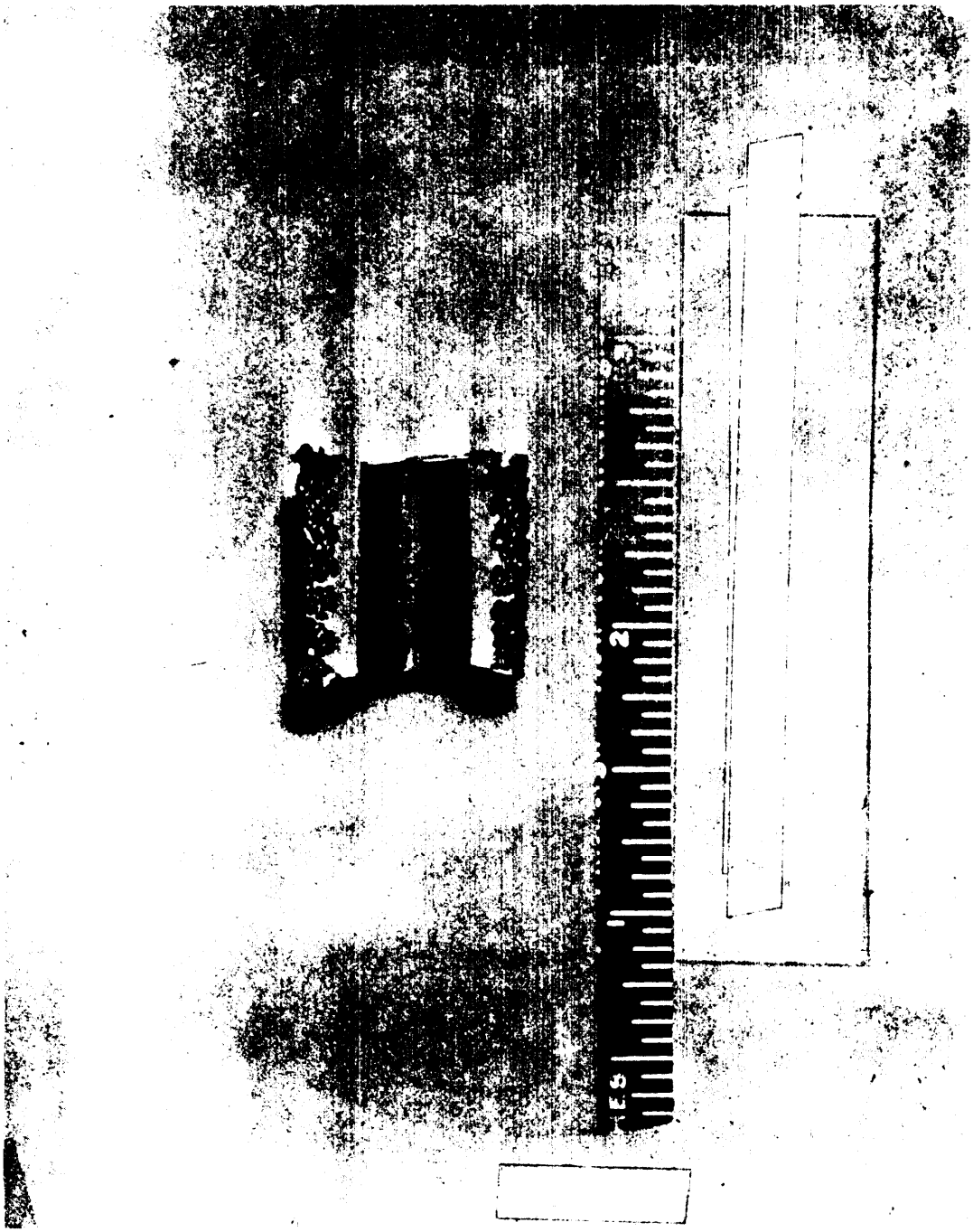


FIGURE 114. Cement in tube—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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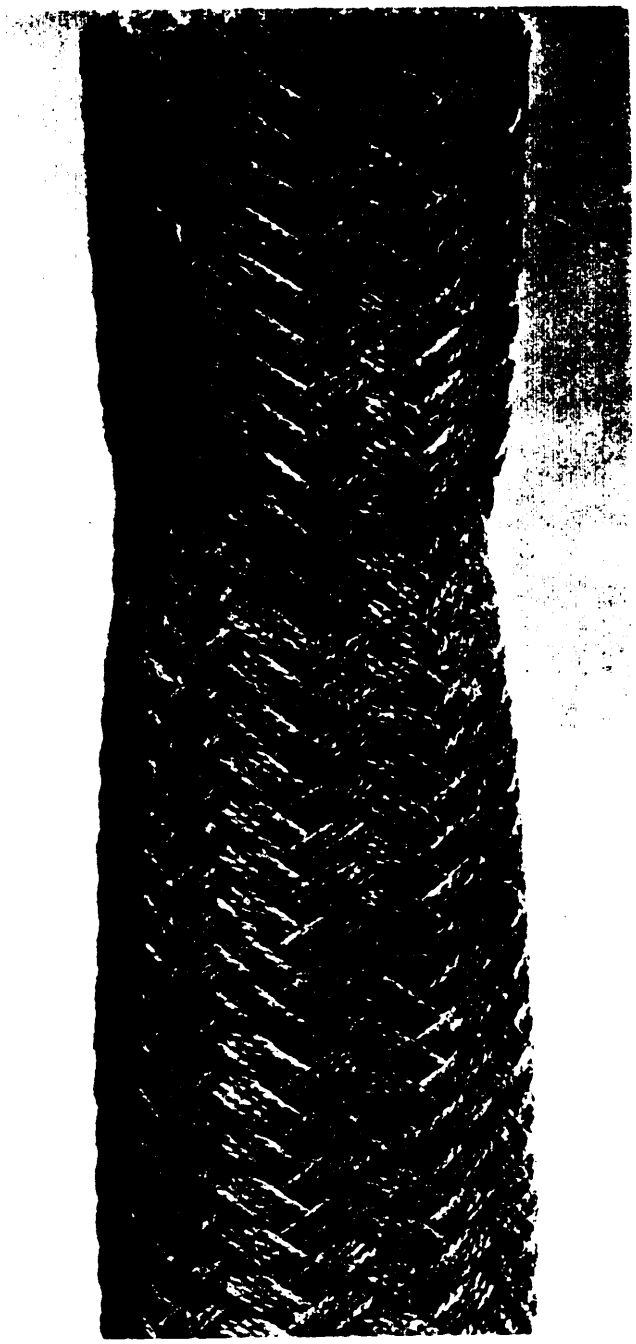


FIGURE 115. Clamp mark—major.

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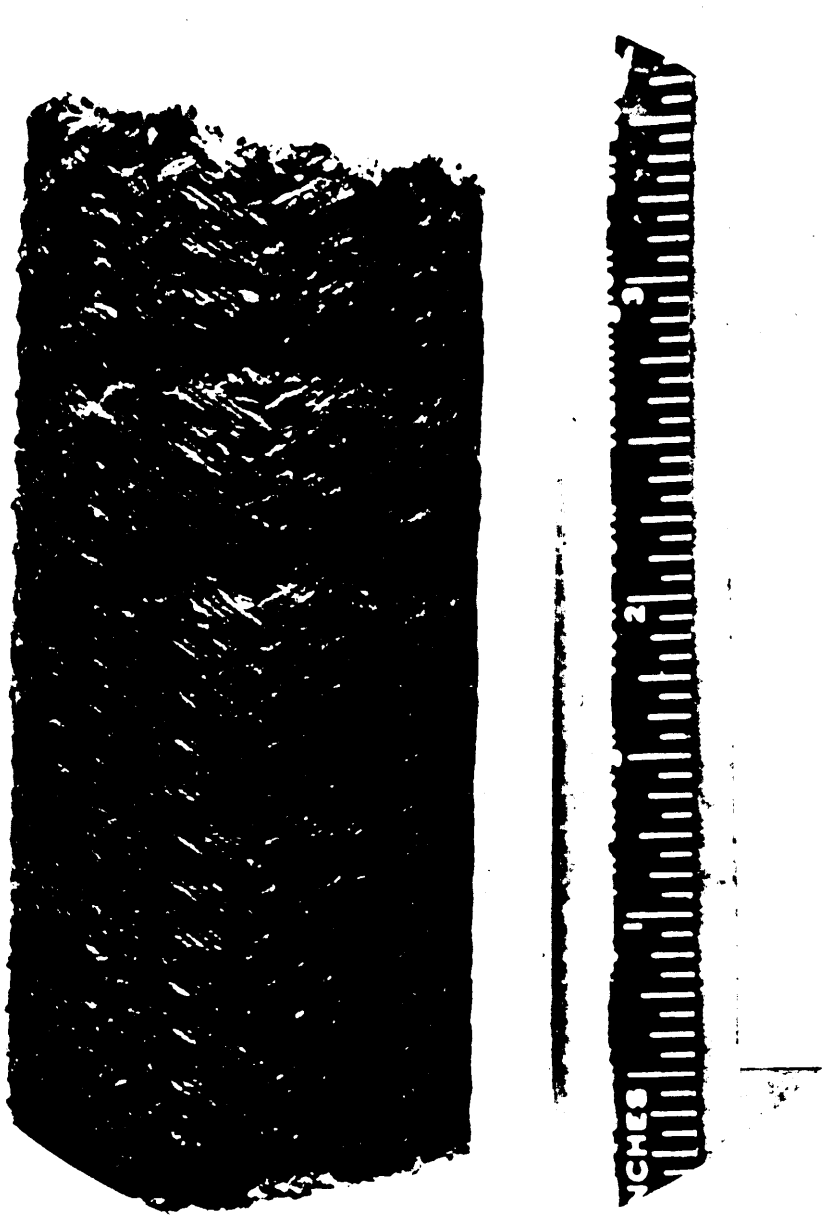


FIGURE 116. Clamp mark—major.

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FIGURE 117. Clamp mark—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

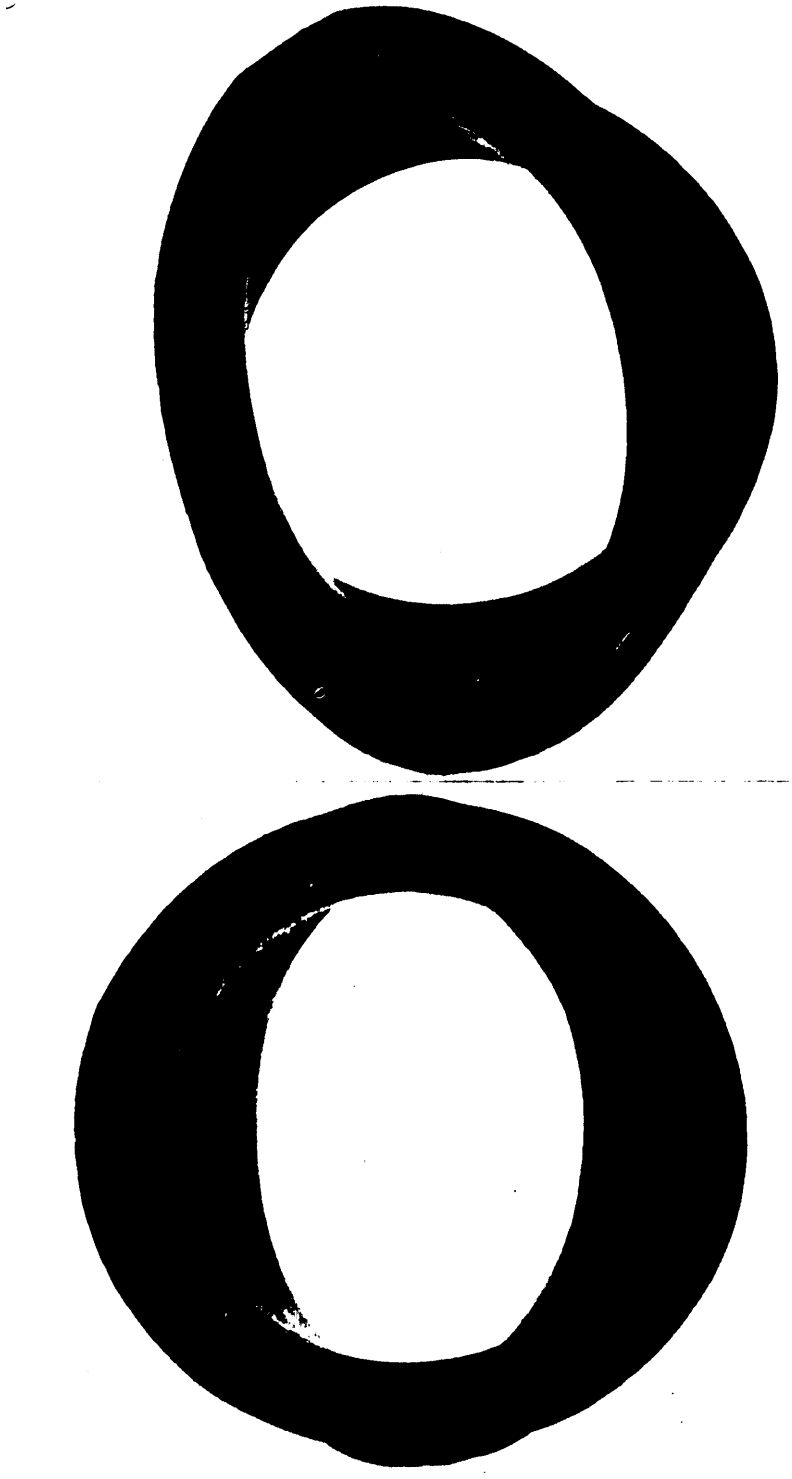


FIGURE 118. Crushed, wire reinforced—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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FIGURE 119. Fabric pulled out from coupling—major.

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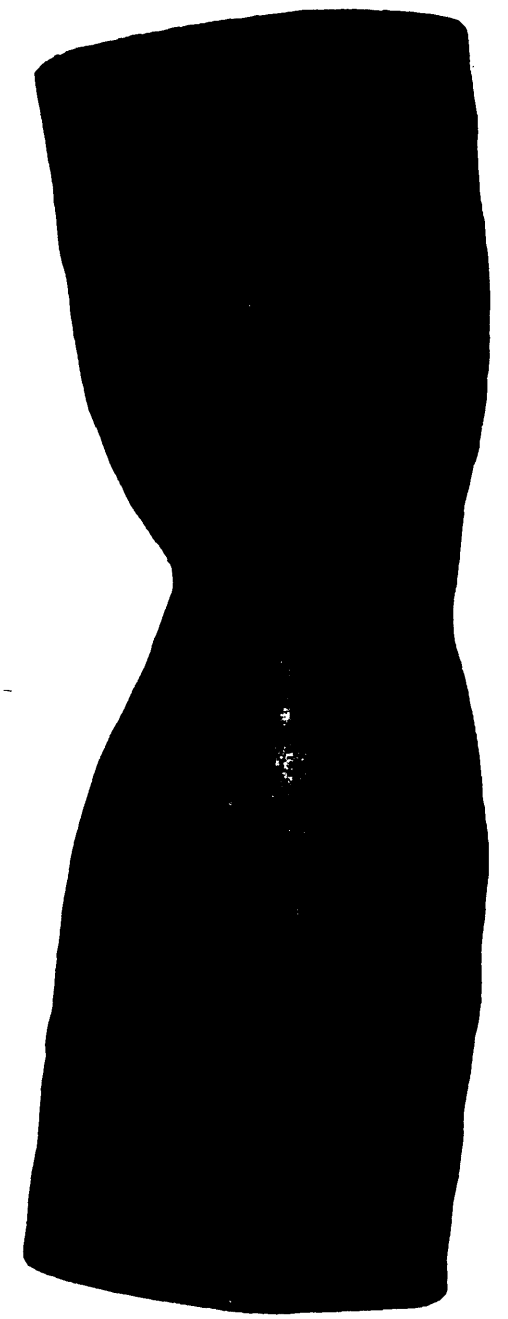


FIGURE 120. Kinked, wire reinforced—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

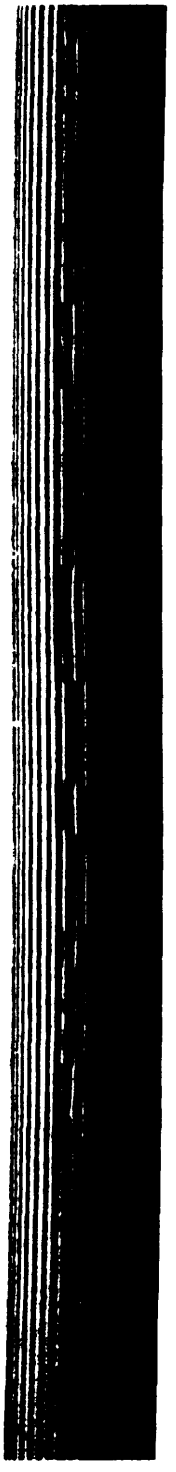


FIGURE 121. Skips, corrugated surface—not a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

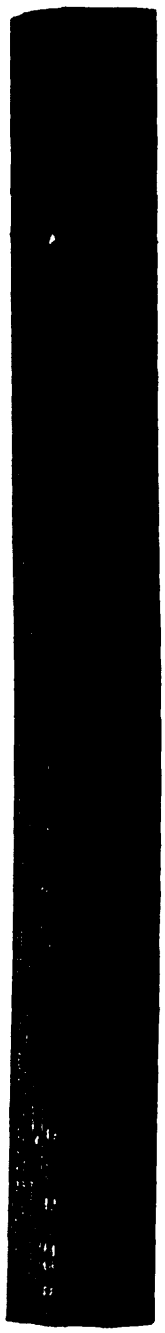


FIGURE 122. Smooth, corrugated surface—not a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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FIGURE 123. Twisted, nonwire reinforced—major.

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Check the source to verify that this is the current version before use.

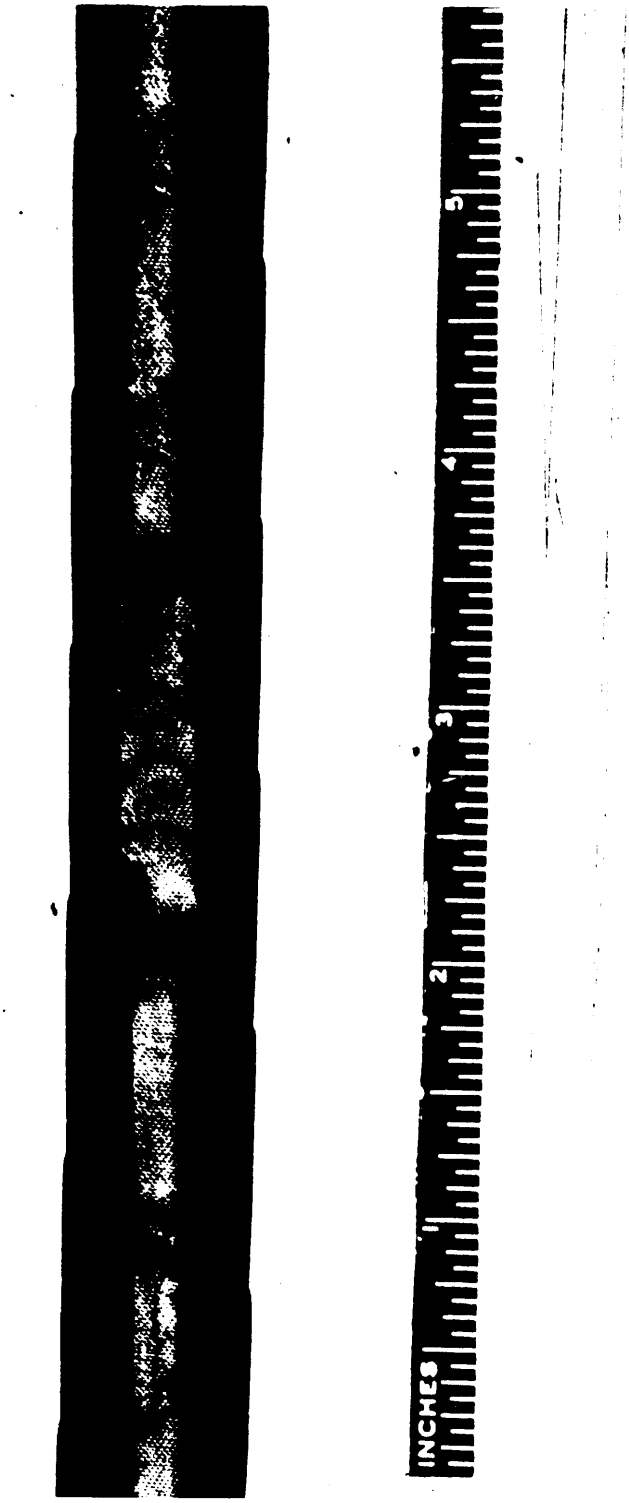


FIGURE 124. Illegible—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

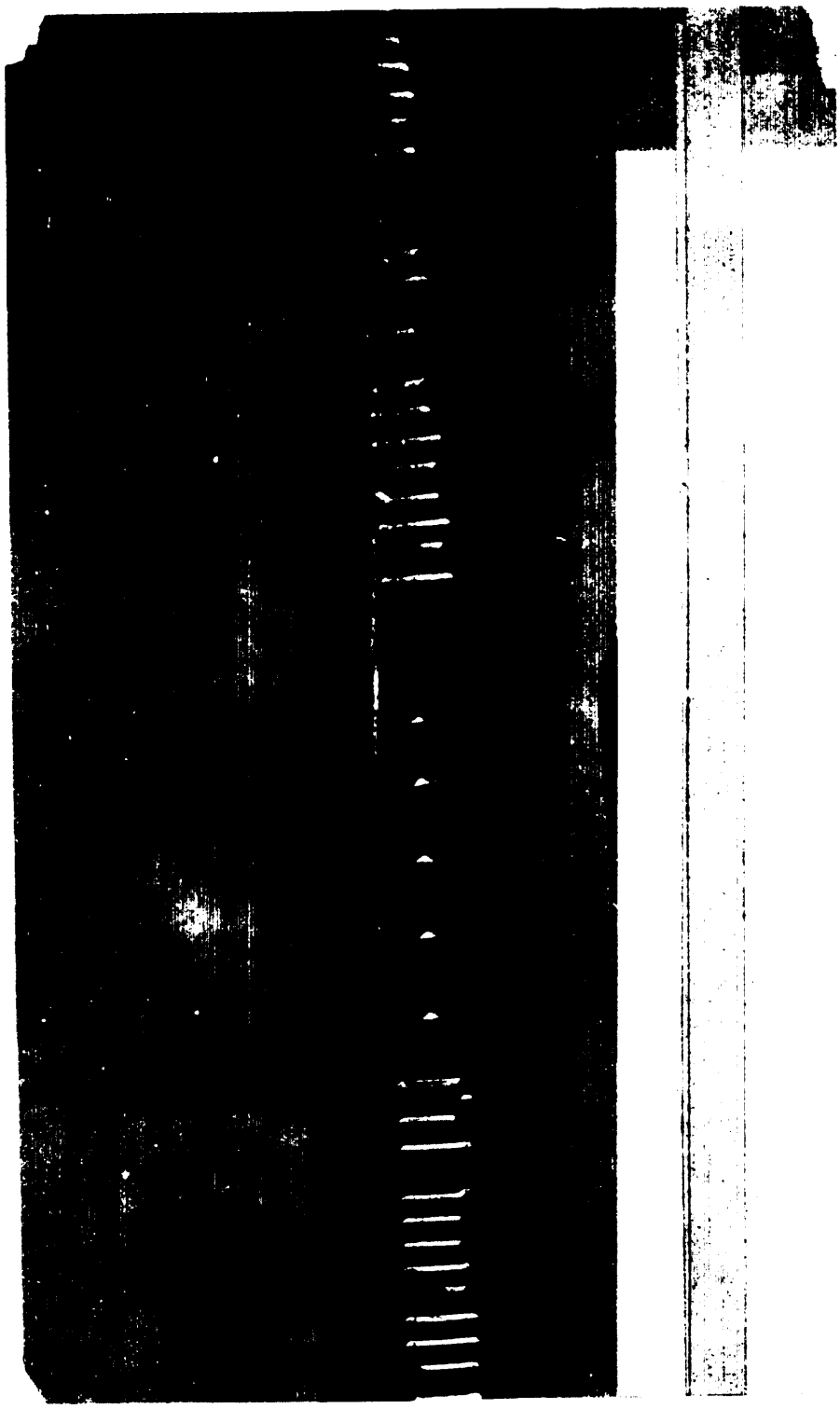


FIGURE 125. Illegible—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

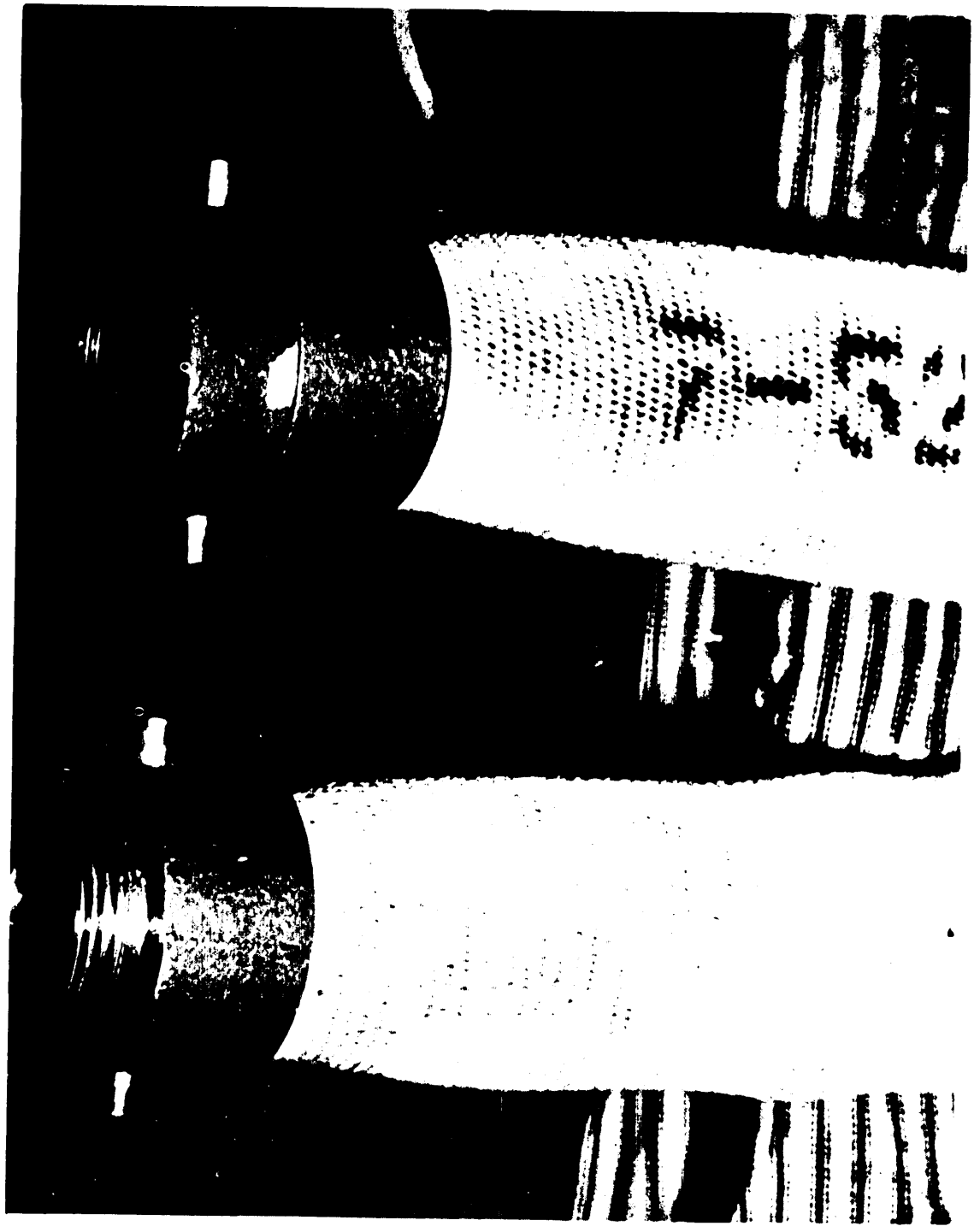


FIGURE 126. Location, improper—minor.

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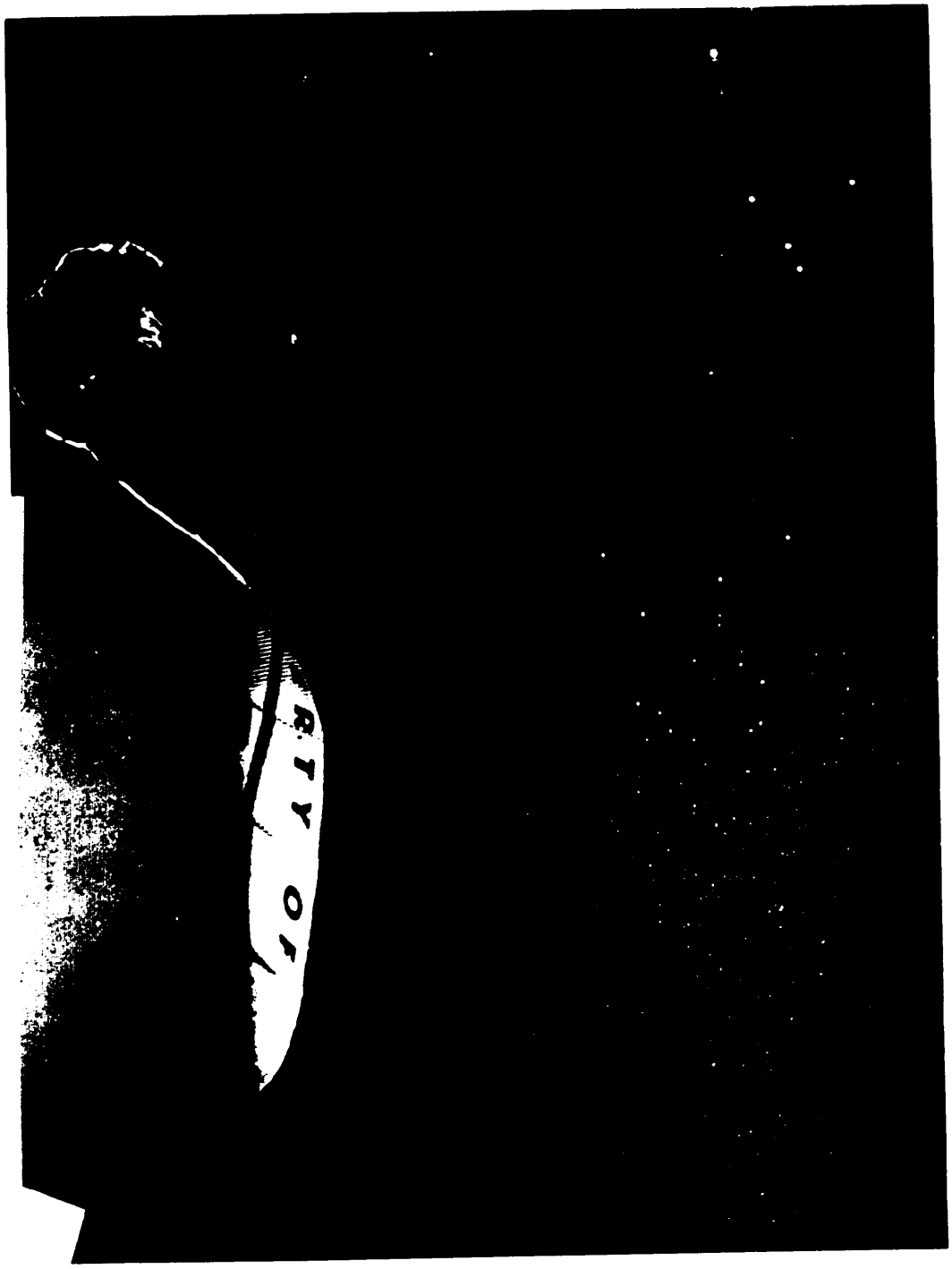


FIGURE 127. Loose, large area—major.

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FIGURE 128. Slipped brand, legible—minor,

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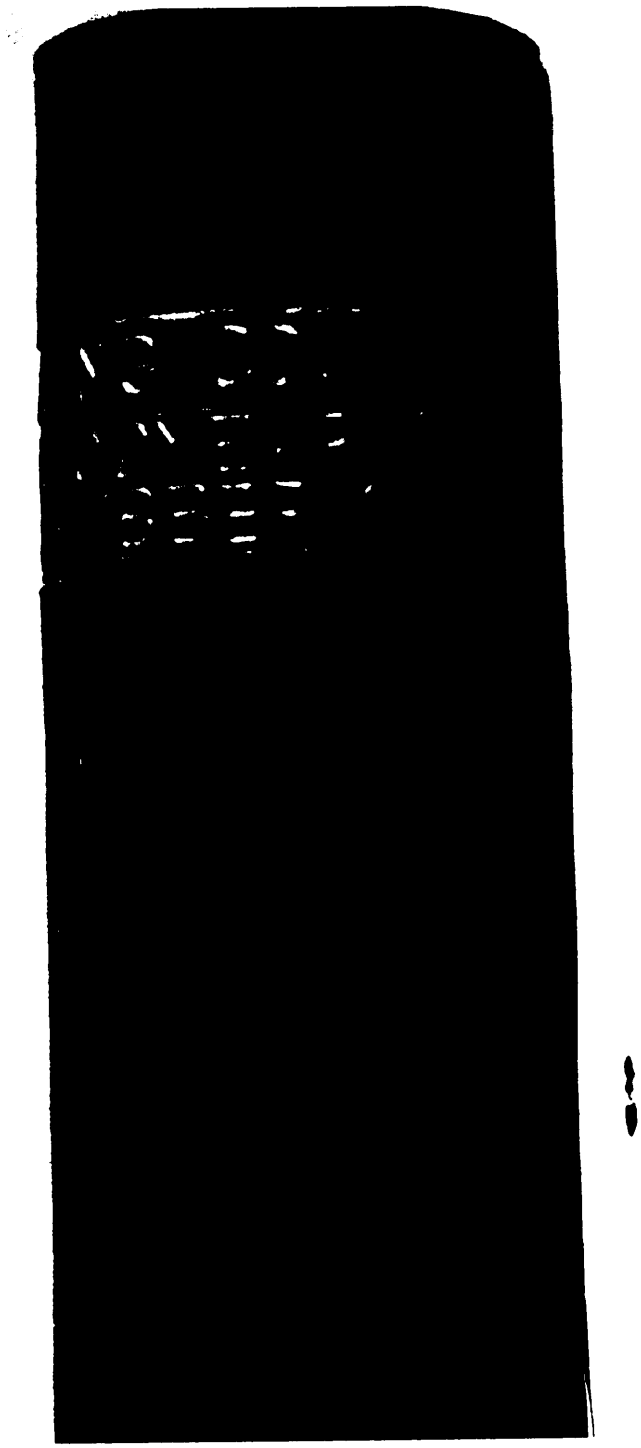


FIGURE 129. Slipped brand, legible—not a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
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FIGURE 130. Ball-up:
I—Yarn ball-up—major.
II—Normal hose.

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Check the source to verify that this is the current version before use.

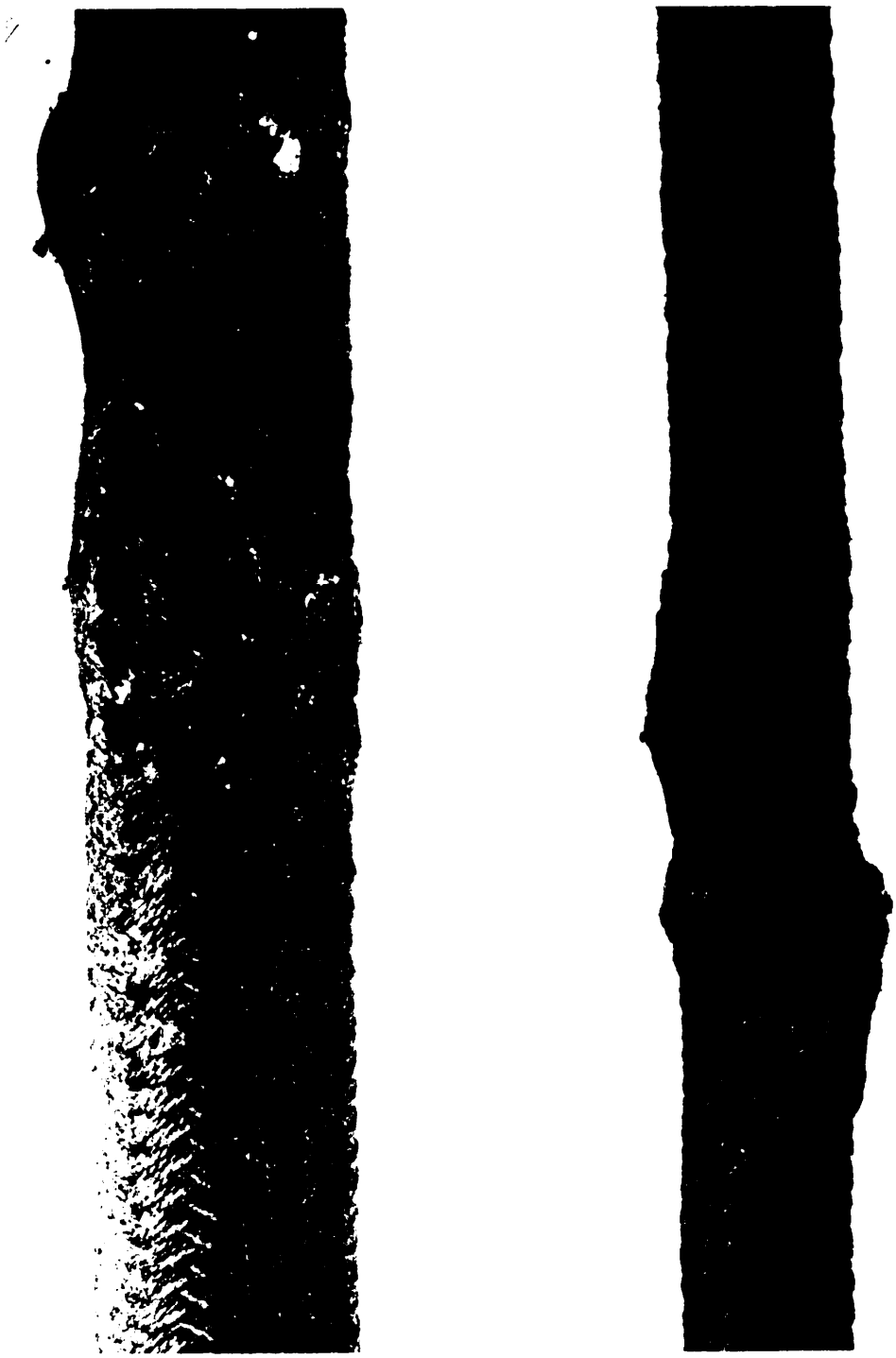


FIGURE 131. Ball-up—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

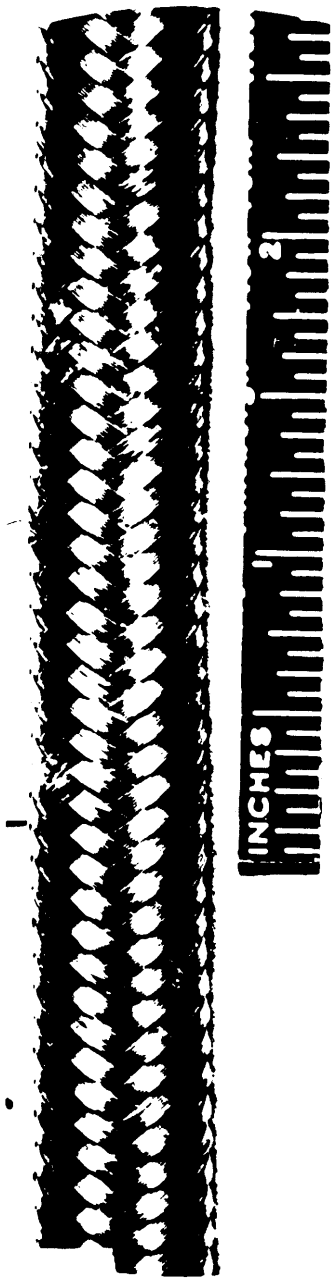


FIGURE 132. Broken strand, more than two per plait or two in 2 inches—major.

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Check the source to verify that this is the current version before use.

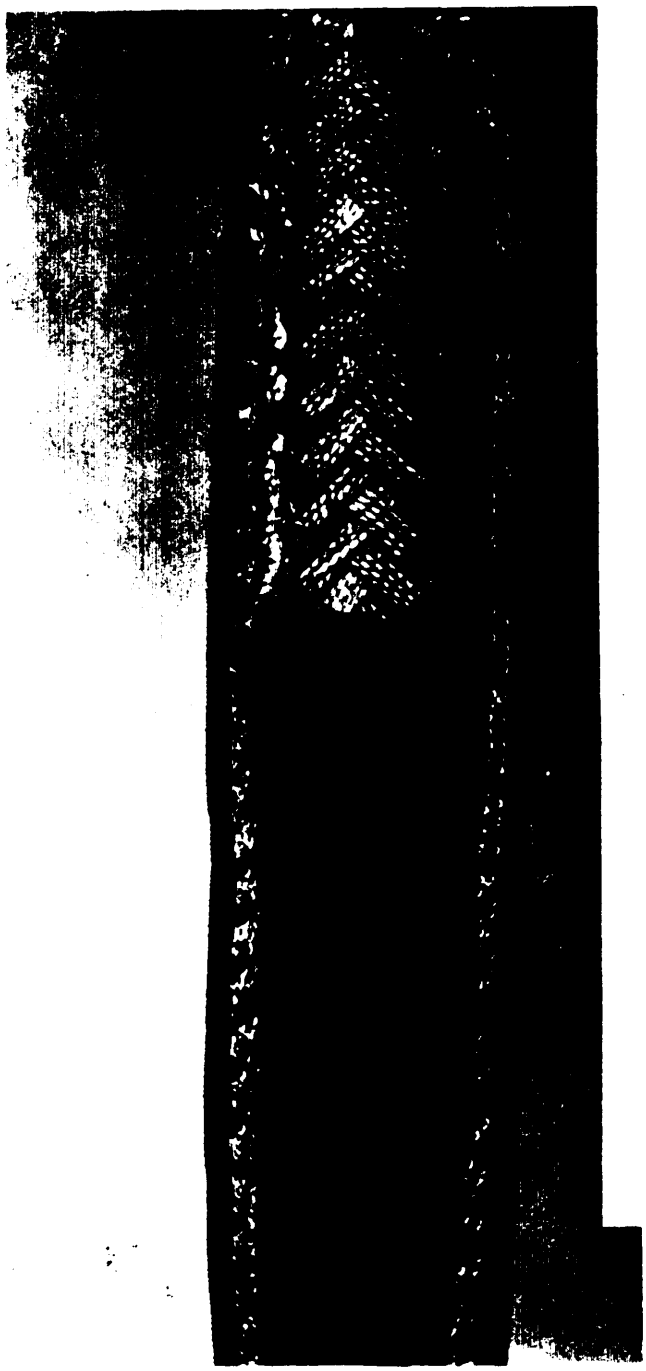


FIGURE 133. Crossed strand: first braid—major.

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FIGURE 134. Crossed strand: other than first braid—not a defect.

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FIGURE 135. Dough (filler). where specified, absent or missing—major.

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FIGURE 136. Dough (filler), where specified, excessive—major.

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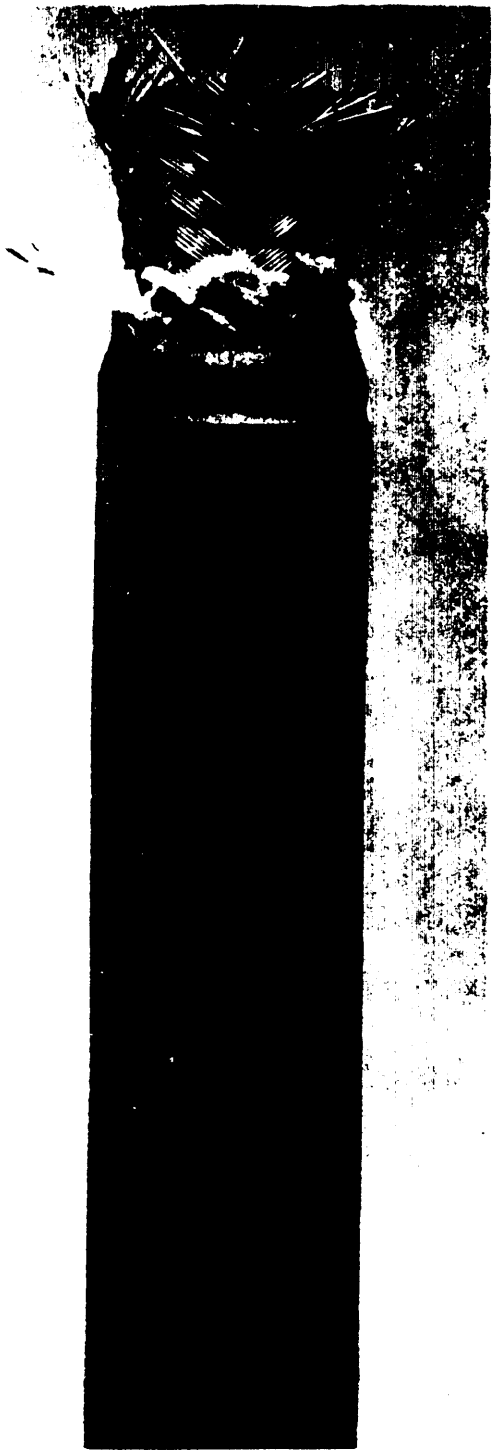


FIGURE 137. Flare, wire at 1/8 inch greater than braid O. D.—major.

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Check the source to verify that this is the current version before use.

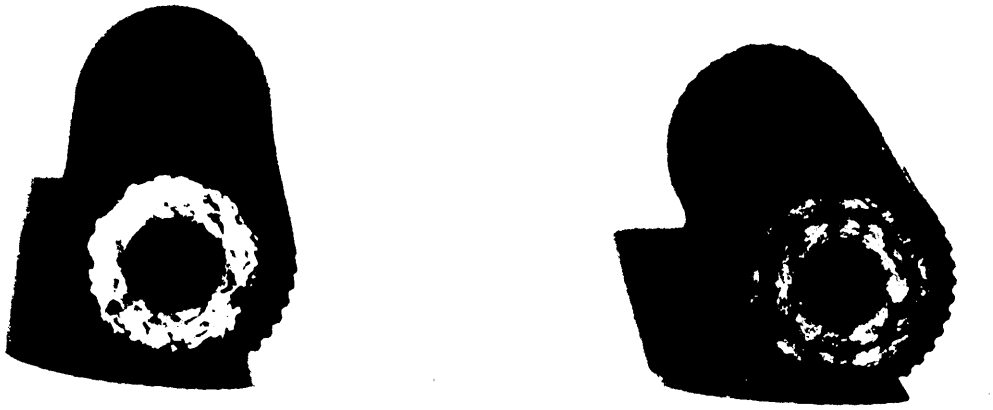


FIGURE 138. Friction between braids, where specified, missing:
I—Friction missing—major.
II—Normal hose.

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FIGURE 139. Knot in yarn:
I—Knot in yarn—major.
II—Normal hose.

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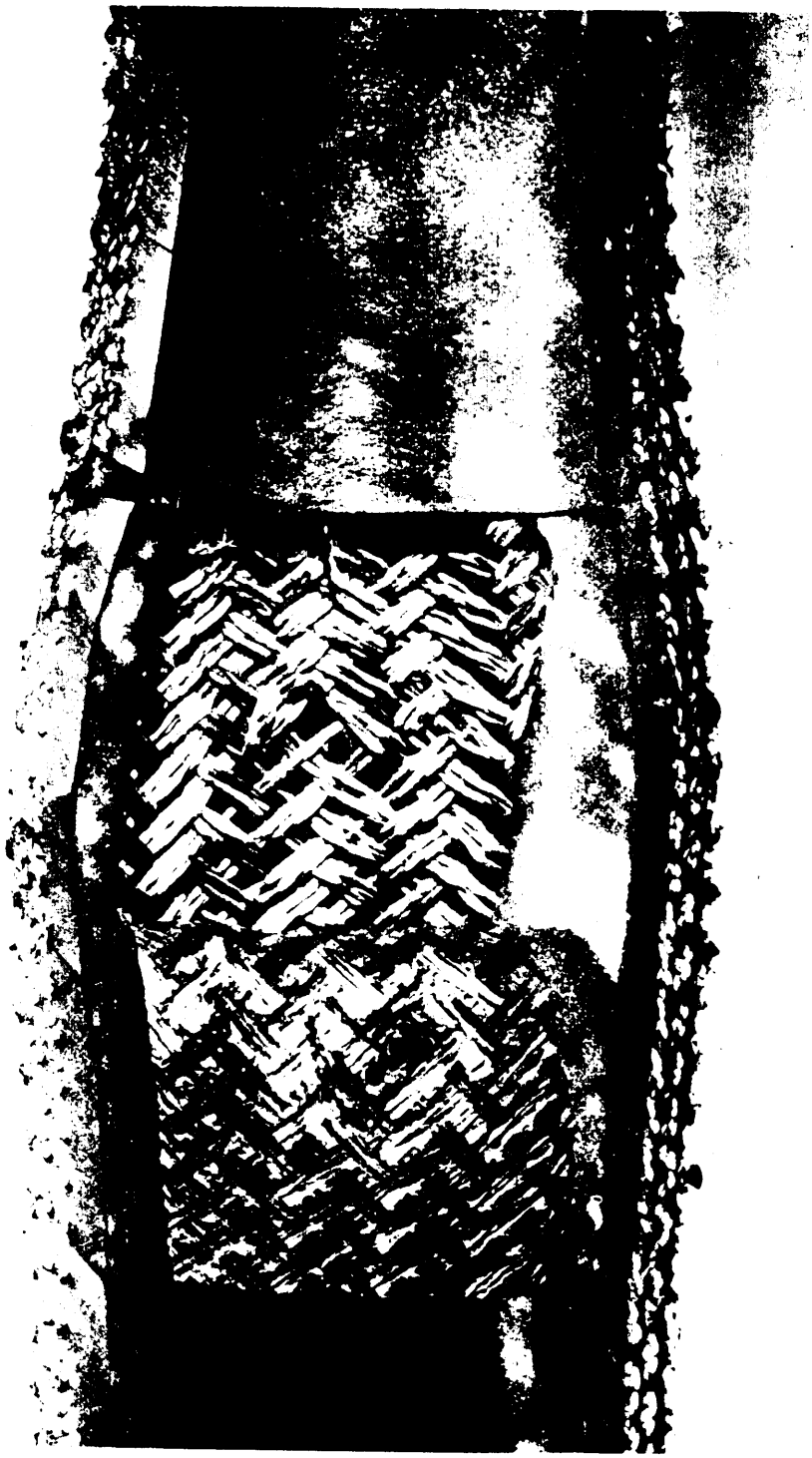


FIGURE 140. Looped strand: first braid, class A—major; first braid, classes B and C—minor; other than first braid—minor

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Check the source to verify that this is the current version before use.

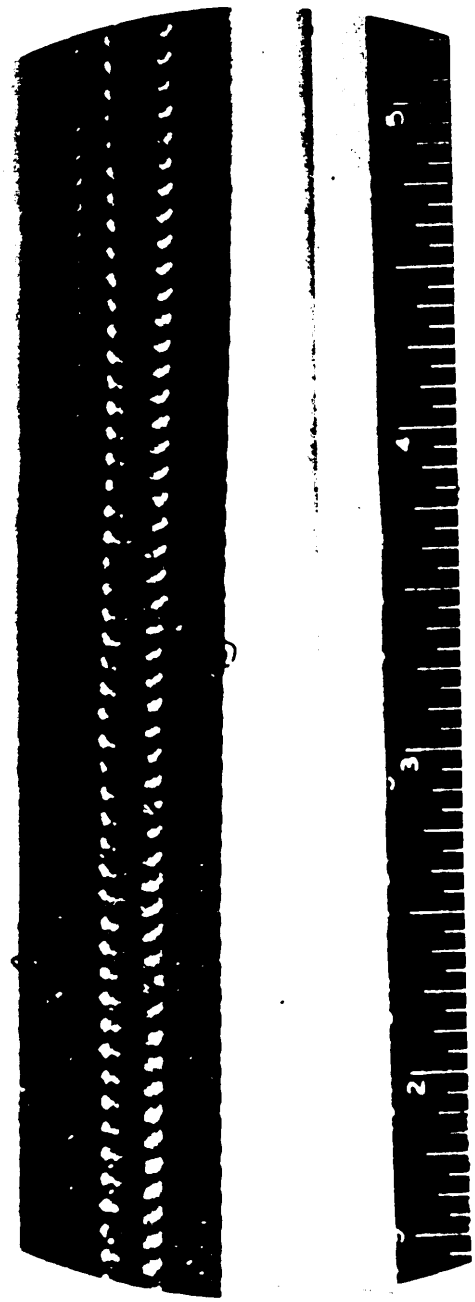


FIGURE 141. Looped wires, more than two per plait (carrier) or two in 2 inches—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

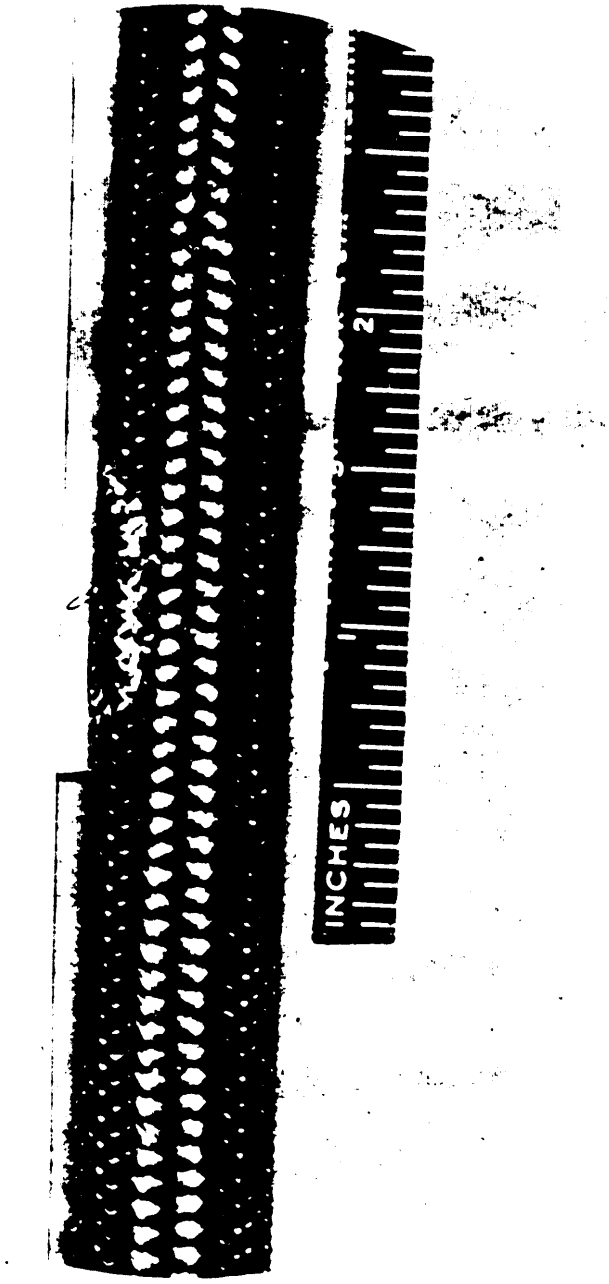


FIGURE 142. Looped wires. less than two per plait (carrier) or less than two in 2 inches—minor.

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FIGURE 143. Loose braid, more than two plaits (carrier) in either direction—minor.

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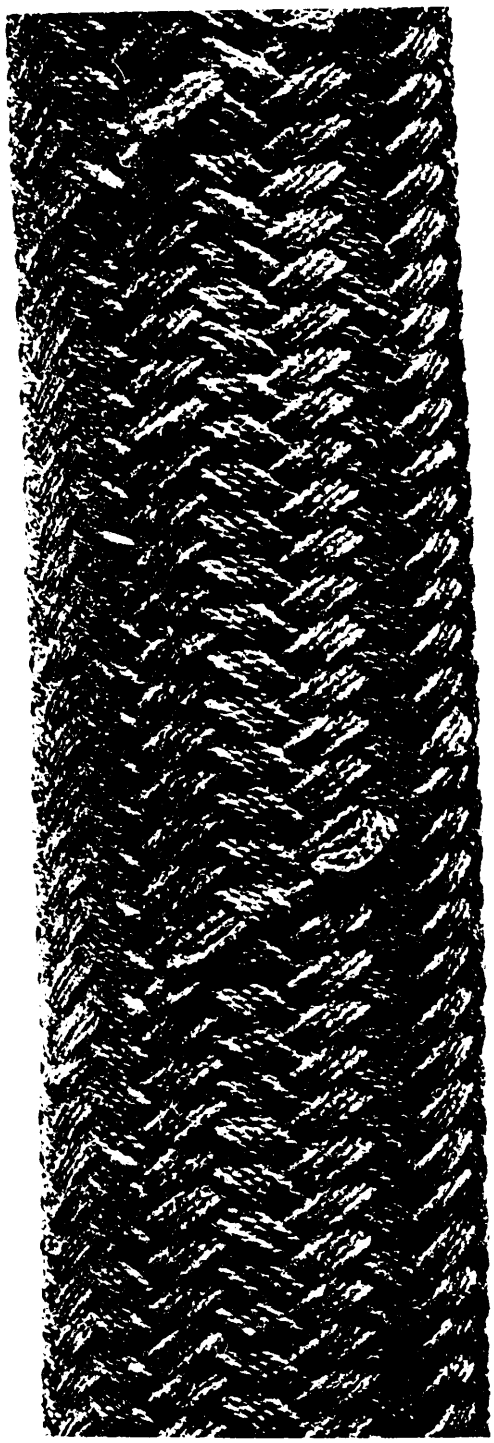


FIGURE 144. Loose braid, one plait (carrier) in either direction—not a defect.

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Check the source to verify that this is the current version before use.



FIGURE 145. Loose braid, one plait (carrier) in either direction—not a defect.

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Check the source to verify that this is the current version before use.

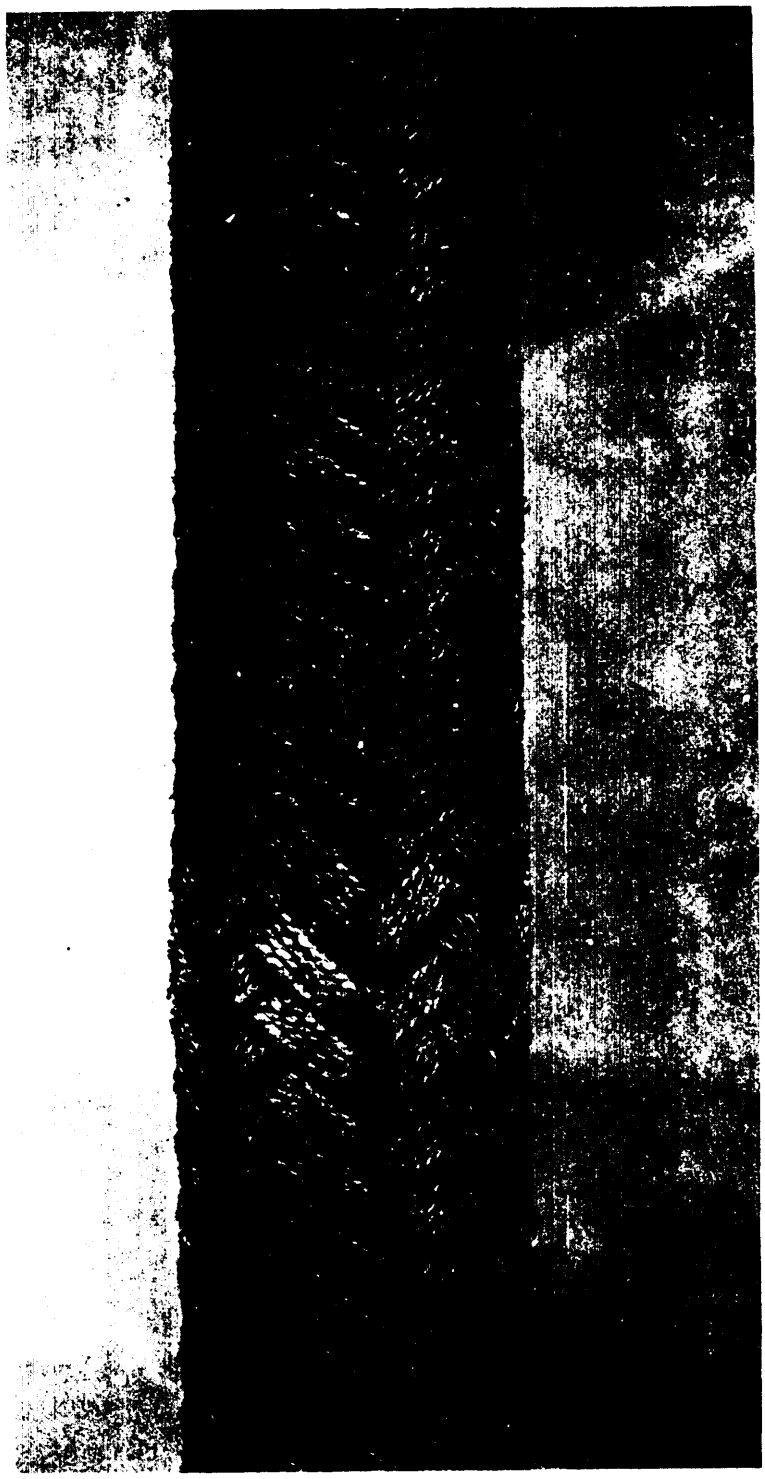


FIGURE 146. Pitch, out of tolerance—major.

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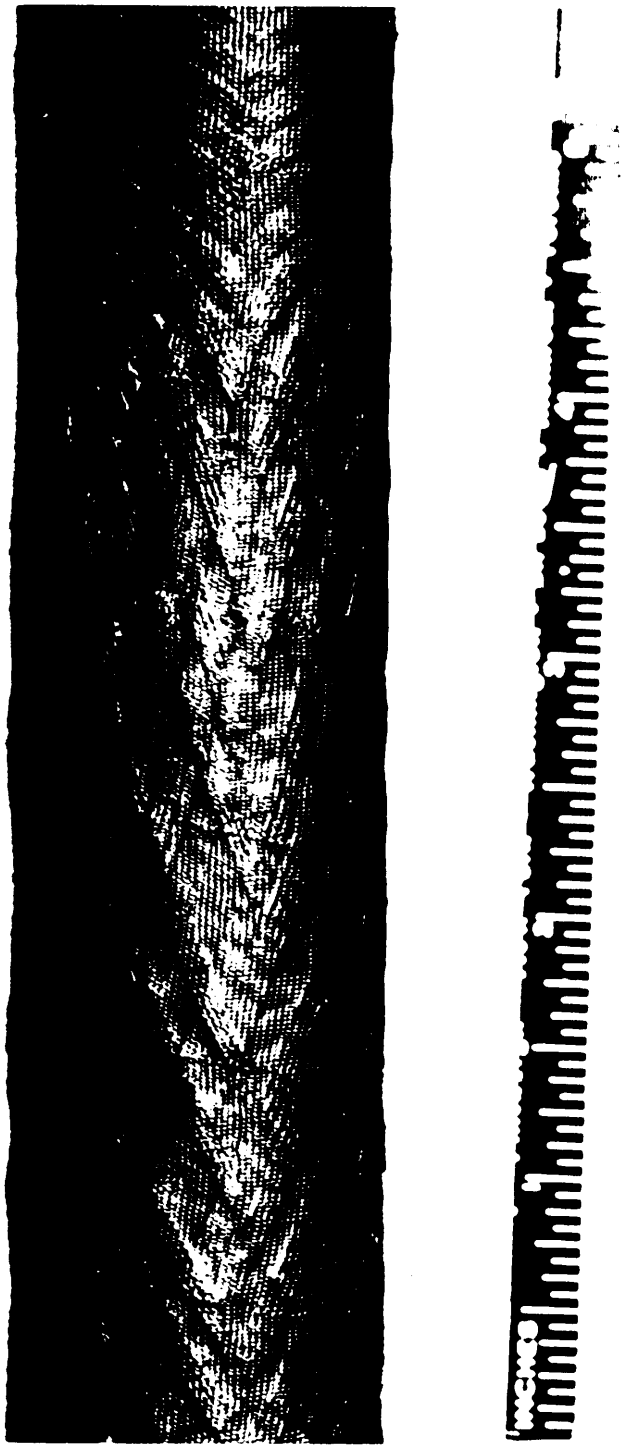


FIGURE 147. Pitch, out of tolerance—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

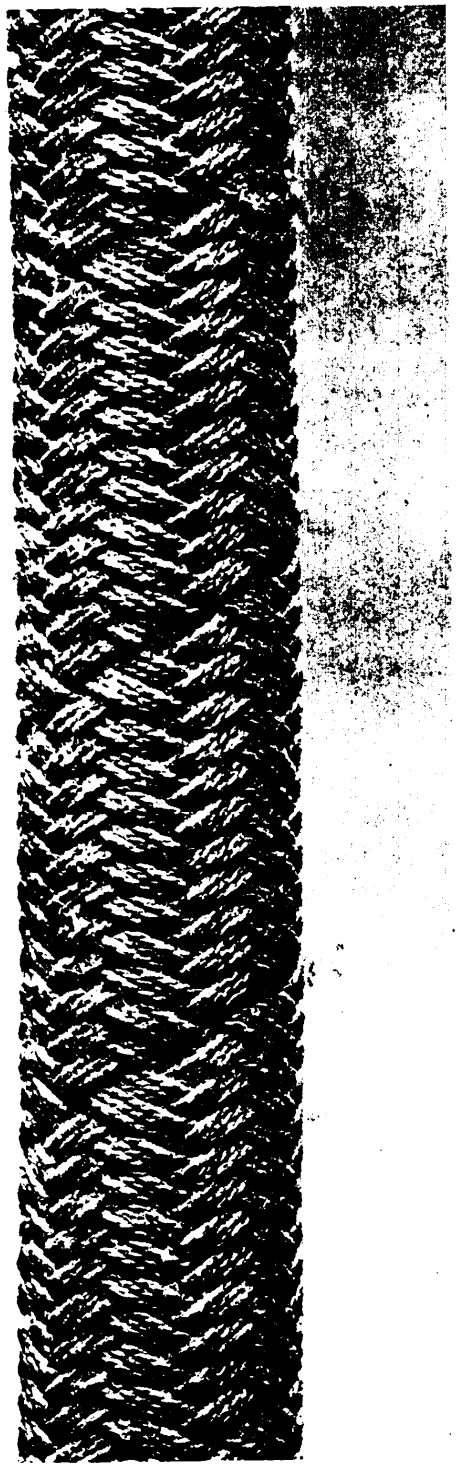


FIGURE 148. Missing plait (carrier or strand), one only: class A—major; classes B and C—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

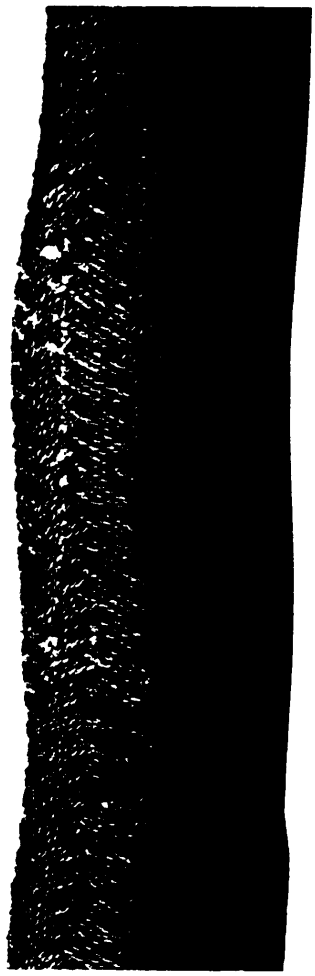


FIGURE 149. Wavy braid—minor.

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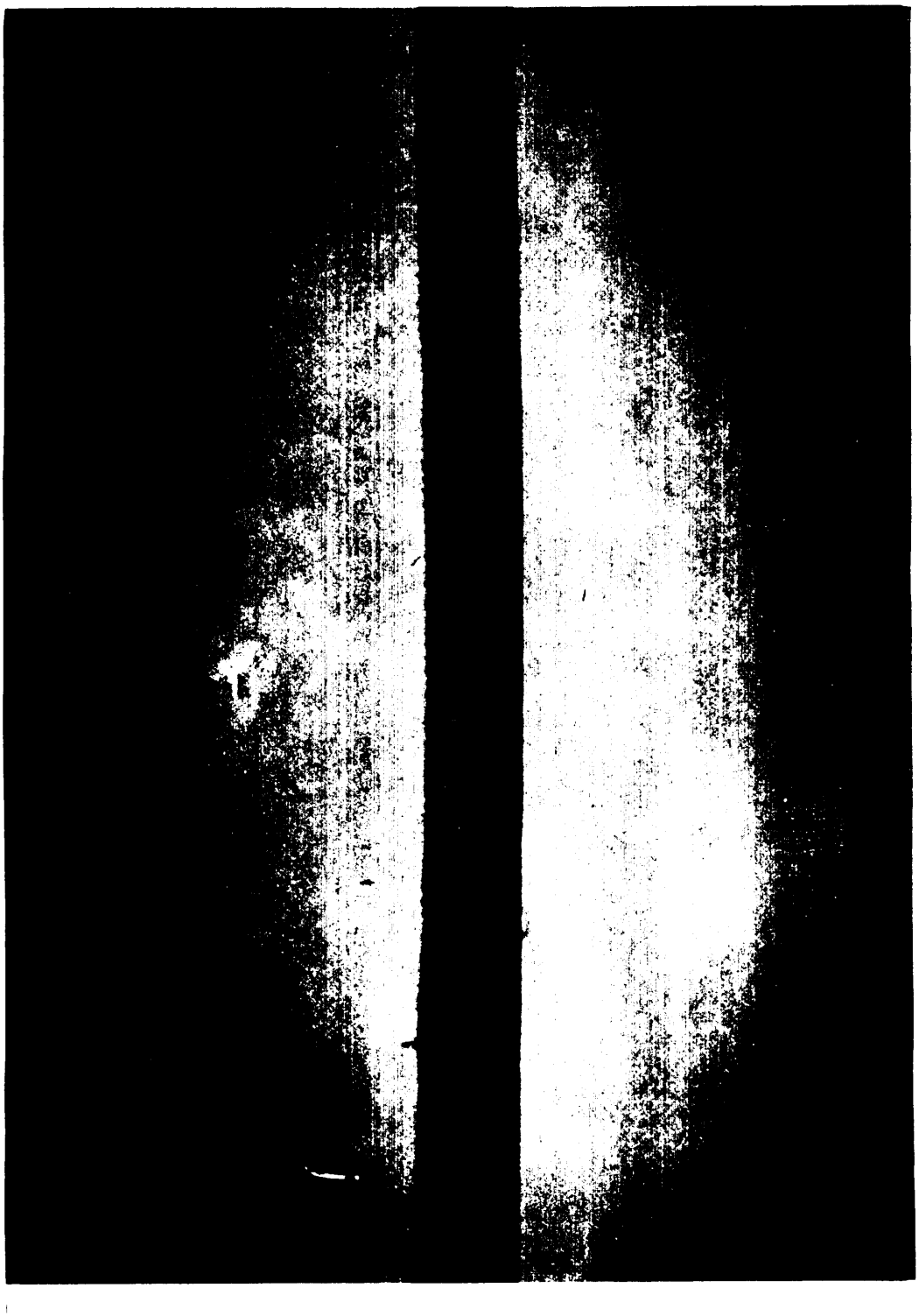


FIGURE 150. Wire through cover—major.

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Check the source to verify that this is the current version before use.

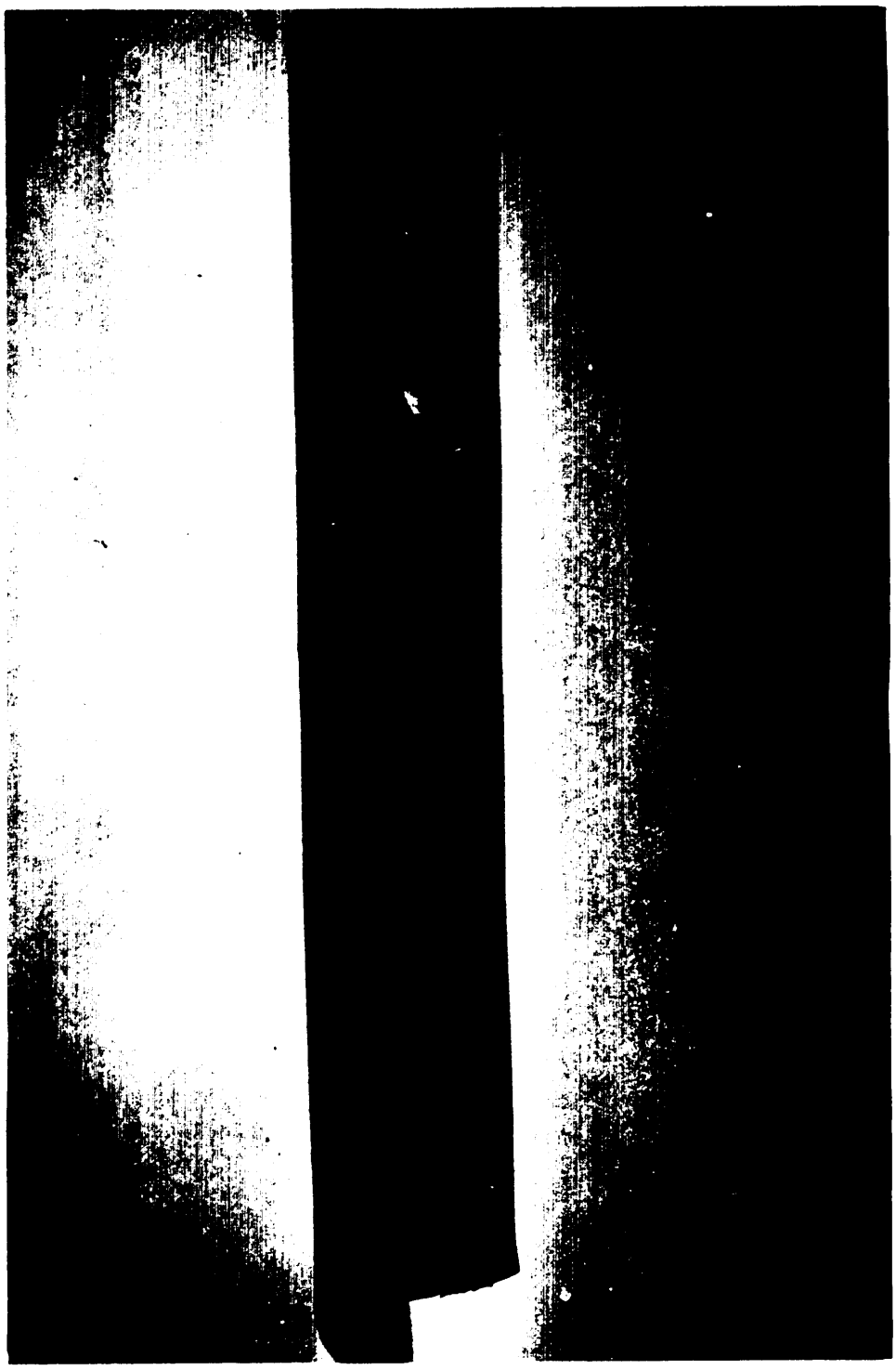


FIGURE 151. Wire through cover—major.

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FIGURE 152. Wire through cover—major.

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Check the source to verify that this is the current version before use.



FIGURE 153. Wire through tube—major.

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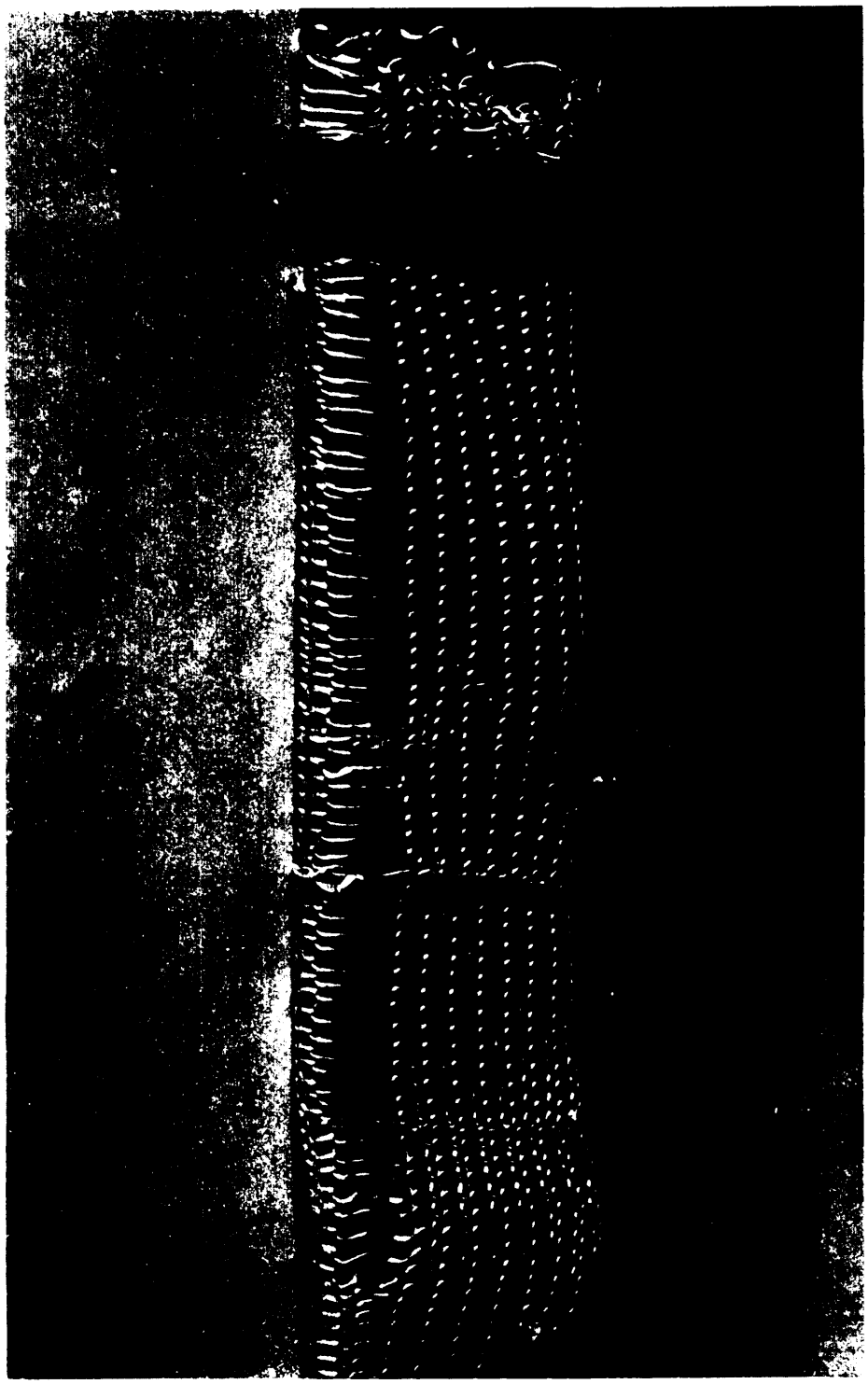


FIGURE 154. Knitted, skip—major.

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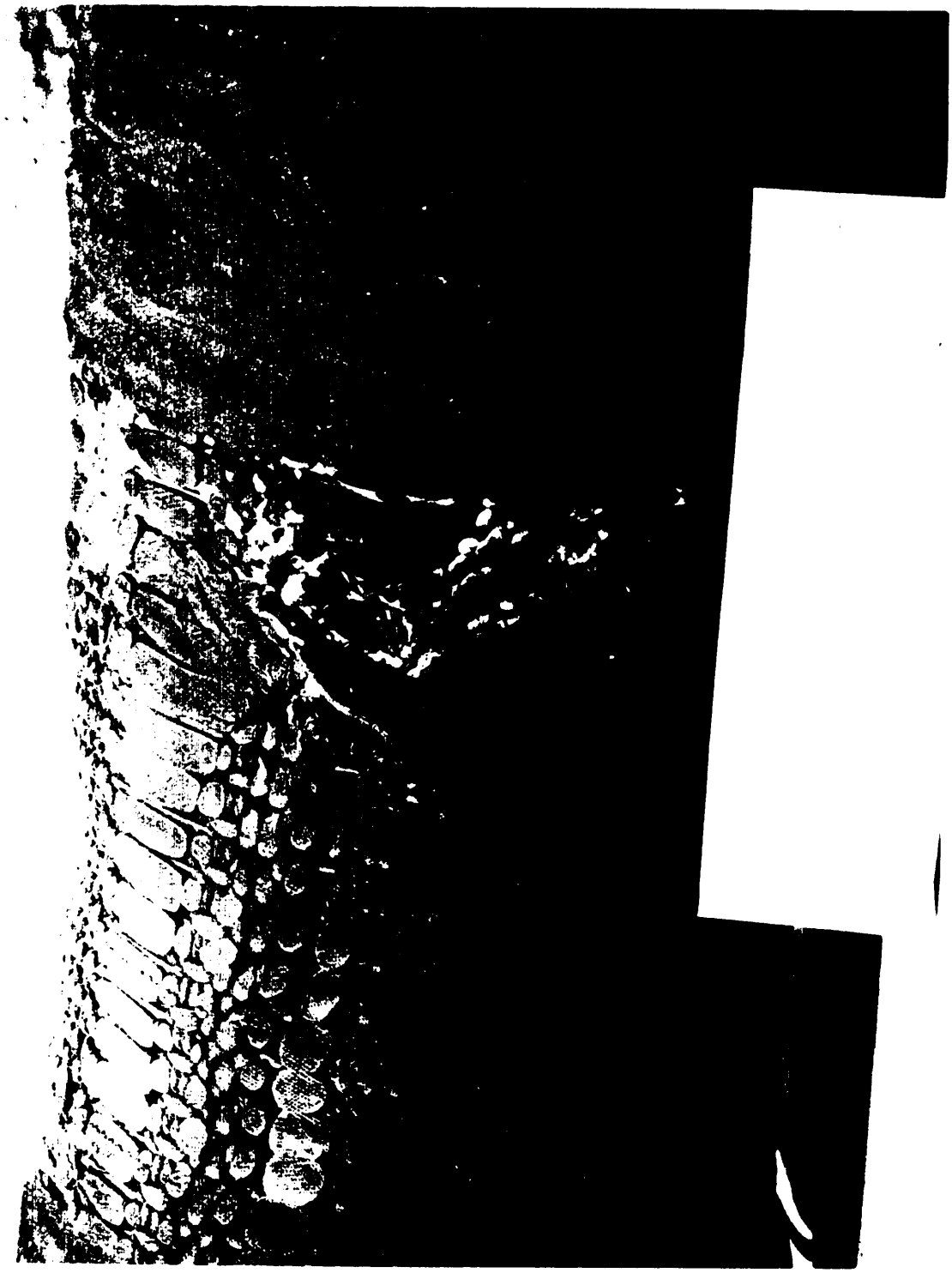


FIGURE 155. Woven jacket, rubber covered, ball-up—major.

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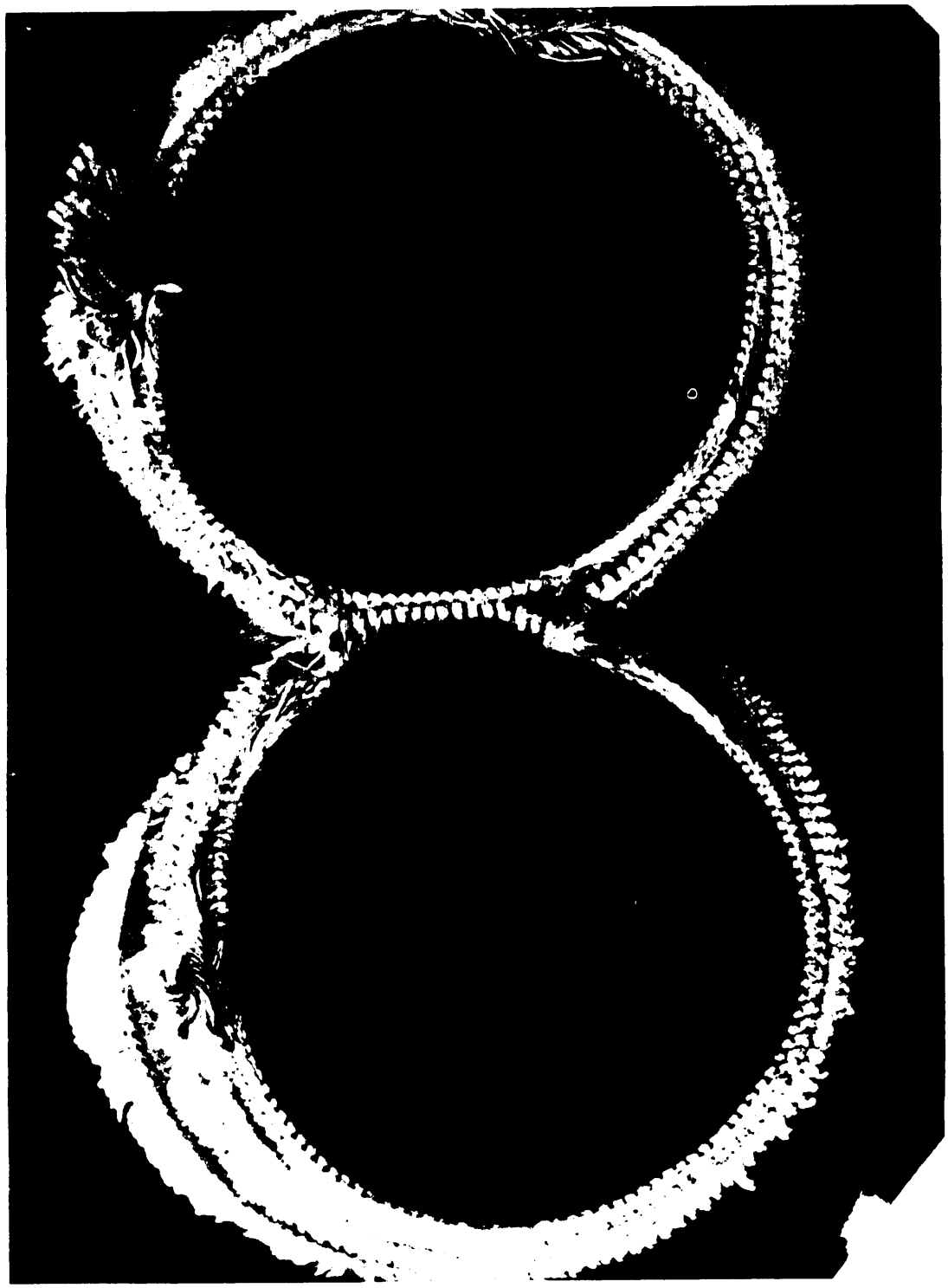


FIGURE 156. Yarn imbedded between jacket: more than 2 feet long—major; less than 6 inches from end—major; less than 2 feet long—minor; 6 inches or more from end—minor.

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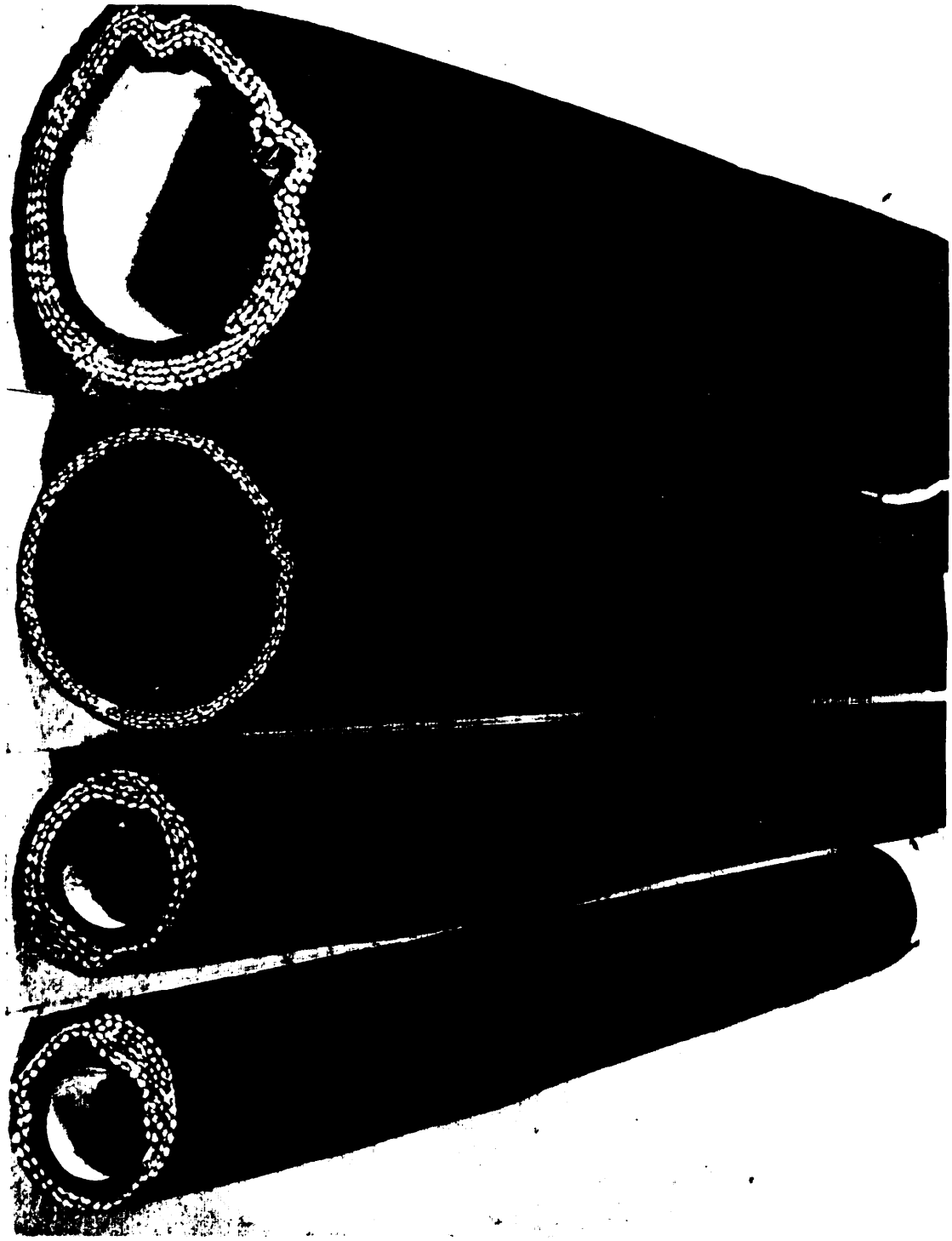


FIGURE 157. Wrapped fabric, buckled or wrinkled—major.

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Check the source to verify that this is the current version before use.



FIGURE 158. Wrapped fabric, buckled or wrinkled—major.

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Check the source to verify that this is the current version before use.



FIGURE 159. Wrapped fabric, buckled or wrinkled—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

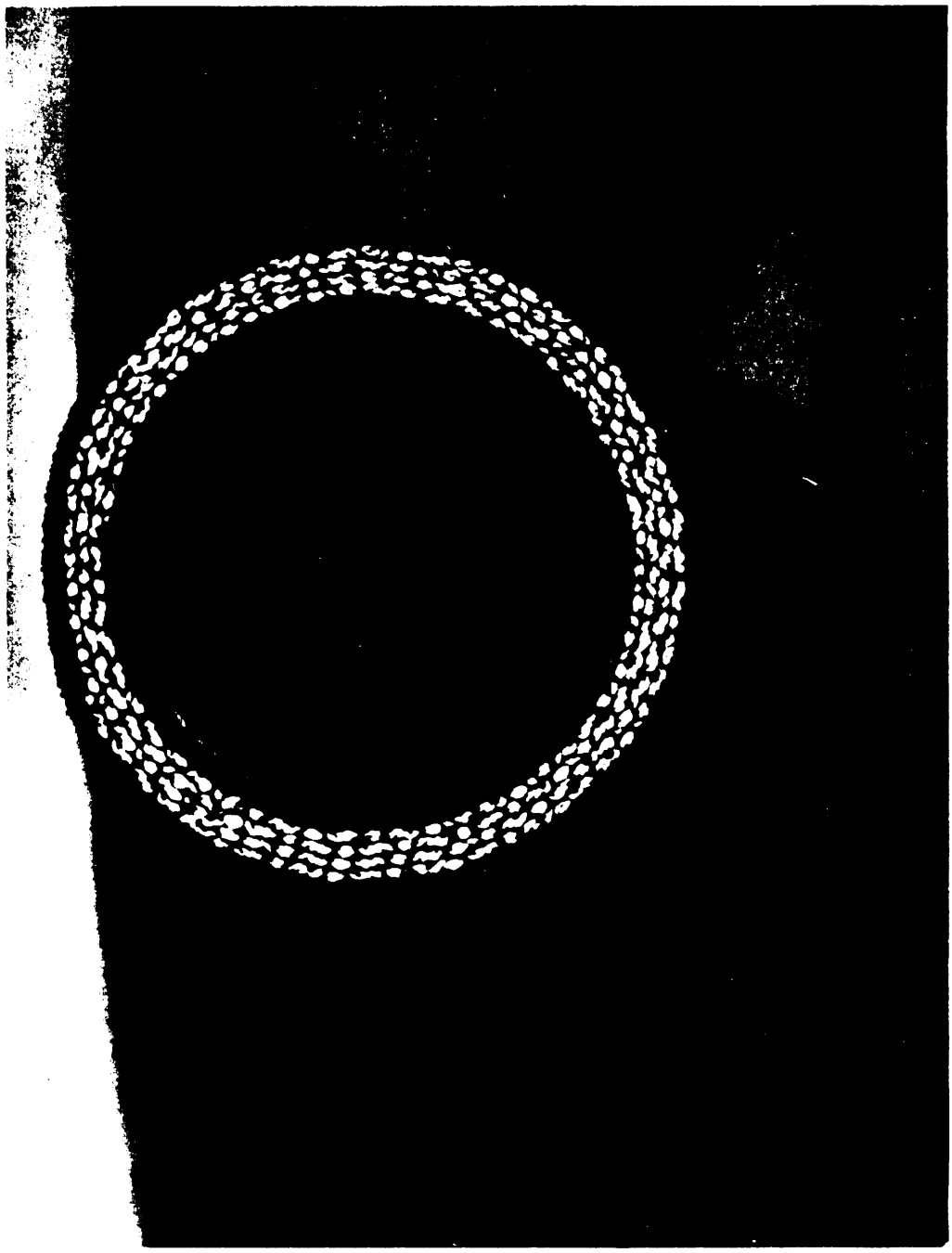


FIGURE 160. Incorrect number of plies or open lap—major (4 ply example).

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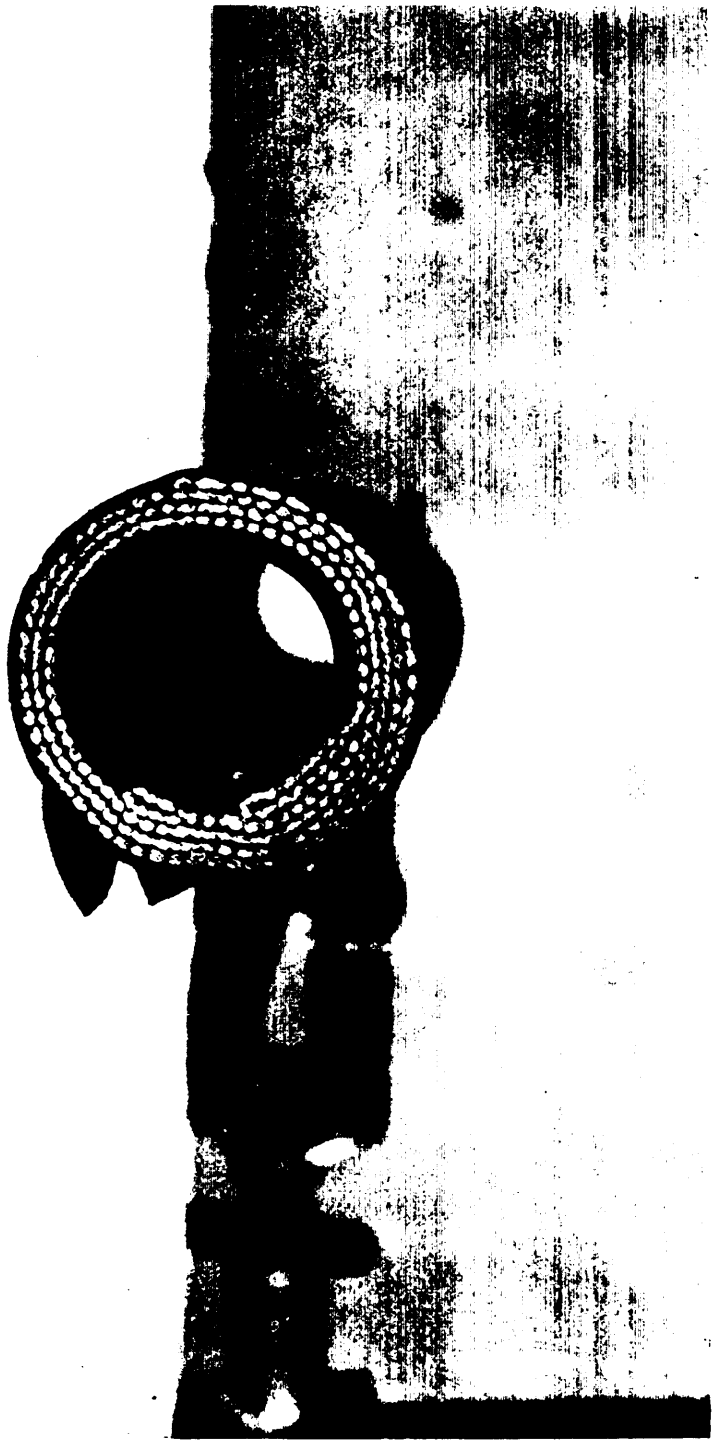


FIGURE 161. Ply turnback at leading edge, improper ply distribution—major.

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FIGURE 162. Corrugations bad—major.

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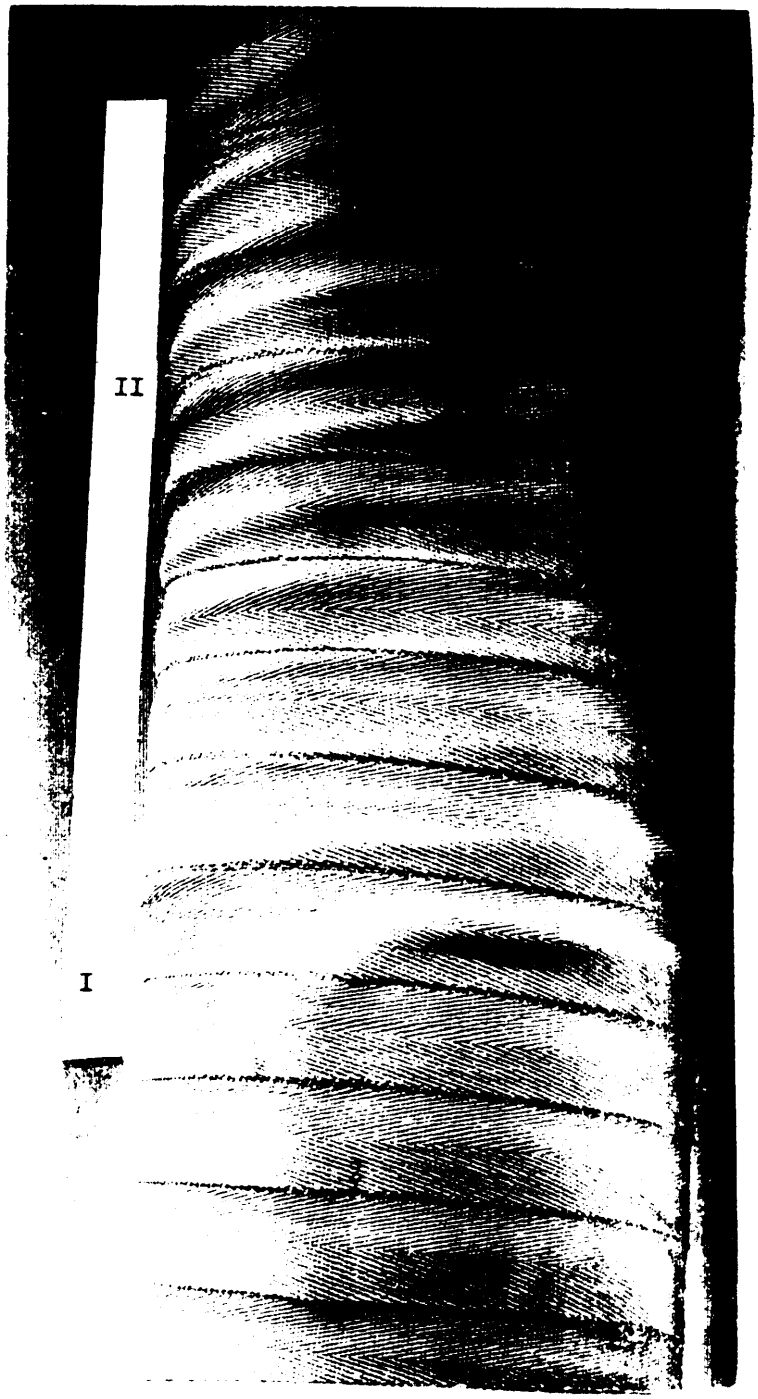


FIGURE 163. Filler, insufficient:
I—Wire not properly tied down at end—minor.
II—Filler, insufficient (noncorrugated surface hose)—minor; for Air Force procurement only—major.

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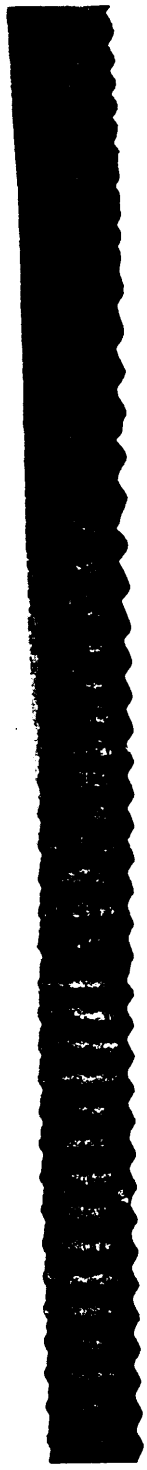


FIGURE 164. Pitch incorrect—major.

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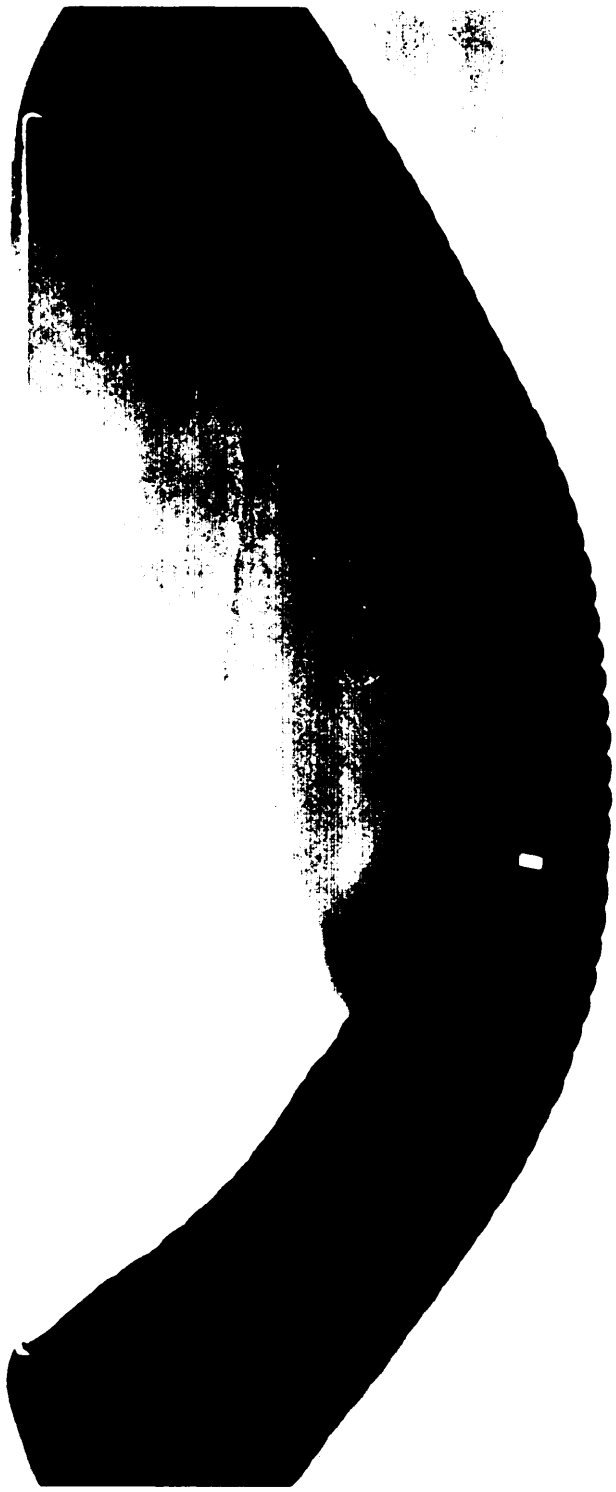


FIGURE 165. Pitch uneven—minor.

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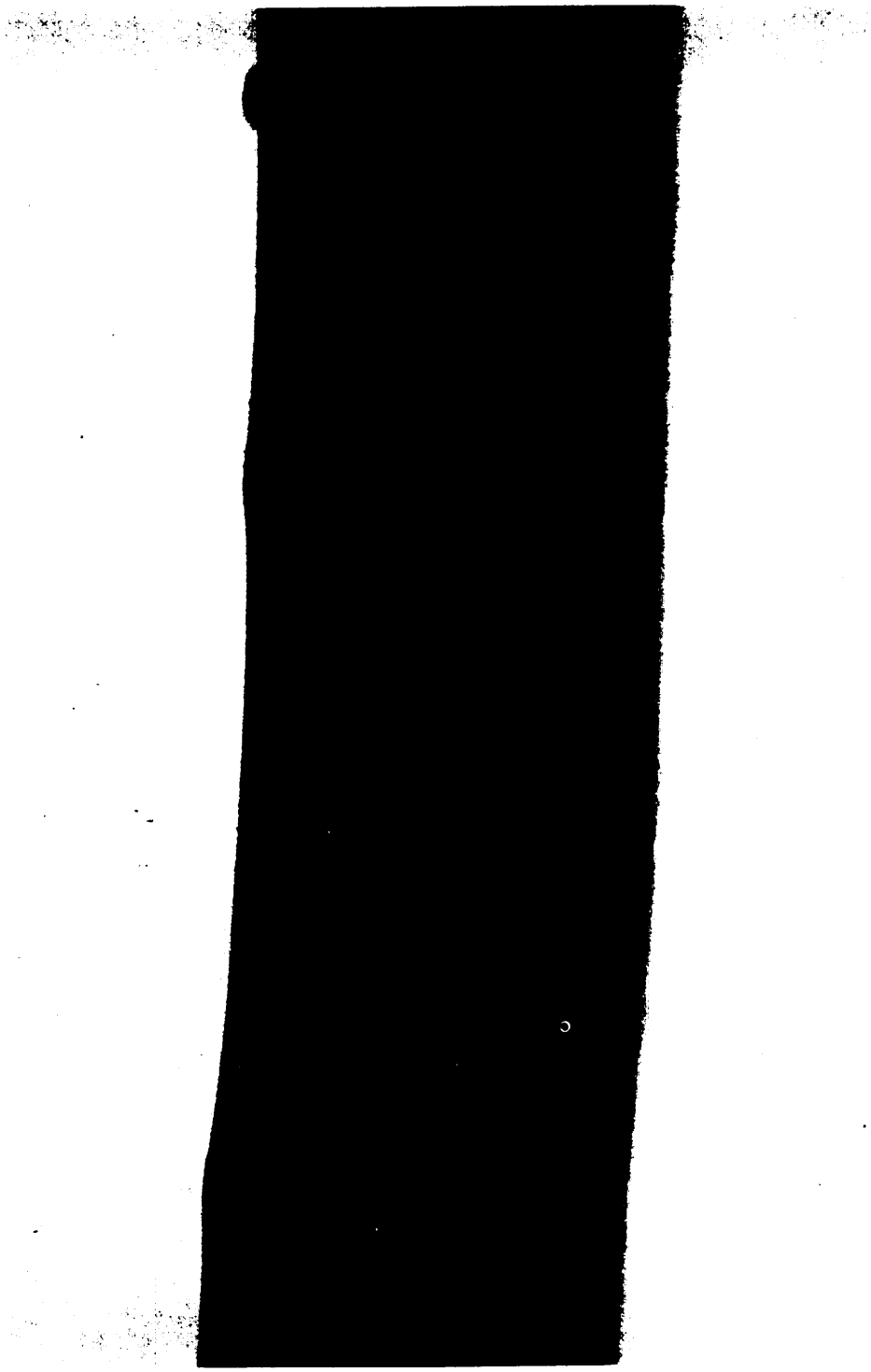


FIGURE 166. Wire, static, bulge: with loose cover—major; dynamic use (Air Force procurement only)—major; with adhered cover—not a defect.

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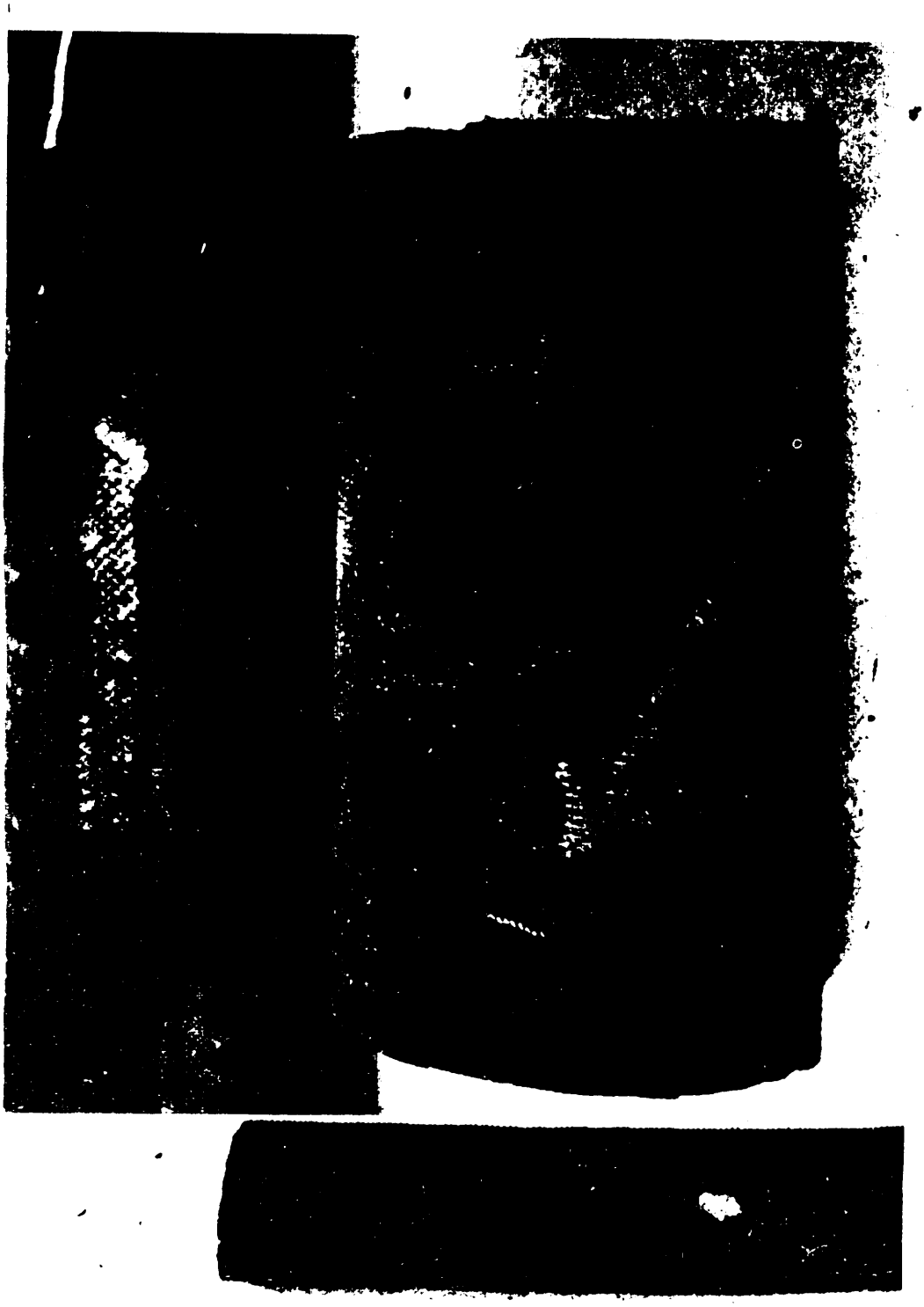


FIGURE 167. Yarn through cover—major.

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FIGURE 168. Yarn through cover—major.

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FIGURE 169. Yarn in cover: class A—major; classes B and C—minor.

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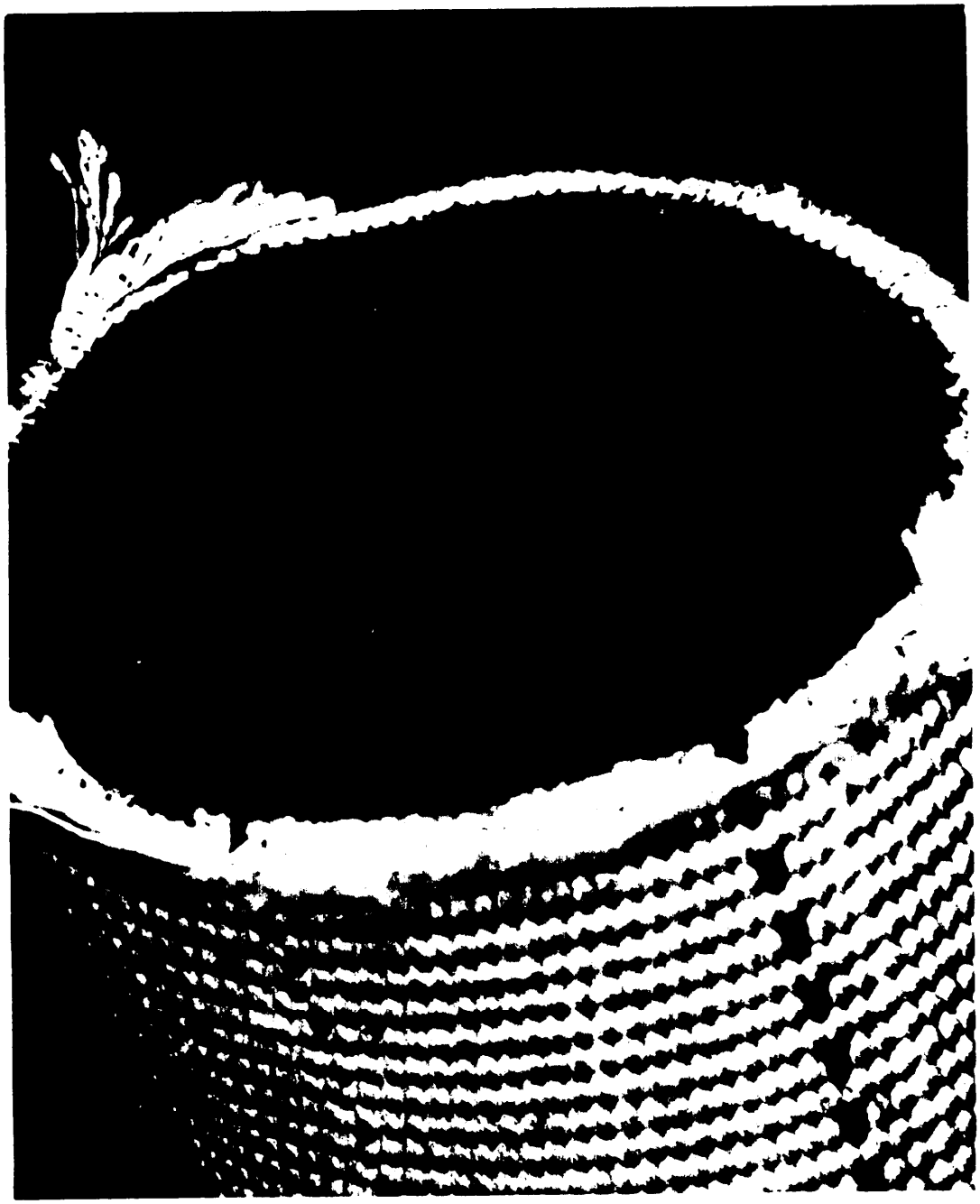


FIGURE 170. Buckle—major.

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FIGURE 171. Cement, excess: class A—major; classes B and C—minor.

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FIGURE 172. Cement, excess: class A—major; classes B and C—minor.

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FIGURE 173. Fold—major.

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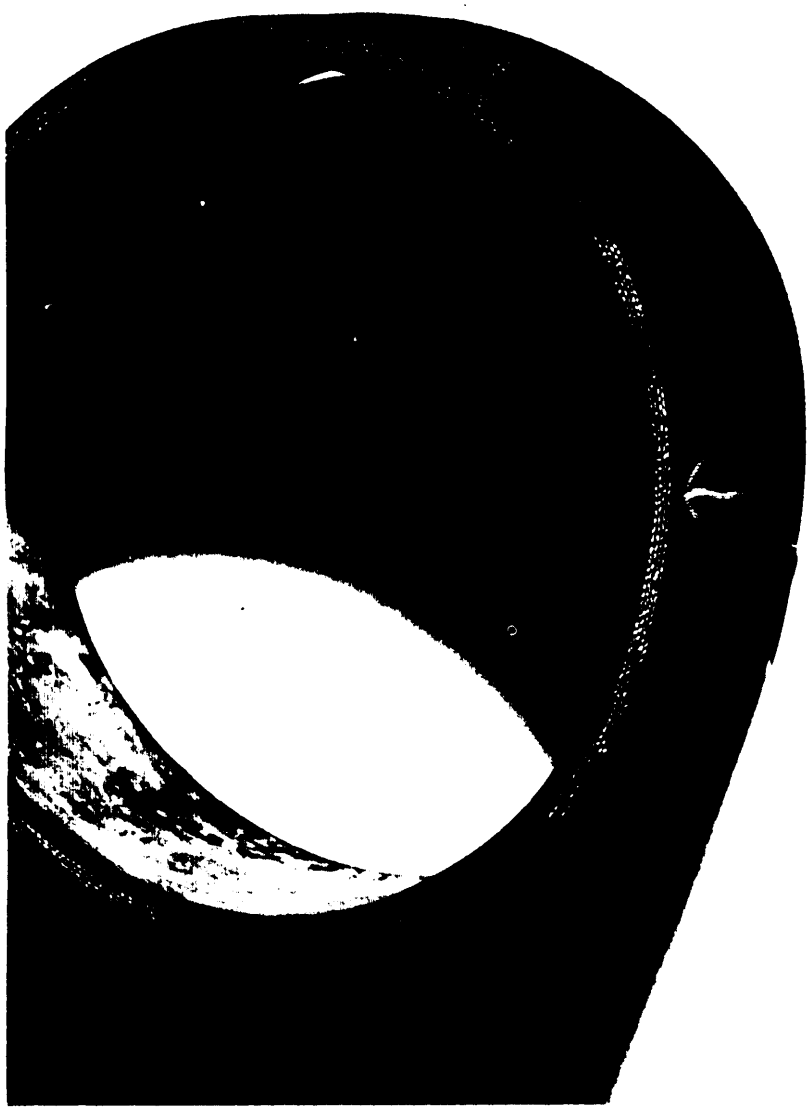


FIGURE 174. Hole—major.

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FIGURE 175. Hole—major.

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FIGURE 176. Hole—major.

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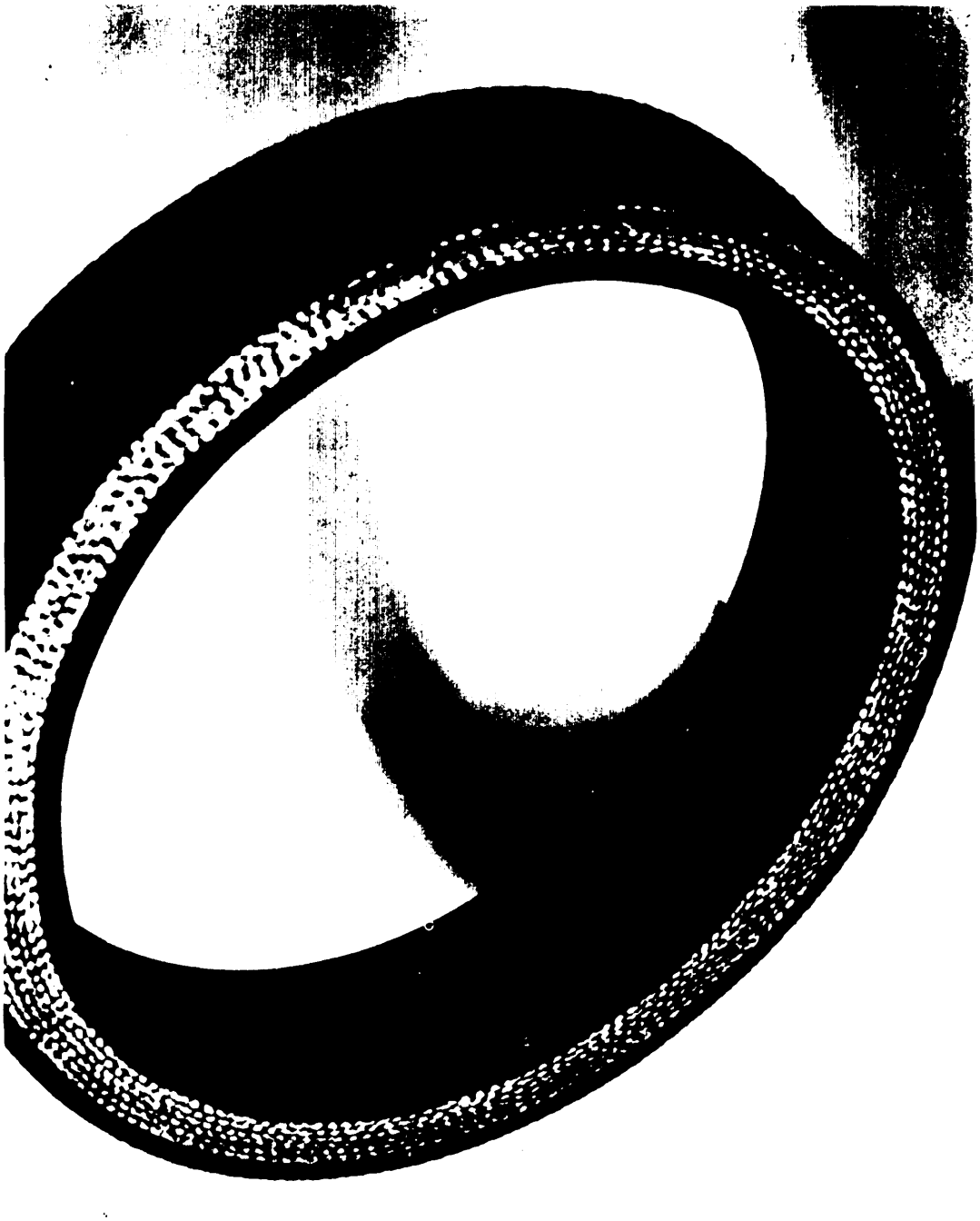


FIGURE 177. Lap, open or loose—major.

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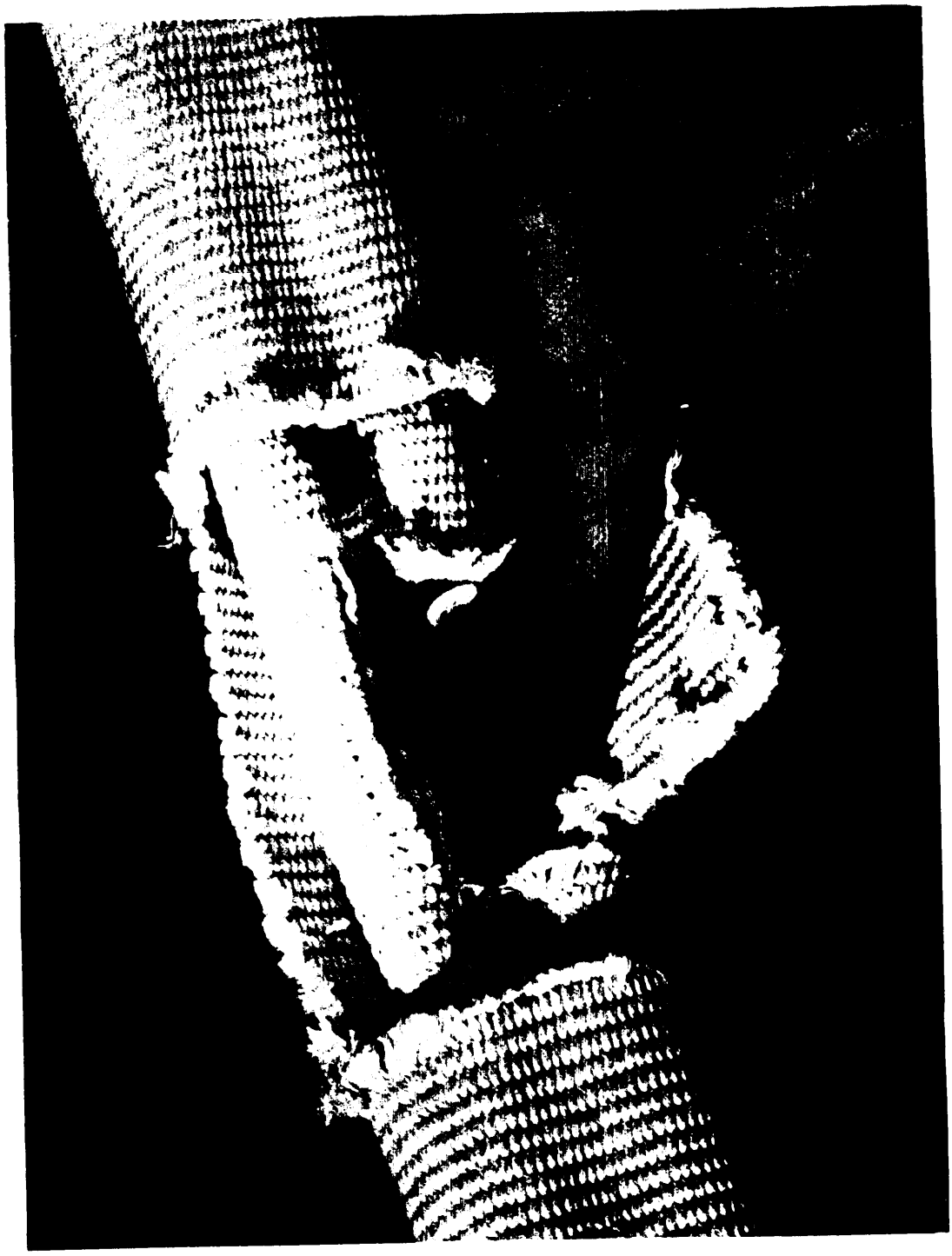


FIGURE 178. Loose (separation)—major. (See fig. 179.)

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FIGURE 179. Loose (separation)—major. (Hose does not round out ahead of inspector's right hand); (see fig. 178.)

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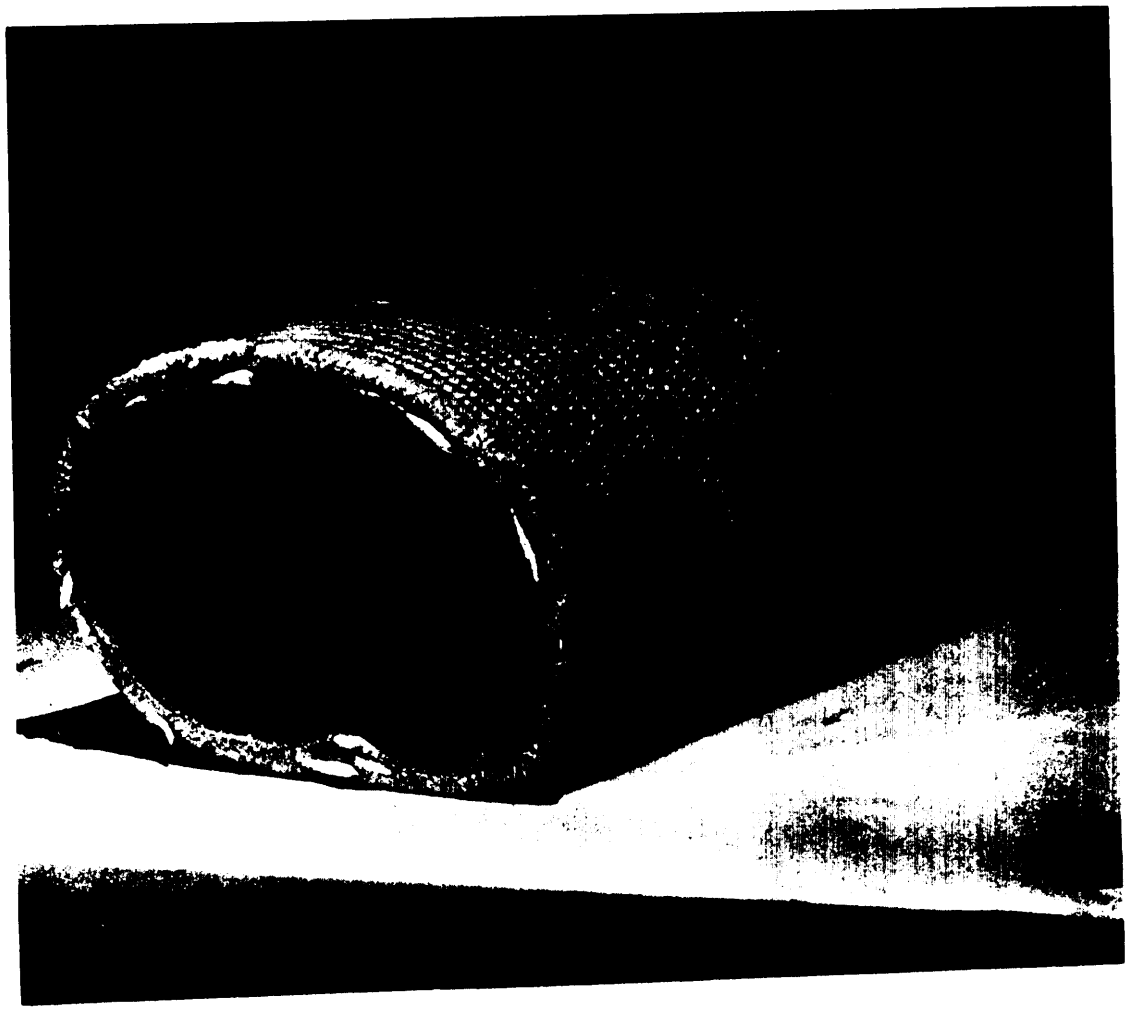


FIGURE 180. Loose (separation)—major.

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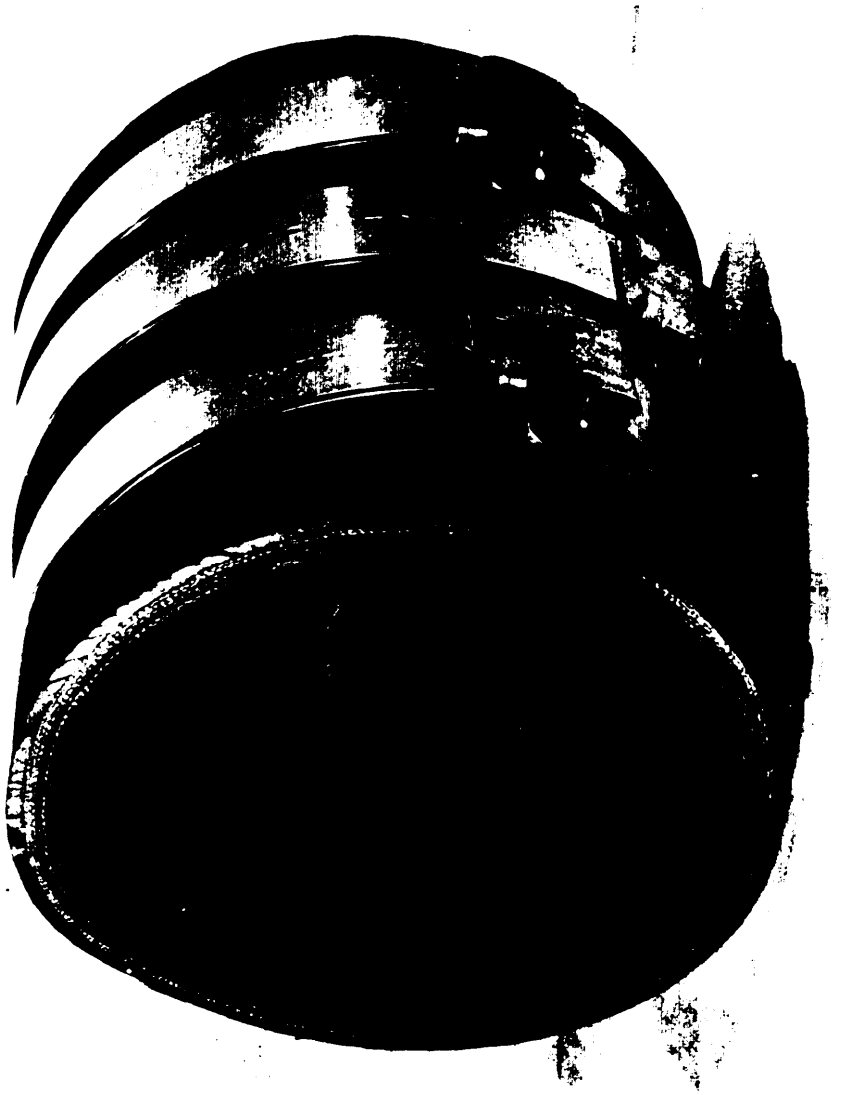


FIGURE 181. Loose (separation)—major.

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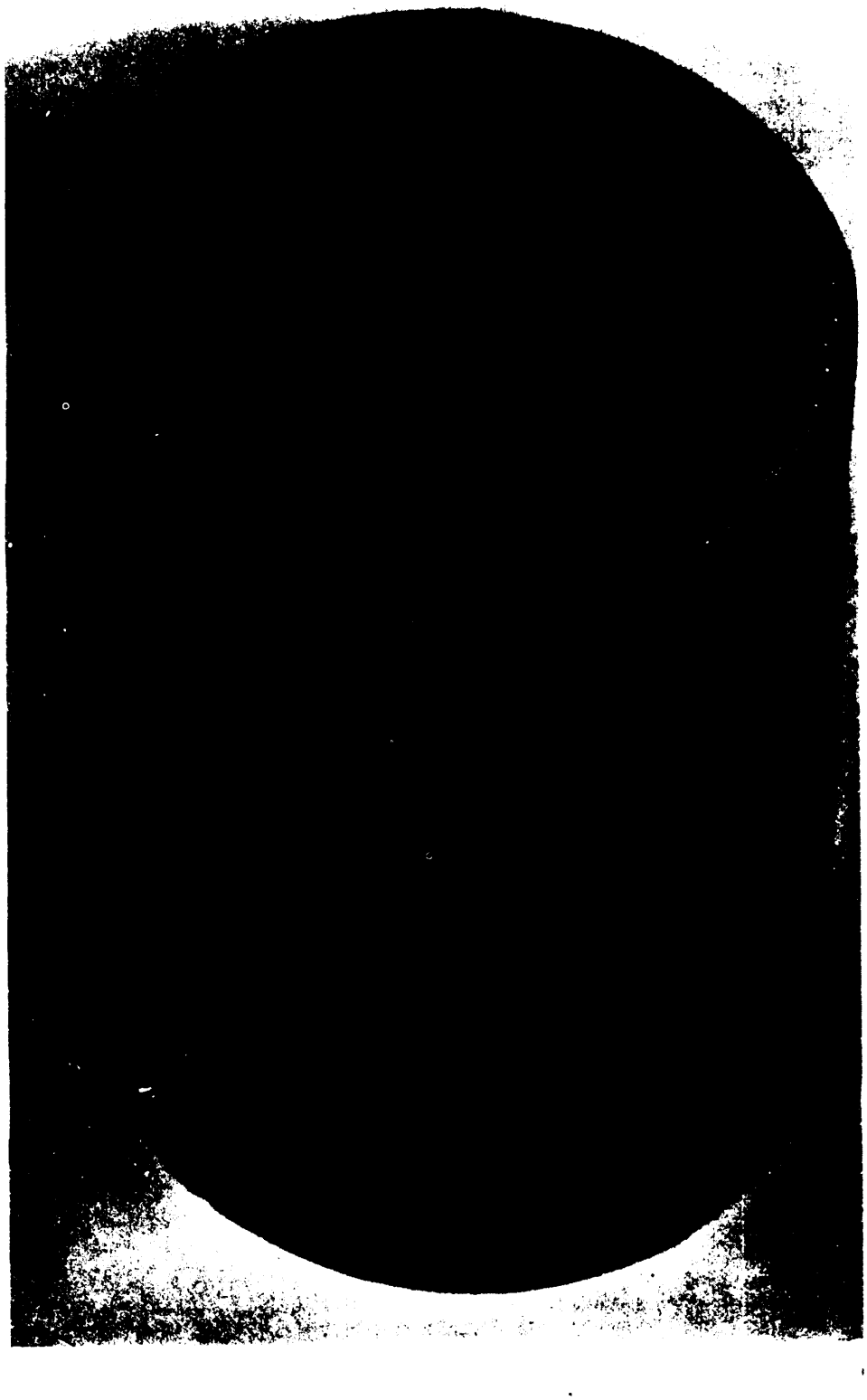


FIGURE 182. Loose (separation)—major.

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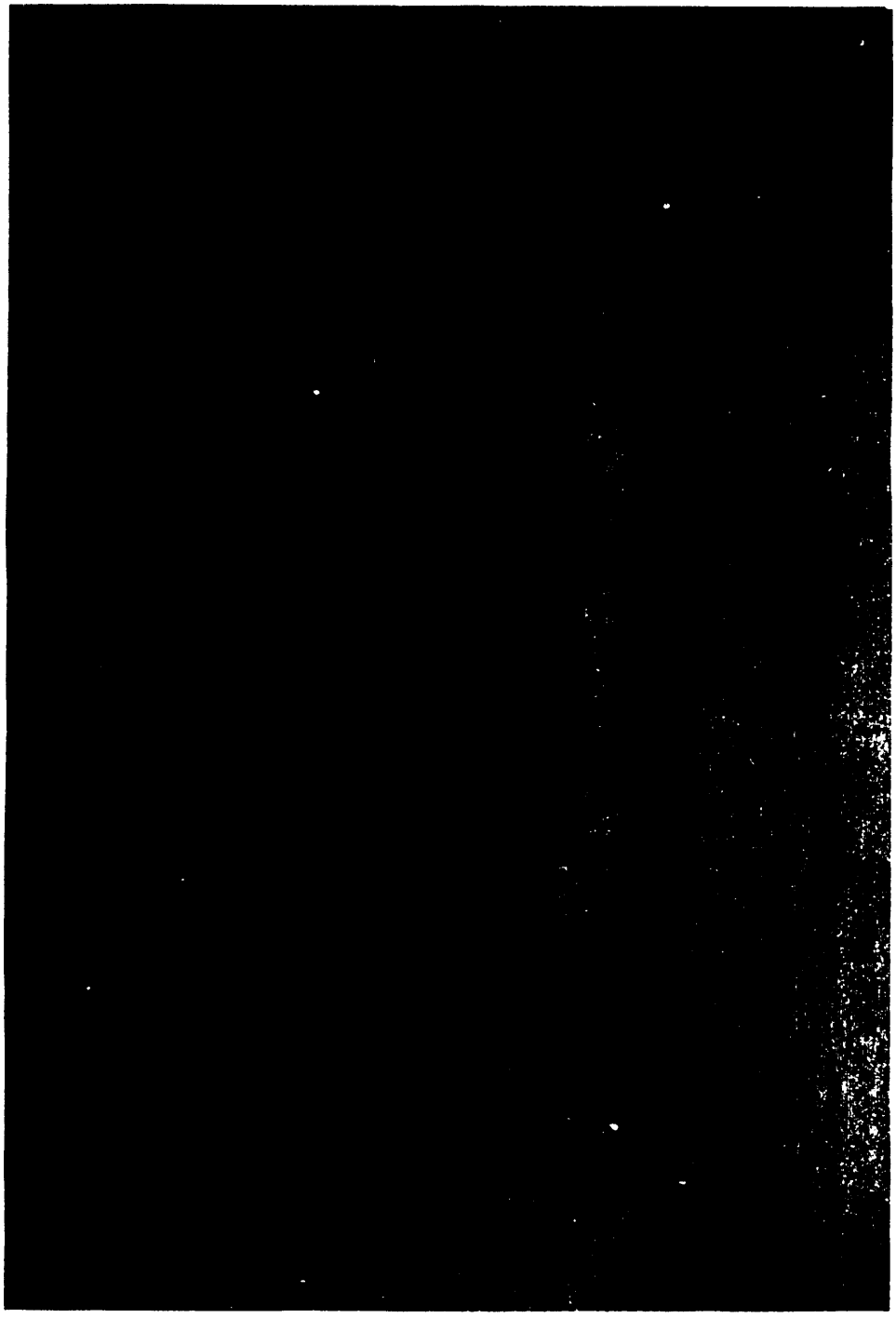


FIGURE 183. Lubricant, excess or loose: class A—major; classes B and C—minor.

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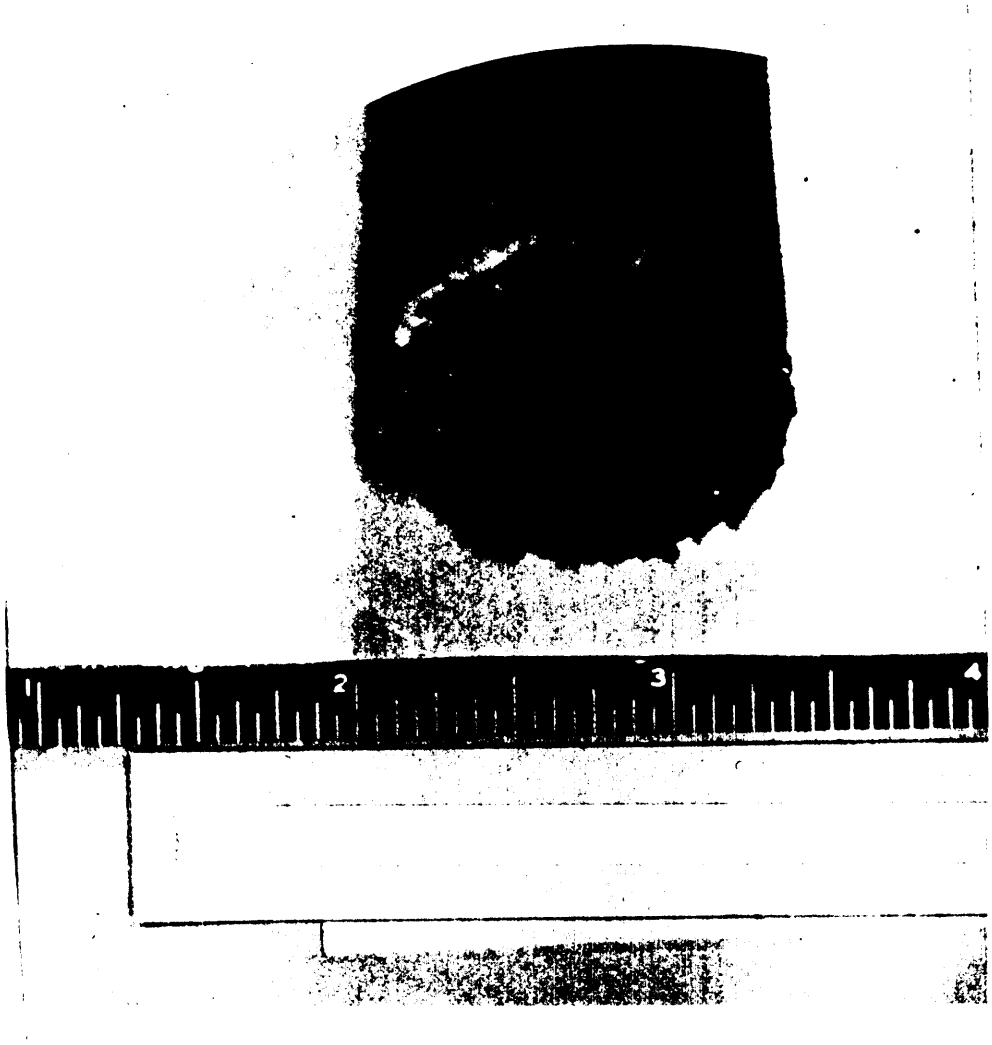


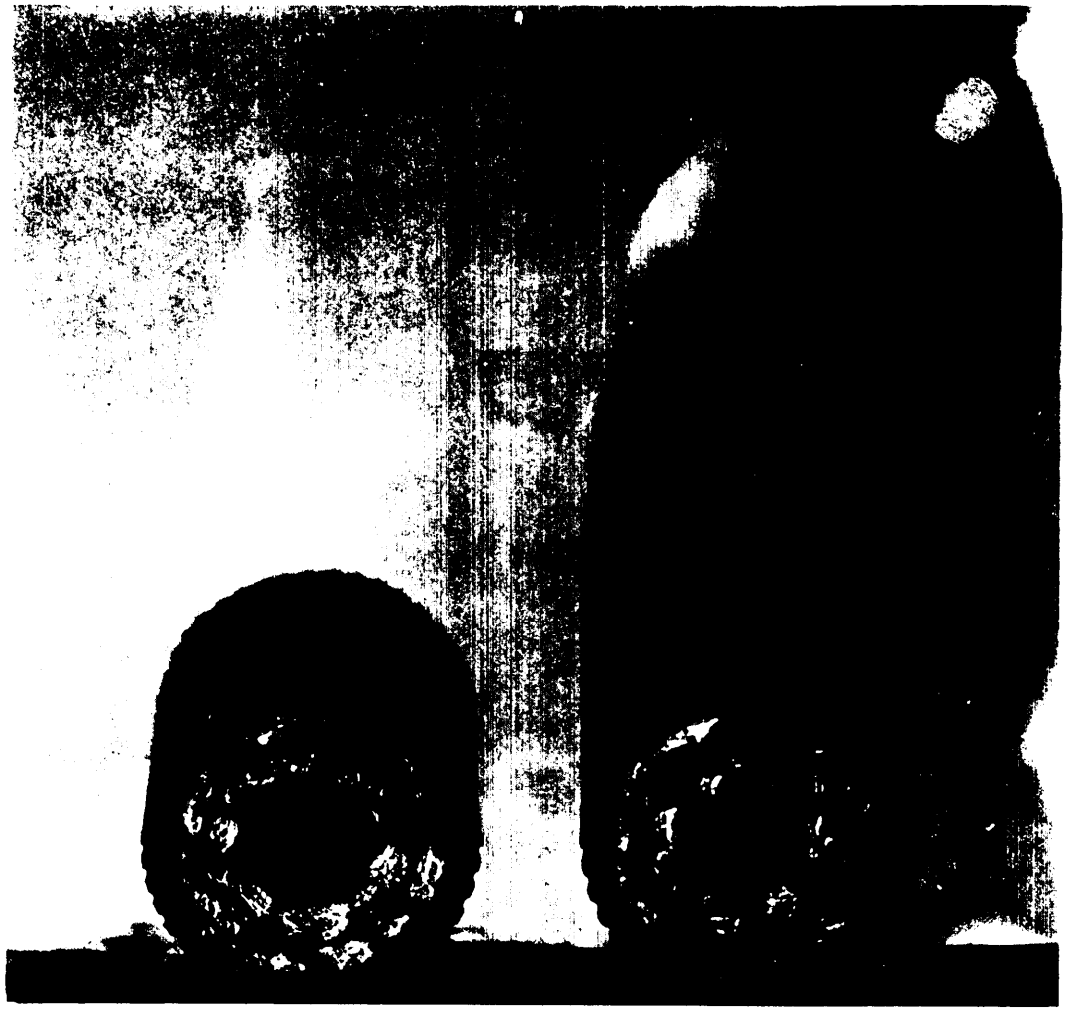
FIGURE 184. Lubricant, normal amount—not a defect.

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FIGURE 185. Tube missing—major.

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I II

FIGURE 186. Tube missing:
I—Normal hose.
II—Major.

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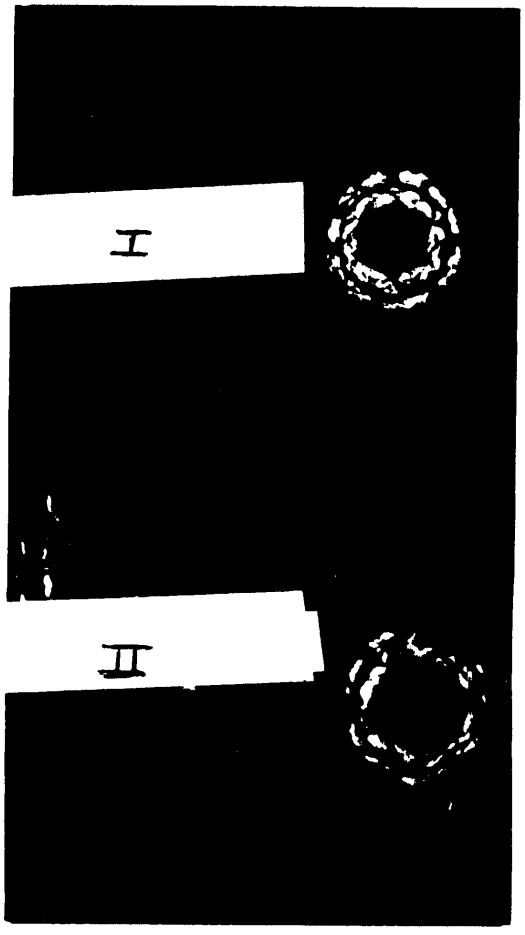


FIGURE 187. Off-center tube:
I—Thickness meets tolerance—not a defect.
II—Thickness below tolerance—major.

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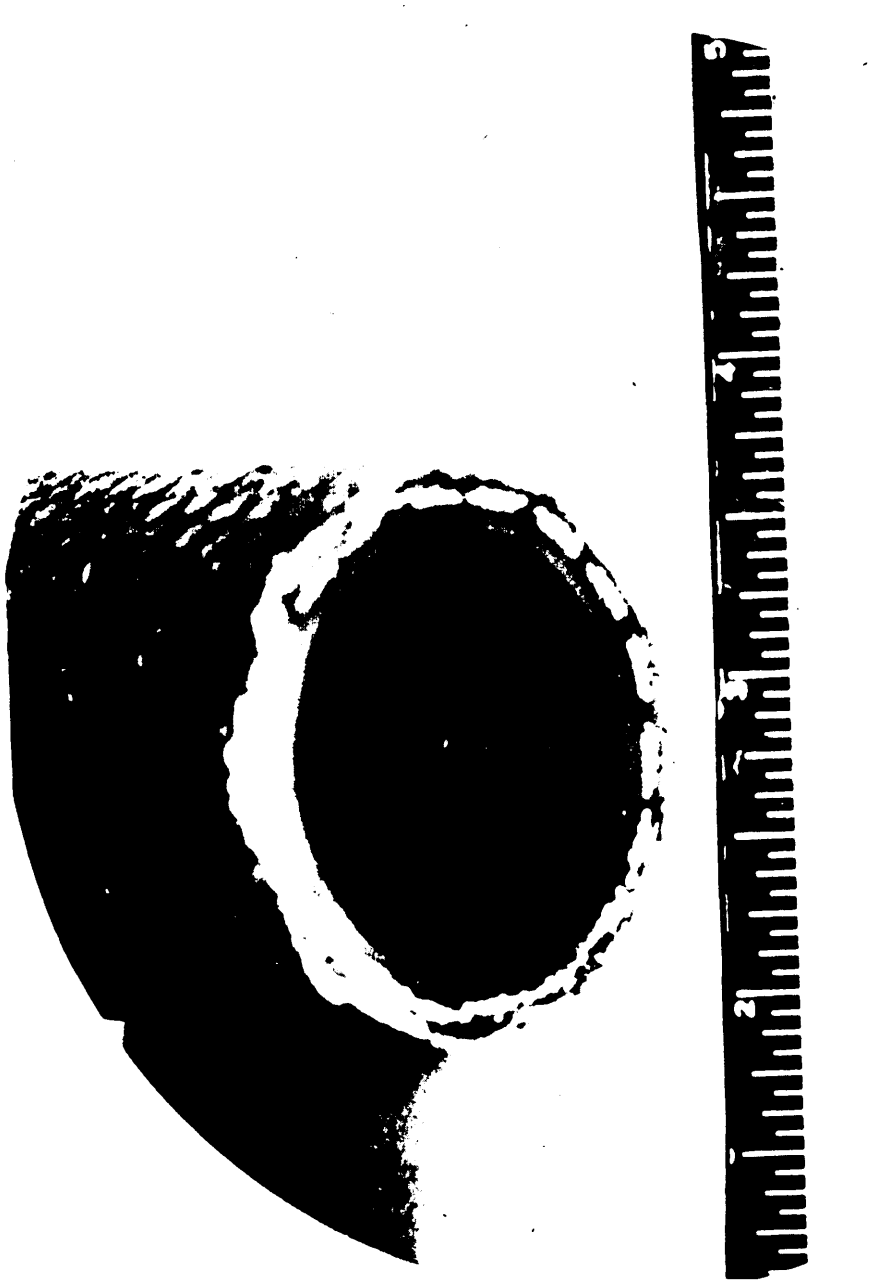


FIGURE 188. Pits deep—major.

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FIGURE 189. Pits deep—major.

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FIGURE 190. Pits, small, less than 15 percent of tube thickness: class A—major; classes B and C—minor.

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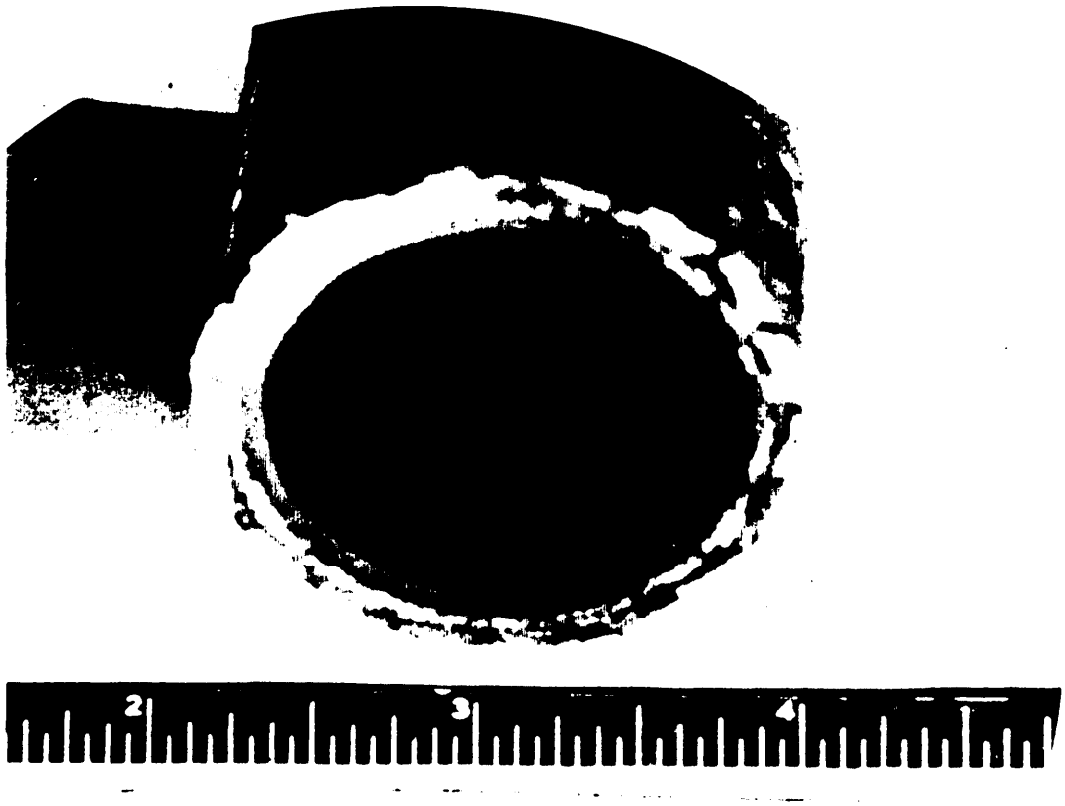


FIGURE 191. Pits, small irregularities—not a defect.

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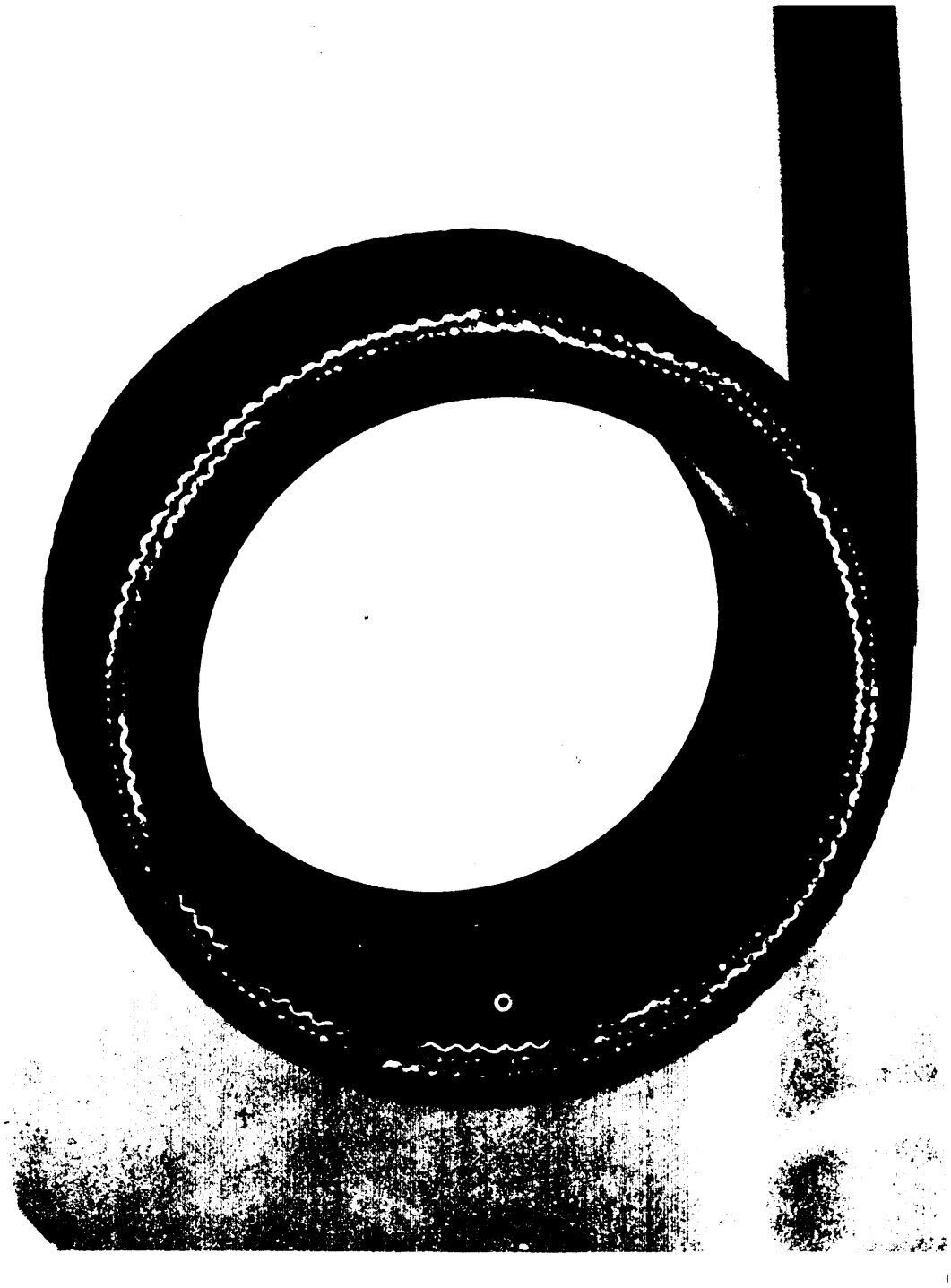


FIGURE 192. Porosity, more than 15 percent of tube thickness—major.

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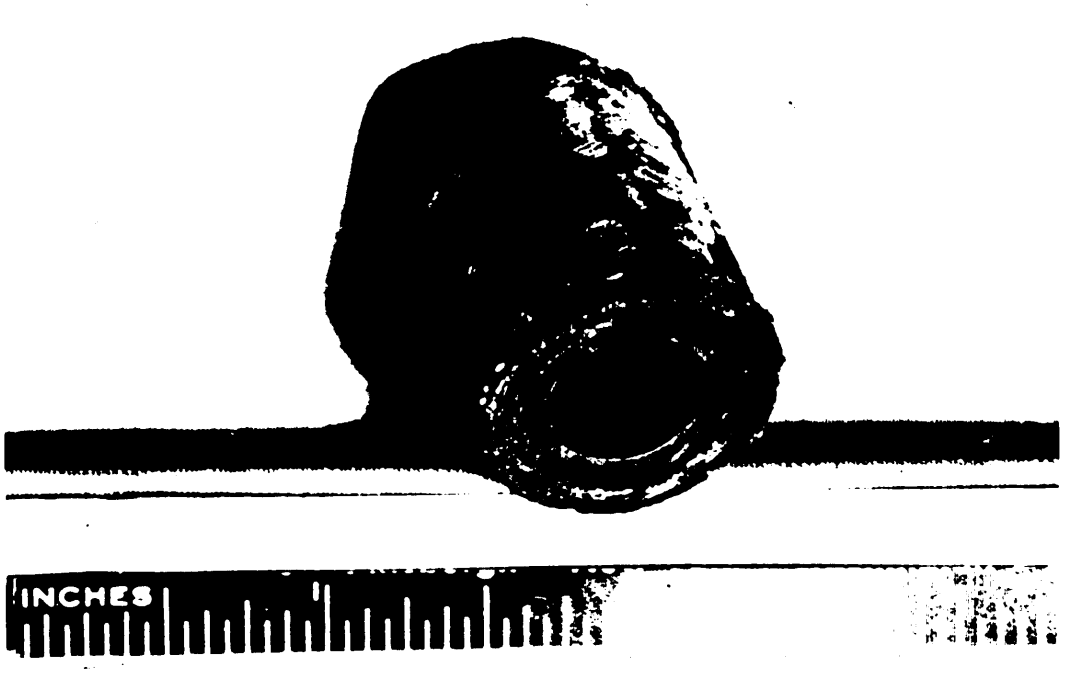


FIGURE 193. Ring: class A—major; classes B and C—not a defect.

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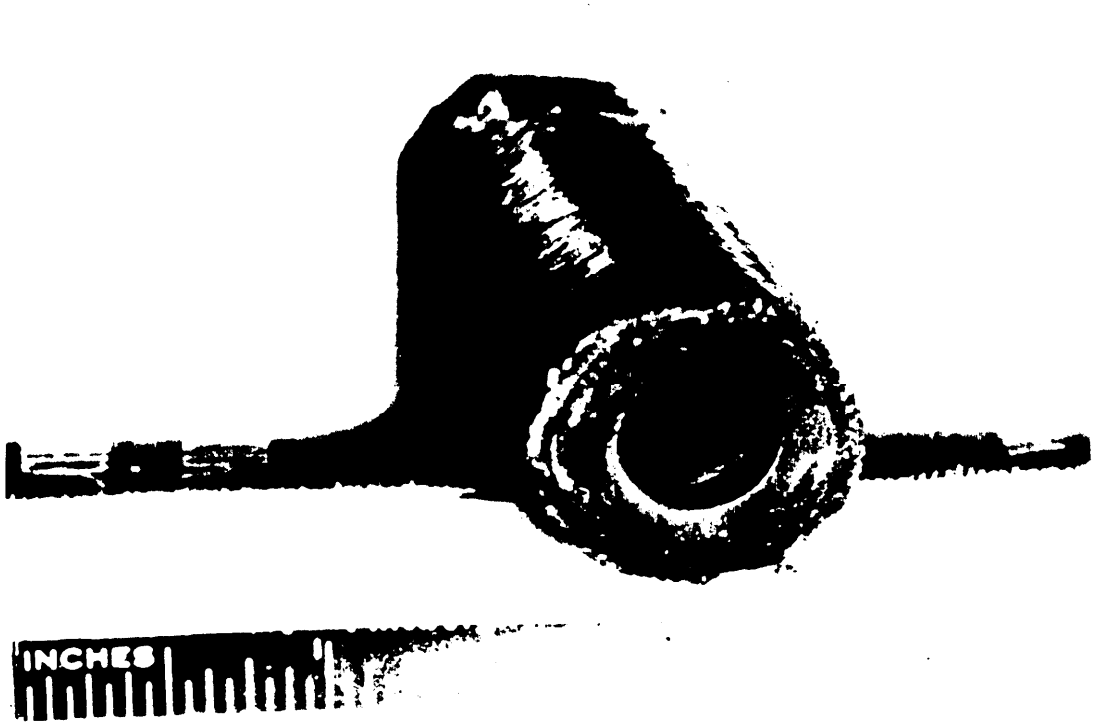


FIGURE 194. Ring: class A—major; classes B and C—not a defect.

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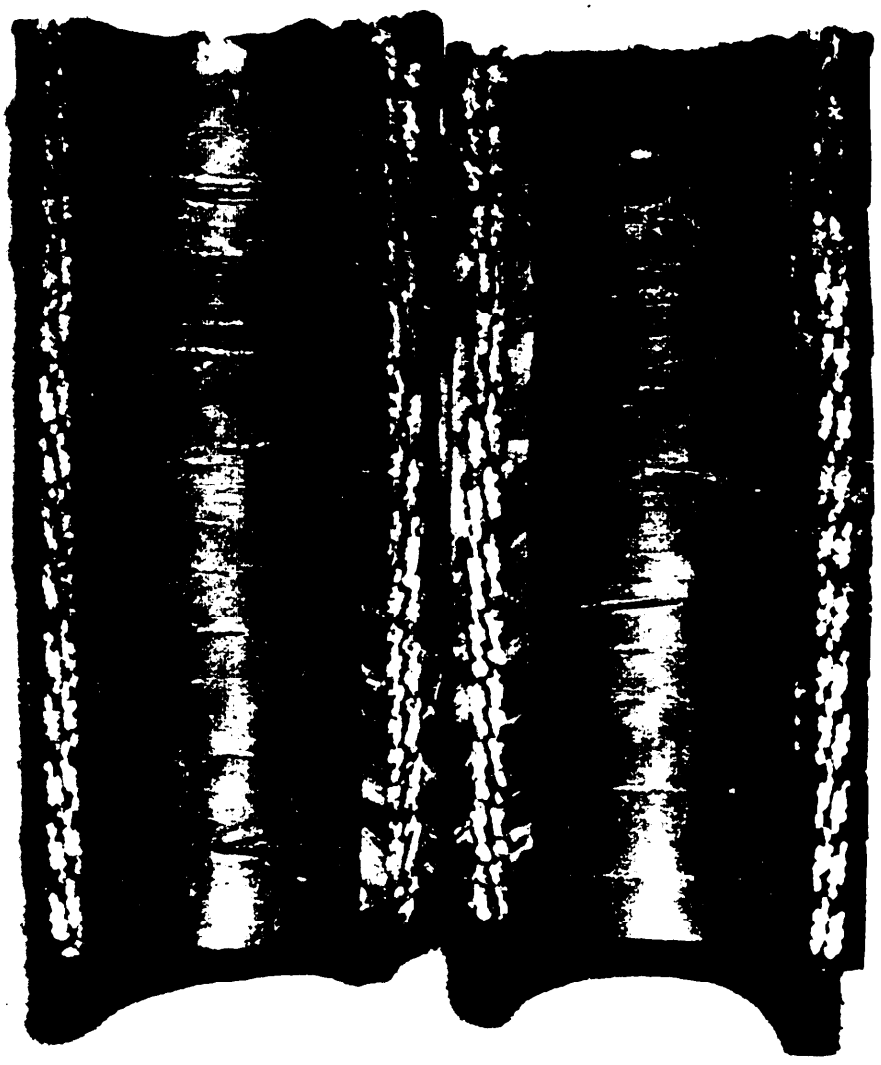


FIGURE 195. Ridge: over 1/64 inch or over 10 percent of tube thickness, whichever is greater—major; less than 1/64 inch or 10 percent of tube thickness, whichever is greater: class A—minor: classes B and C—not a defect.

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FIGURE 196. Ridge: over 1/64 inch or over 10 percent of tube thickness, whichever is greater—major; less than 1/64 inch or 10 percent of tube thickness, whichever is greater: class A—minor; classes B and C—not a defect.

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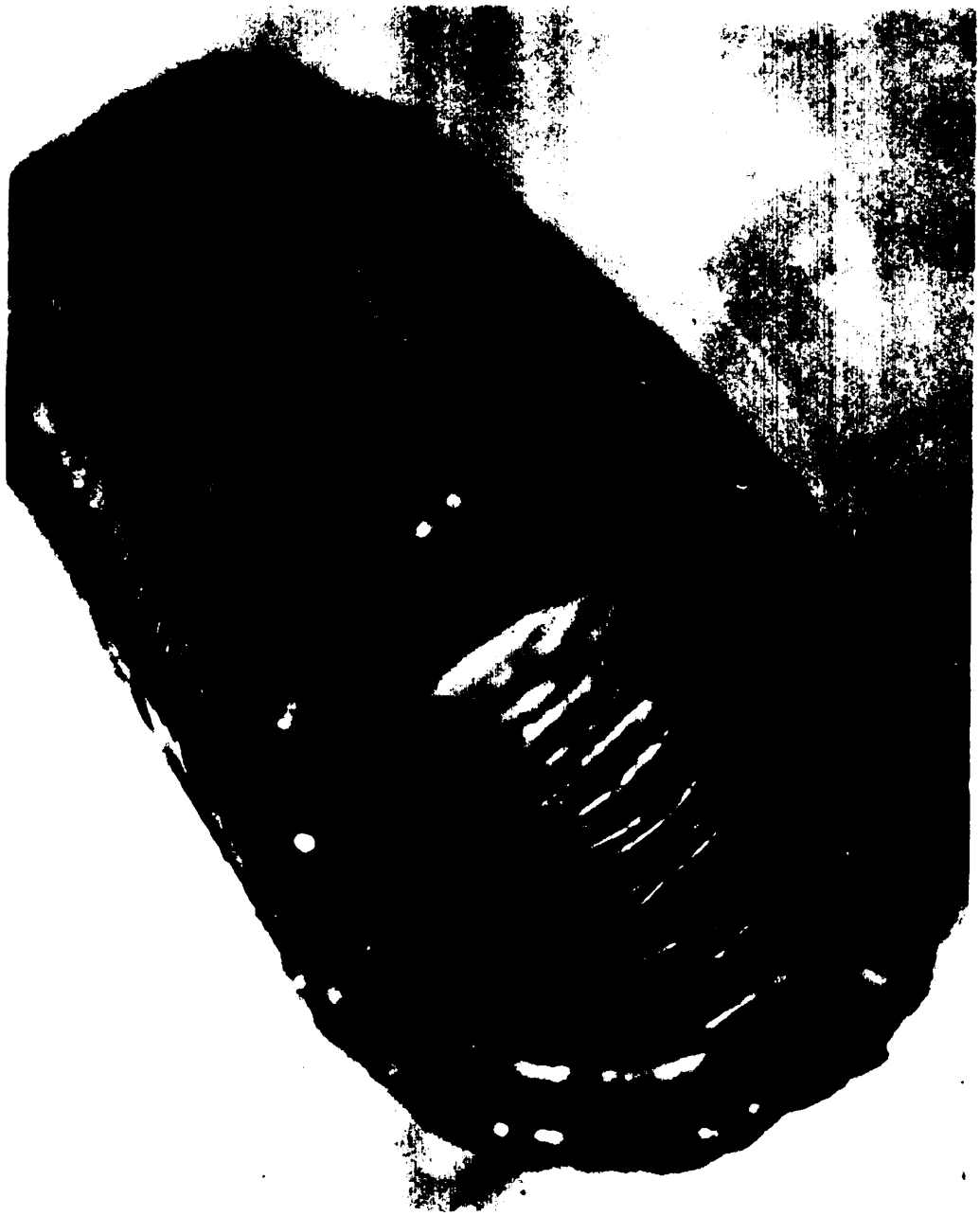


FIGURE 197. Rough, over $^{1}/_{64}$ inch or over 10 percent of tube thickness, whichever is greater—major.

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FIGURE 198. Rough, less than $^{1}/_{64}$ inch or less than 10 percent of tube thickness, whichever is greater:
I—Acceptable.
II—Rough (corrugated)—minor.
III—Roughness—minor.

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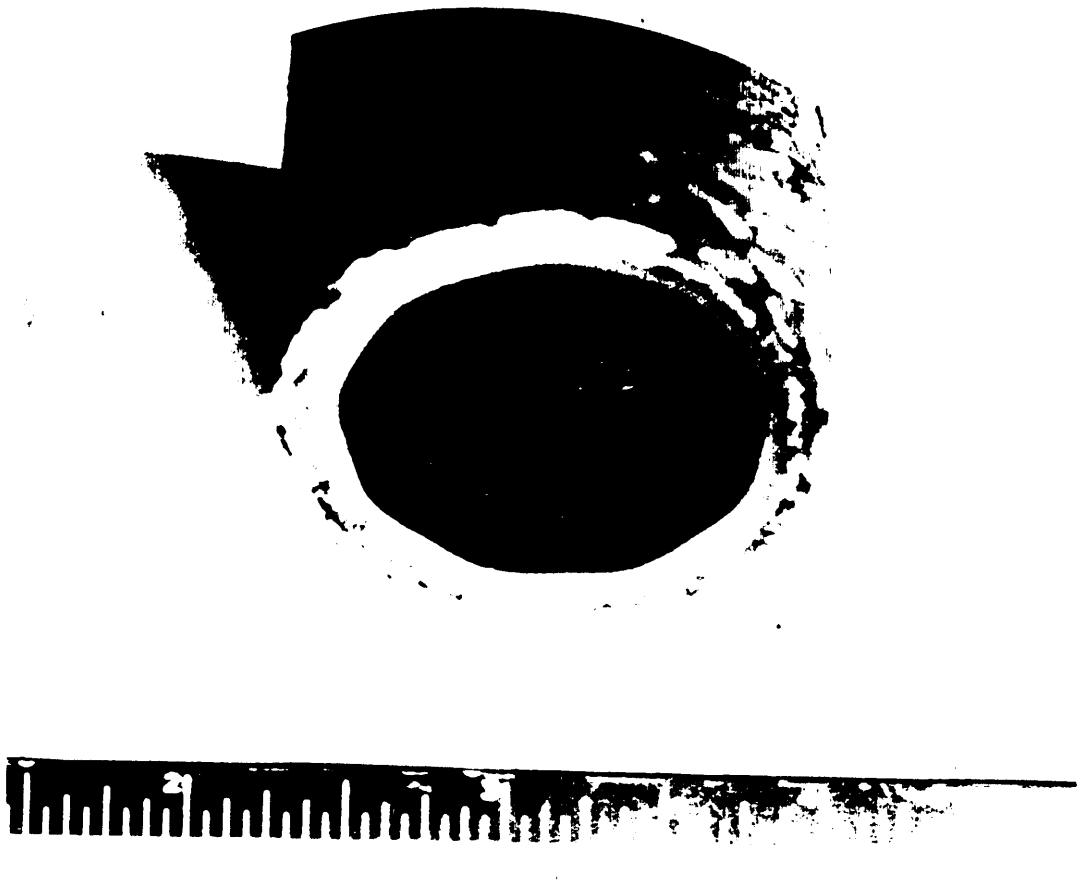


FIGURE 199. Scored (slit)—major.

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FIGURE 200. Surface irregularity: when assembled with swaged fittings—major; all others—minor.

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FIGURE 201. Surface irregularity: when assembled with swaged fittings—major; all others—minor.

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FIGURE 202. Surface irregularity: when assembled with swaged fittings—major; all others—minor.

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FIGURE 203. Wire through tube—major.

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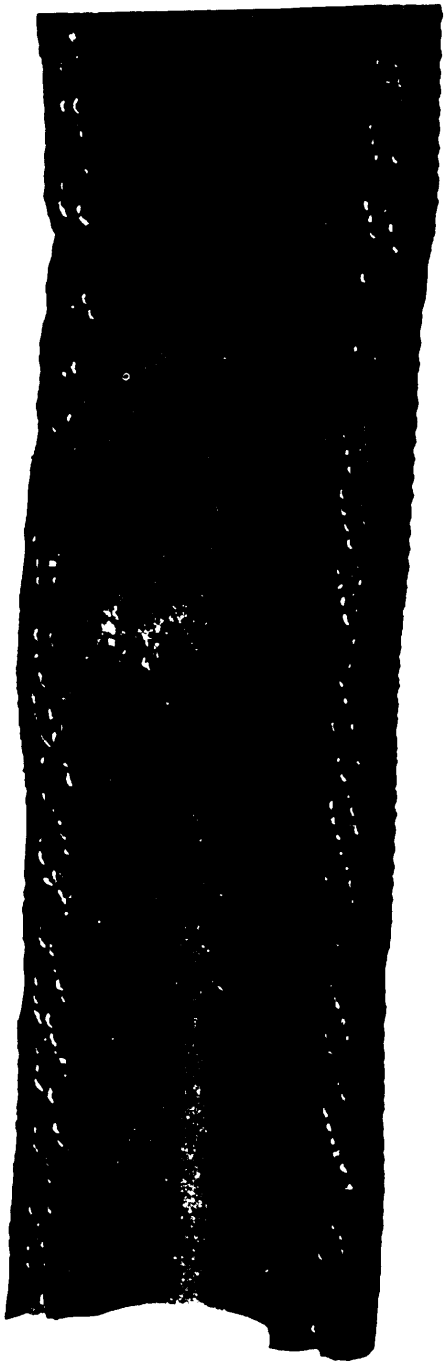


FIGURE 204. Yarn through tube—major.

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